

BIOL 250 — Systemic Comprehensive Study Notes

Endocrine (Ch 19)

19.1 An Overview of the Endocrine System

- Endocrine system = all endocrine cells/tissues producing hormones; endocrine cells are glandular secretory cells that release hormones directly into interstitial fluid, lymph, or blood (no ducts).
- Hormones = chemical messengers ("to excite") that stimulate specific target cells/tissues; effects are longer-lasting (days) vs. nervous system's short-term (seconds) responses.
- Nervous vs endocrine: nervous system = short-term crisis management via electrical/chemical signals at synapses; endocrine = long-term regulation of ongoing processes (growth, development) via bloodborne chemicals.
- Overlap example: adrenal medulla = modified sympathetic ganglion (neurons secrete epinephrine/norepinephrine into blood) — anatomically nervous, functionally endocrine. Hypothalamus is anatomically part of brain but functionally part of endocrine system (secretes hormones).
- Three chemical classes of hormones:
 1. **Amino acid derivatives** — small molecules structurally similar to amino acids.
 - Tyrosine derivatives: thyroid hormones (thyroid gland), catecholamines (epinephrine, norepinephrine, dopamine — adrenal medulla).
 - Tryptophan derivatives: melatonin (pineal gland).
 2. **Peptide hormones** — largest group; chains of amino acids; e.g., all pituitary hormones.
 3. **Lipid derivatives** — two subtypes:
 - Eicosanoids — small molecules with a five-carbon ring at one end; released by most body cells; coordinate local cellular activities, affect enzymatic processes (e.g., blood clotting).
 - Steroid hormones.
- Hormones act on **target cells** = specific cells bearing receptors for that hormone; hormones alter cellular operations by changing types/activity/quantity of key cytoplasmic enzymes (enzymes control all cellular/metabolic activities).
- Endocrine activity is triggered by three stimulus types:
 1. Humoral stimuli — changes in extracellular fluid composition (e.g., blood Ca^{2+} or glucose levels).
 2. Hormonal stimuli — arrival or removal of a specific hormone.
 3. Neural stimuli — neurotransmitter arrival at neuroglandular junctions.
 - Most endocrine responses are regulated by **negative feedback**; a few (must be rushed to completion) use **positive feedback** — hormone secretion produces an effect that further stimulates more hormone release. Example: oxytocin during labor — uterine contractions stimulate more oxytocin release.

19.2 Hypothalamus and the Pituitary Gland

- Hypothalamus = highest level of endocrine control; integrates nervous and endocrine systems via neural stimuli; closely linked to both pituitary lobes.
- **Pituitary gland (hypophysis)**: weighs ~1/5 oz (~6 g), size/weight of a small grape; lies inferior to hypothalamus within the **sella turcica** (depression in sphenoid bone).

- **Infundibulum** connects hypothalamus to the posterior/superior surface of the pituitary. **Diaphragma sellae** = dural sheet encircling the infundibulum, holding pituitary in place.
- Hypothalamus controls pituitary in 3 ways: (1) produces ADH & oxytocin, released at posterior lobe; (2) controls anterior lobe via releasing/inhibiting hormones; (3) controls adrenal medulla output as part of sympathetic ANS.
- **Hypophyseal portal system**: At the **median eminence** (swelling near infundibulum attachment), hypothalamic neurons release regulatory hormones into a primary capillary plexus; portal vessels carry these to a secondary capillary plexus in the anterior lobe, delivering regulatory hormones directly to anterior lobe endocrine cells before they reach general circulation. Ensures concentrated local hormone delivery.
- Two classes of regulatory hormones: **releasing hormones (RH)** — increase anterior pituitary hormone secretion; **inhibiting hormones (IH)** — prevent anterior pituitary hormone secretion. Each targets specific anterior-lobe endocrine cells.

Anterior Lobe (adenohypophysis)

- Also called adenohypophysis; contains varied endocrine cells; 3 regions: **pars distalis** (major portion), **pars tuberalis** (wraps infundibulum), **pars intermedia** (narrow band).
 - Extensive capillary network gives every endocrine cell access to bloodstream.
 - Hormones regulating other glands = **tropic hormones** (names indicate targets/activity).
 - Six hormones from pars distalis + one from pars intermedia (total 7 anterior lobe hormones):
1. **TSH (thyroid-stimulating hormone)** — secreted by **thyrotropes**; targets thyroid gland; triggers release of thyroid hormones.
 2. **ACTH (adrenocorticotropic hormone)** — secreted by **corticotropes**; stimulates release of steroid hormones (glucocorticoids) from adrenal cortex; glucocorticoids affect glucose metabolism.
 3. **FSH (follicle-stimulating hormone)** — secreted by **gonadotropes**; in women promotes oocyte development within ovarian follicles and stimulates estrogen (estradiol) secretion by follicular cells; in men, stimulates sperm maturation.
 4. **LH (luteinizing hormone)** — induces ovulation in women; promotes progesterone (progesterone) secretion by the ovary/corpus luteum; in men, stimulates testosterone (androgen) production by interstitial cells of testes. FSH + LH = **gonadotropins** (regulate gonads).
 5. **Prolactin (PRL)** — secreted by **lactotropes**; stimulates mammary gland development and milk production; mammary regulation also involves estrogen, progesterone, GH, glucocorticoids, placental hormones, but PRL has greatest effect on glandular cells; male functions poorly understood.
 6. **Growth hormone (GH)** — also called human growth hormone (HGH) or somatotropin; secreted by **somatotrophs**; stimulates cell growth/replication by accelerating protein synthesis; every tissue responds, but GH especially promotes protein synthesis/growth in bone and muscle. Liver cells respond to GH by synthesizing/releasing **somatomedins** (peptide hormones stimulating protein synthesis and growth in skeletal muscle, cartilage, other targets). Deficient GH in children → **pituitary growth failure / pituitary dwarfism** (no normal growth/maturation before/during puberty).
 7. **MSH (melanocyte-stimulating hormone)** — only hormone from pars intermedia; increases melanin production/distribution in skin melanocytes; secreted by corticotropes (ACTH cells) only during fetal development, in young children, pregnant women, and some disease states.
 - Endocrine disorder example — **diabetes insipidus** (clinical note): caused by inadequate ADH release from posterior pituitary (neurohypophysis); impairs renal water conservation → excessive water loss in urine, constant thirst without fluid retention; severe cases → fluid loss up to 10 L/day, fatal dehydration if untreated; treated with **desmopressin acetate (DDAVP)** nasal spray (synthetic ADH), also used for bed-wetting.

Posterior Lobe (neurohypophysis)

- Extension of the CNS; contains axons of hypothalamic neurons (~50,000), with cell bodies in **supra-optic nucleus** (produces ADH) and **paraventricular nucleus** (produces OXT) of hypothalamus.
- No portal system; blood supplied by inferior hypophyseal artery; hypophyseal veins carry hormones away.
- Stores/releases 2 hormones made in hypothalamus ("neurosecretions" because produced/released by neurons):

1. **ADH (antidiuretic hormone / vasopressin)** — released in response to (1) rise in blood electrolyte concentration or (2) fall in blood volume/pressure; decreases urine volume produced by kidneys (promotes water reabsorption); also constricts peripheral blood vessels, raising blood pressure.
2. **Oxytocin (OXT)** — stimulates smooth muscle contraction of (1) uterus and (2) myoepithelial (contractile) cells around mammary secretory cells. Required for labor/childbirth (positive feedback loop); suckling triggers release → milk ejection. In males, causes smooth muscle contraction in ductus deferens and prostate.

19.3 The Thyroid Gland

- Location: anterior surface of trachea, inferior to thyroid cartilage ("shield-shaped") and cricoid cartilage ("ring-shaped") of larynx; palpable.
- Shape: butterfly-like, two lobes joined by a slender **isthmus**; superior portions extend to inferior border of thyroid cartilage; inferiorly extends to 2nd–3rd tracheal cartilage ring.
- Deep red color due to rich blood supply. Blood supply: paired **superior thyroid arteries** (from external carotid arteries) and paired **inferior thyroid arteries** (from thyrocervical trunks). Venous drainage: superior & middle thyroid veins → internal jugular veins; inferior thyroid veins → brachiocephalic veins.
- Connective tissue capsule attaches gland to trachea; extends inward, dividing gland into lobules and surrounding thyroid follicles.
- **Thyroid follicles** = functional unit of thyroid gland; spherical (like a tennis ball) — wall = follicular epithelium (simple cuboidal, made of **T thyrocytes**/follicular cells), hollow interior = **follicle cavity** containing **colloid** (viscous fluid with suspended thyroglobulin protein).
- Epithelium height reflects activity: inactive gland → low simple cuboidal; highly active gland → simple columnar.
- Capillary network surrounds each follicle, delivering nutrients/hormones and removing secretory products/wastes.
- **T thyrocytes**: protein secretors — many mitochondria, extensive RER; synthesize **thyroglobulin** (contains tyrosine, the building block of thyroid hormones), secreted into colloid. T thyrocytes actively transport iodide ions (I^-) from interstitial fluid into the cell; iodide converted to iodine (I^0) at apical membrane, attaches to tyrosine residues on thyroglobulin within the follicle cavity.
- **Thyroxine (T4 / tetraiodothyronine)** = 4 iodine atoms; **triiodothyronine (T3)** = related molecule (fewer iodines). Thyroid is the only endocrine gland that stores its hormone product extracellularly (in colloid).
- Regulation: **TSH** (from anterior pituitary) stimulates active transport of iodide into follicle cells and controls synthesis/release of T3/T4; TSH release is triggered by **TRH (thyrotropin-releasing hormone)** from hypothalamus. Under TSH, follicle cells remove thyroglobulin from follicles by endocytosis; lysosomal enzymes break it down, releasing T3 and T4, which diffuse into bloodstream.
- ~90% of thyroid secretion is T4; T3 secreted in much smaller amounts. Both increase rate of cellular metabolism and O₂ consumption in almost every body cell.
- **C thyrocytes (parafollicular / clear cells)**: second endocrine cell type in thyroid; sandwiched between cuboidal follicle cells and their basement membrane; larger than follicular cells, don't stain as clearly. Produce **calcitonin (CT)**: helps regulate Ca²⁺ concentration in body fluids, especially important during childhood (stimulates skeletal growth); also reduces bone mass loss during (1) prolonged starvation and (2) late pregnancy.
- Summary table values: T thyrocytes → T4, T3 → target most cells → increase energy utilization, O₂ consumption, growth. C thyrocytes → calcitonin (CT) → target bone & kidneys → decreases blood Ca²⁺ (uncertain overall significance stated in text, though it lowers calcium).

19.4 The Parathyroid Glands

- Four pea-sized, reddish-brown glands, typically on posterior surface of thyroid gland, embedded in thyroid connective tissue; surrounded by their own connective tissue capsule, which extends inward forming small irregular lobules.

- Blood supply: superior parathyroids ← superior thyroid arteries; inferior parathyroids ← inferior thyroid arteries; venous drainage same as thyroid.
- Two cell types:

1. **Parathyroid (principal) cells** — produce **parathyroid hormone (PTH)**.

2. **Oxyphil cells** — function unknown; absent until after puberty, then increase in number with age.

- Parathyroid cells monitor blood Ca^{2+} concentration (like C thyrocytes but opposite response): when Ca^{2+} falls below normal, parathyroid cells secrete PTH.
- PTH actions: increases blood Ca^{2+} by (1) stimulating osteoblasts to secrete a growth factor that activates osteoclasts (increasing bone resorption/turnover), (2) reduces Ca^{2+} loss in urine, (3) stimulates production of **calcitriol** (kidney hormone) which promotes intestinal Ca^{2+} absorption. PTH levels stay elevated until blood Ca^{2+} normalizes.
- Clinical: PTH shown effective in reducing progression of osteoporosis in elderly.
- Concept check: parathyroid removal → decreased blood calcium.

19.5 The Thymus

- Location: posterior to sternum, within the mediastinum of thoracic cavity.
- Size changes with age: relatively large in newborns/young children, extending from base of neck to superior border of heart; continues to slowly enlarge; reaches max size just before puberty (~40 g); after puberty gradually shrinks — by age 50, weighs less than 12 g.
- Produces hormones important for development/maintenance of immune defenses. "**Thymosin**" originally named for an extract promoting lymphocyte (white blood cell) development/maturation; now known to be a blend of several hormones: thymosin-1, thymopoietin, thymopentin, thymulin, thymic humoral factor, and IGF-1.
- Hypothesis: gradual decline in thymus size/secretory ability with age may increase disease susceptibility in the elderly.
- (Detailed histology/thymosin functions covered in Ch 23.)

19.6 The Adrenal Glands

- Paired adrenal (suprarenal) glands: yellow, pyramid-shaped, attached to superior border of each kidney via dense fibrous capsule; nestled among kidney, diaphragm, major abdominal vessels; anterior surface covered by parietal peritoneum; highly vascularized.
- Blood supply: branches of renal artery, inferior phrenic artery, and a direct branch of aorta (middle adrenal artery); adrenal veins drain blood away.
- Each gland weighs ~7.5 g; usually heavier in men than women; size varies with secretory demand.
- Divided into: superficial **cortex** and inner **medulla**.

Adrenal Cortex

- Yellowish color from stored lipids (cholesterol, fatty acids). Produces >24 different steroid hormones (**corticosteroids/adrenocortical steroids**).
- Three zones (superficial to deep):

1. **Zona glomerulosa** — outermost; densely packed clusters of cells ("glomerulus" = little ball); occupies ~15% of cortex; extends from capsule to zona fasciculata. Produces **mineralocorticoids (MCs)** — affect electrolyte composition of body fluids. Main MC = **aldosterone**: stimulates Na^{+} conservation and K^{+} elimination; causes retention of Na^{+} by kidneys, sweat glands, salivary glands, pancreas; prevents Na^{+} loss in urine/sweat/saliva/digestive secretions; K^{+} loss accompanies Na^{+} retention. Aldosterone secretion triggered by: (1) decreased blood Na^{+} , (2) increased blood K^{+} , (3) arrival of angiotensin II (produced via kidney renin).

2. **Zona fasciculata** — middle zone; cells have more lipid droplets (pale, foamy cytoplasm) than zona glomerulosa; cells form cords radiating outward like a sunburst from zona reticularis, separated by fenestrated sinusoids. ACTH (anterior pituitary) stimulates steroid production here. Produces **glucocorticoids (GCs)** (named for glucose

metabolism effects): **cortisol** (hydrocortisone) and **corticosterone** = most important; liver converts some circulating cortisol to cortisone (another active GC). These hormones speed glucose synthesis and glycogen formation, especially in liver.

3. **Zona reticularis** — deepest, smallest zone (~5% of cortex); cells smaller than others, form folded branching network with extensive capillary supply; easily distinguished from adjacent adrenal medulla. Secretes small amounts of **androgens**: stimulate pubic hair development in boys and girls before puberty; in women, promote muscle mass, blood cell formation, support sex drive; less important in men (testes make much more androgen).

- All cortical cells have extensive smooth ER (SER) for steroid/lipid synthesis (contrasts with RER-rich protein-secreting cells like anterior pituitary or thyroid).

Adrenal Medulla

- Pale gray/pink (many blood vessels). Contains **chromaffin cells** — large, rounded cells resembling sympathetic ganglion neurons; innervated by preganglionic sympathetic fibers; sympathetic activation via splanchnic nerves triggers secretion.
- Two secretory cell populations: one secretes **epinephrine (adrenaline)**, other **norepinephrine (noradrenaline)** — both catecholamines. Medulla secretes ~3x more epinephrine than norepinephrine (chapter review states 75-80% epinephrine, 20-25% norepinephrine).
- Catecholamine release speeds cellular energy use and mobilizes energy reserves → increases muscular strength/endurance. Effects peak 30 seconds after adrenal medulla stimulation and last several minutes — outlasting direct sympathetic nervous effects.
- Concept check: disorder preventing Na⁺ retention → zona glomerulosa affected, aldosterone deficient.

19.7 Endocrine Functions of the Kidneys and Heart

- Kidneys produce the enzyme **renin** and two hormones: peptide **erythropoietin (EPO)** and steroid **calcitriol**.
- **Renin-angiotensin pathway**: renin converts circulating **angiotensinogen** (inactive liver protein) → **angiotensin I**; angiotensin I converted to **angiotensin II** in lung capillaries; angiotensin II stimulates aldosterone secretion by adrenal cortex.
- **Erythropoietin (EPO)**: released in response to low O₂ levels in kidneys; stimulates red blood cell production in red bone marrow → increases blood volume and improves O₂ delivery. Clinical: used in "blood doping" by endurance athletes for competitive advantage (increases RBCs/O₂-carrying capacity).
- **Calcitriol**: steroid hormone secreted by kidneys in response to PTH; synthesis depends on availability of **cholecalciferol (vitamin D3)** (synthesized in skin or absorbed from diet); liver converts cholecalciferol to an intermediate, released into circulation, absorbed by kidneys, converted to calcitriol. Best-known function: stimulates absorption of calcium and phosphate ions along digestive tract. Since PTH stimulates calcitriol release, PTH has an indirect effect on intestinal calcium absorption.
- **Atrial natriuretic peptide (ANP)**: produced by cardiac muscle cells in response to increased blood pressure or blood volume. Effects oppose angiotensin II: promotes loss of Na⁺ and water by kidneys; inhibits renin release, ADH, and aldosterone secretion.

19.8 The Pancreas and Other Endocrine Tissues of the Digestive System

- Digestive organs (pancreas, digestive tract lining, liver) produce exocrine secretions essential for digestion; also communicate via hormones (detailed in Ch 25). Autonomic nervous system influences pace, but digestion mostly locally controlled.
- **Pancreas**: both exocrine and endocrine gland; lies in abdominopelvic cavity within first fold of small intestine (duodenum), close to stomach; slender, pale, lumpy organ; ~20–25 cm (8–10 in.) long, weighs ~80 g (2.8 oz).
- Exocrine pancreas = ~99% of pancreatic volume; produces enzyme-rich fluid delivered via secretory ducts to digestive tract.
- Endocrine pancreas = small clusters called **pancreatic islets** (islets of Langerhans); only ~1% of pancreatic cell population but ~2 million islets in a normal pancreas.
- Blood supply: pancreaticoduodenal and pancreatic arteries; venous blood returns via hepatic portal vein. Islets innervated by ANS via celiac plexus branches.

- Four cell types per islet:

1. **Alpha cells** — produce **glucagon**; increases blood glucose by increasing glycogen breakdown and glucose release by liver.

2. **Beta cells** — produce **insulin**; lowers blood glucose; increases rate of glucose uptake/utilization by most body cells (except brain, kidneys, digestive tract epithelium, RBCs, per summary table — actually table says insulin targets all cells except brain, kidneys, digestive tract epithelium, RBCs).

3. **Delta cells** — produce **somatostatin** (growth hormone–inhibiting hormone); inhibits production/secretion of glucagon and insulin; slows rate of food absorption and enzyme secretion by digestive tract.

4. **F cells** — produce **pancreatic polypeptide (PP)**; inhibits gallbladder contractions; regulates production of some pancreatic enzymes; may control rate of nutrient absorption by digestive tract.

- Alpha/beta cells directly sensitive to blood glucose concentration (not under direct nervous/other endocrine control): high glucose → beta cells secrete insulin; low glucose → alpha cells secrete glucagon.
- Clinical: **Diabetes mellitus** — glucose concentrations high enough to overwhelm kidney reabsorption capacity; **hyperglycemia** = abnormally high blood glucose; **glycosuria** = glucose in urine; **polyuria** = excessive urine production. Causes: genetic abnormalities (inadequate insulin production, abnormal insulin molecules, defective receptor proteins), other pathologies, injuries, immune disorders, hormonal imbalances; obesity accelerates onset/severity.
 - **Type 1 (insulin-dependent)** diabetes — managed with insulin injection or pump infusion.
 - **Type 2 (non-insulin-dependent)** diabetes — dietary restriction most effective treatment; peripheral cells don't respond normally to insulin (abnormal receptors).
 - Chronic complications (tissues in energy crisis, breaking down lipids/proteins as in starvation):
 - **Diabetic retinopathy** — capillary proliferation/hemorrhage at retina → partial/complete blindness.
 - **Cataracts** — changes in lens clarity.
 - **Diabetic nephropathy** — small hemorrhages/inflammation in kidneys → degenerative changes, primary cause of kidney failure; blood-flow-improving drugs can slow progression.
 - **Diabetic neuropathy** — peripheral neuropathies, abnormal autonomic function, likely from disturbed blood supply to neural tissue.
 - Cardiac: degenerative circulatory changes → early heart attacks (3–5x more likely in diabetics vs nondiabetics of same age).
 - Peripheral vascular changes → reduced blood flow to limbs (e.g., feet) → tissue death, ulceration, infection, toe/foot loss.

19.9 Endocrine Tissues of the Reproductive System

- Reproductive endocrine tissue restricted mainly to gonads (testes, ovaries).
- **Testes:**
 - **Interstitial cells** produce androgens, especially **testosterone**: (1) promotes production of functional sperm, (2) maintains secretory glands of male reproductive tract, (3) influences secondary sexual characteristics, (4) stimulates muscle growth. During embryonic development, testosterone affects anatomical development of male structures.
 - **Nurse cells** support differentiation/maturation of sperm; under FSH stimulation, secrete **inhibin**, which inhibits FSH secretion at anterior pituitary; inhibin and FSH interact throughout adult life to maintain normal sperm production levels.
- **Ovaries:**
 - Oocytes mature within **follicles**; FSH triggers maturation.
 - **Follicular cells** surrounding oocytes produce **estrogens**, especially **estradiol**: support oocyte maturation, stimulate growth of uterine lining. FSH also stimulates active follicles to secrete **inhibin**, which decreases FSH release from anterior pituitary (negative feedback).

- After ovulation, remaining follicular cells reorganize into **corpus luteum**, which releases a mixture of estrogens and progestins, especially **progesterone**: accelerates oocyte movement along uterine tube, prepares uterus for embryo arrival/implantation.
- Summary table also lists **relaxin** (corpus luteum) — loosens pubic symphysis, relaxes uterine/cervical muscles, stimulates mammary gland development.

19.10 The Pineal Gland

- Small, red, pinecone-shaped; part of the epithalamus (of CNS).
- Two cell types: **pinealocytes** (secretory) and **interstitial cells** (resemble neuroglia).
- Pinealocytes synthesize **melatonin**, derived from the neurotransmitter serotonin.
- Melatonin slows maturation of sperm, oocytes, and reproductive organs by inhibiting hypothalamic release of factors that stimulate FSH/LH secretion.
- Collaterals from visual pathways enter the pineal gland and affect rate of melatonin production (light-sensitive).
- Melatonin production increases at night, decreases during day — regulates **circadian rhythms** (natural sleep-wake cycles).
- Melatonin is also a powerful antioxidant, potentially protecting CNS tissue from free radicals generated by active neurons/neuroglia.
- Concept check example: melatonin tablets suggested to counteract jet lag by resetting circadian rhythm.

19.11 Hormones and Aging

- Endocrine system shows relatively few functional changes with age overall.
- Most dramatic exceptions: (1) rise in reproductive hormone levels at puberty, (2) decline in reproductive hormone concentration at menopause in women.
- Age-related changes in other (non-endocrine) tissues can reduce their ability to respond to hormonal stimulation — so many tissues become less responsive to circulating hormones even though hormone concentrations remain normal.

Clinical Correlations Summary (Ch 19)

- **Diabetes insipidus**: inadequate ADH from posterior pituitary → polyuria, severe thirst, risk of fatal dehydration (up to 10 L/day fluid loss); treated with DDAVP nasal spray.
- **Pituitary growth failure (pituitary dwarfism)**: GH deficiency in children.
- **Gigantism** (children) / **Acromegaly** (adults): GH overproduction; acromegaly shows broadened facial features, enlarged jaw, cartilage/bone shape changes after growth plates fuse.
- **SIADH** (syndrome of inappropriate ADH secretion): ADH overproduction → increased body water content.
- **Myxedema** (adult) / **Infantile hypothyroidism**: thyroxine/T3 underproduction — low metabolic rate/body temp; infantile form from thyroid hormone insufficiency in infancy.
- **Graves disease**: thyroid hormone overproduction — high metabolic rate, tachycardia, weight loss.
- **Goiter**: enlarged thyroid gland, usually from thyroid hyposecretion due to dietary iodine insufficiency.
- **Hypoparathyroidism**: low PTH — muscular weakness, neurological problems, tetany (from low blood Ca^{2+}).
- **Hyperparathyroidism**: high PTH — neurological/mental/muscular problems from high blood Ca^{2+} ; weak/brittle bones.
- **Diabetes mellitus** (insulin underproduction/resistance): high blood glucose, impaired glucose utilization, dependence on lipids for energy, glucosuria, ketosis.
- **Excess insulin**: low blood glucose, possibly coma.
- **Hypoaldosteronism**: polyuria, low blood volume, high blood K^{+} .
- **Aldosteronism**: increased body weight (water retention), low blood K^{+} .

- **Addison disease:** glucocorticoid hyposecretion — inability to tolerate stress, mobilize energy reserves, maintain normal blood glucose; pigment changes from melanocyte stimulation by excess ACTH (structurally similar to MSH).
- **Cushing disease:** glucocorticoid hypersecretion — excessive tissue protein/lipid breakdown, impaired glucose metabolism; lipid reserves mobilized/redistributed to cheeks and base of neck.
- **Pheochromocytoma:** epinephrine/norepinephrine overproduction — high metabolic rate, body temp, heart rate, elevated blood glucose.
- **Hypogonadism** (female): estrogen underproduction — sterility, lack of secondary sexual characteristics.
- **Androgenital syndrome:** androgen overproduction by zona reticularis of adrenal cortex → masculinization.
- **Precocious puberty:** early hormone production (androgens or estrogens/follicle development).
- **Hypogonadism/eunuchoidism** (male): androgen underproduction — sterility, lack of secondary sexual characteristics.
- **Gynecomastia:** abnormal estrogen overproduction in males (sometimes adrenal or intestinal tumors) → breast enlargement.
- General naming convention: **hyper-** = excessive hormone production; **hypo-** = inadequate hormone production. Causes of dysfunction: gland abnormality, neural/endocrine regulatory mechanism problems, target tissue unresponsiveness (e.g., receptor issues), tumors, metabolic substrate deficiency (e.g., iodine deficiency → hypothyroidism), physical damage/circulatory interruption, thyroiditis, radiation damage, autoimmune destruction (e.g., antibodies attacking follicle cells), congenital disorders (gland too small, abnormal enzymes, insensitive/absent receptors).
- **Type 2 diabetes** = best example of abnormal target-tissue hormone receptors (peripheral cells don't respond normally to insulin) despite normal gland/regulatory mechanisms.
- Related clinical terms: **adrenalectomy** (surgical adrenal gland removal); **Hashimoto's disease** (chronic lymphocytic thyroiditis — autoimmune attack on thyroid; most common cause of hypothyroidism in the US); **thyroidectomy** (surgical removal of all/part of thyroid); **thyroid function tests** (blood/radionuclide tests of thyroid activity); **virilism** (development of secondary male characteristics, e.g., hirsutism, lowered voice, in females due to hormone regulation disorders).
- Clinical Case (Kathy): hypothyroidism from likely Hashimoto's thyroiditis — low T4, elevated TRH and TSH (pituitary/hypothalamus responding normally to low T4 — indicates primary thyroid failure, not pituitary/hypothalamic failure); treated with daily levothyroxine (synthetic T4).

Blood (Ch 20)

20.1 Functions and Composition of the Blood

- Blood = fluid connective tissue distributing nutrients, oxygen, hormones to ~75 trillion body cells; cells deprived of circulation may die within minutes.
 - Normally confined to the cardiovascular system; two components:
1. **Plasma** — liquid component; density slightly greater than water; contains dissolved proteins.
 2. **Formed elements** — RBCs, WBCs, platelets (cell fragments) suspended in plasma.
 - Eight major functions of blood:
1. Transport dissolved gases (O₂ from lungs to tissues; CO₂ from tissues to lungs).
 2. Distribute nutrients (from digestive tract or storage in adipose tissue/liver).
 3. Transport metabolic wastes to excretion sites (especially kidneys).
 4. Deliver enzymes and hormones to target tissues.
 5. Stabilize pH and electrolyte composition of interstitial fluids (buffers neutralize acids, e.g., lactic acid from muscle).

6. Prevent fluid loss via clotting reaction at injury sites.
7. Defend against toxins and pathogens (WBCs migrate into tissues; antibodies attack invaders; toxins delivered to liver/kidneys for inactivation/excretion).
8. Stabilize body temperature by absorbing/redistributing heat (regulated by cardiovascular centers in medulla oblongata); vasodilation to skin increases heat loss when hot, warm blood redirected to vital organs when cold.
 - Whole blood = plasma + formed elements; can be fractionated for clinical purposes.
 - **Viscosity**: pure water = 1.0; plasma = 1.5; whole blood = 5x as viscous as water (from dissolved proteins, formed elements, water interactions).
 - **Blood volume estimate**: ~7% of body weight (kg) = volume in liters; e.g., 75-kg person ≈ 5.25 L (5.4 quarts). Adult males typically have more blood than adult females. Normal range ~4–6 L in an average adult.
 - **pH**: 7.35–7.45 (slightly alkaline); temperature 38°C (100.4°F), slightly above normal body temp of 37°C.
 - Terminology: **hypovolemic** (low blood volume), **normovolemic** (normal), **hypervolemic** (excessive) — excess volume strains heart, which must pump extra fluid.

Plasma

- ~55% of whole blood volume; water = 92% of plasma volume (varies by cardiovascular region).
- Plasma proteins ≈ 7% of plasma; 100 mL plasma normally contains 6–7.8 g soluble protein. Three classes:
 1. **Albumins** — ~60% of plasma proteins; major contributor to osmotic pressure; transport fatty acids, thyroid hormones, some steroid hormones, other substances.
 2. **Globulins** — ~35% of plasma proteins; two types: **immunoglobulins** (antibodies — aid body defense) and **transport globulins** (bind ions, hormones, insoluble/filterable compounds).
 3. **Fibrinogen** — ~4% of plasma proteins; largest plasma protein; essential for clotting; converts to insoluble **fibrin** strands (clot framework) under certain conditions. Removing clotting proteins from plasma leaves **serum**.
 - **Lipoproteins**: albumins/globulins bound to non-water-soluble lipids (triglycerides, fatty acids, cholesterol) — allows lipid transport in plasma.
 - Liver synthesizes/releases >90% of plasma proteins; liver disorders (e.g., inadequate fibrinogen synthesis) can cause uncontrolled bleeding.
 - **Differences between plasma and interstitial fluid**:
 1. Dissolved gas concentration: plasma has higher O₂ (diffuses out to tissues); interstitial fluid has higher CO₂ (diffuses into blood).
 2. Dissolved protein concentration: plasma has significant protein (too large to cross capillary walls); interstitial fluid has little.
 - Other plasma components (table): electrolytes (normal extracellular ion composition, essential for cellular activity), nutrients (energy/growth/maintenance), organic wastes (carried to breakdown/excretion sites, e.g. urea, uric acid).

20.2 Formed Elements

- Three formed elements: **red blood cells (RBCs/erythrocytes)**, **white blood cells (WBCs/leukocytes)**, **platelets**.
- Two major WBC classes: **granular** (cytoplasmic granules) and **agranular** (without granules).

Red Blood Cells (RBCs)

- Anucleate, biconcave discs; contain hemoglobin for O₂ transport; account for a little less than half of total blood volume.
- **Hematocrit** = % of whole blood occupied by formed elements; average adult man = 45 (range 40–54); average adult woman = 42 (range 37–47). Since blood has ~1000x more RBCs than WBCs, hematocrit closely approximates RBC volume — reported as **volume of packed red cells (VPRC)** or **packed cell volume (PCV)**.

- Numbers: man's blood ≈ 5.4 million RBC/ μL (mm^3); woman's blood ≈ 4.8 million RBC/ μL ; ~ 260 million RBCs per drop of whole blood; ~ 25 trillion (2.5×10^{13}) RBCs in an average adult. Summary table lists average 5.2 million/ μL (range 4.2–6.3 million).
- RBC lifespan: 120 days; during differentiation/maturation, RBCs lose organelles, retaining only cytoskeleton — mature circulating RBCs lack mitochondria, ER, ribosomes, nucleus. Energy only via anaerobic metabolism using plasma glucose (ensures absorbed O_2 isn't "stolen" by own mitochondria); cannot replace damaged enzymes/proteins (no nucleus/ER/ribosomes to synthesize new ones).
- Diameter $\sim 7.7 \mu\text{m}$, max thickness $\sim 2.85 \mu\text{m}$, center narrows to $\sim 0.8 \mu\text{m}$. Biconcave shape $\rightarrow$ large surface area for rapid gas diffusion; total surface area of all RBCs in typical adult $\approx 3800 \text{ m}^2$ (2000x total body surface area).
- RBCs stack into **rouleaux** (like dinner plates) — form/dissociate without damage; allow passage through vessels slightly larger than one RBC diameter; individual (non-stacked) RBCs would clump/block vessels. Biconcave/slender shape also gives strength/flexibility to squeeze through small/distorted capillaries.
- Full cardiovascular circuit takes < 30 seconds; RBC survives ~ 120 days, traveling ~ 700 miles total; then plasma membrane ruptures or phagocytes destroy aging cell. $\sim 1\%$ of circulating RBCs replaced daily; ~ 3 million new RBCs enter bloodstream per second.
- RBC composition: plasma membrane + cytoplasm containing water (66%) and proteins ($\sim 33\%$).
- **Hemoglobin (Hb)**: $> 95\%$ of RBC protein; globular protein with nonprotein red pigment; responsible for O_2/CO_2 transport. Oxygenated Hb = bright red; deoxygenated = deep red.
 - Structure: 2 alpha (α) chains + 2 beta (β) chains of polypeptides; each chain has one **heme** molecule (nonprotein pigment ring) holding an iron ion that reversibly binds $\text{O}_2 \rightarrow$ **oxyhemoglobin** (bright red). Iron- O_2 binding is weak/reversible without damaging heme or O_2 . Hb with iron not bound to O_2 = **deoxyhemoglobin** (dark red blood).
 - Each RBC contains ~ 280 million Hb molecules; each Hb has 4 heme units $\rightarrow$ each RBC can carry > 1 billion O_2 molecules.
 - Hb also carries 23% of transported CO_2 — binds to globin amino acids (not iron) forming **carbamino hemoglobin**; binding is reversible like O_2 -heme binding.
 - Gas exchange: low plasma O_2 /high CO_2 (peripheral capillaries) $\rightarrow$ Hb releases O_2 , binds CO_2 . High plasma O_2 /low CO_2 (lung capillaries) $\rightarrow$ Hb binds O_2 , releases CO_2 .
- **Blood types**: determined by presence/absence of RBC surface antigens (**agglutinogens**) — glycoproteins, genetically determined. RBCs have ≥ 50 kinds of surface antigens; 3 most important: A, B, Rh (D).
 - Type A = antigen A; Type B = antigen B; Type AB = both; Type O = neither.
 - **Rh group**: Rh+ = has Rh antigen (Rh factor); Rh- = lacks it. Full type reported e.g. O-, A+.
 - US population % breakdown by ethnicity given in table (e.g., Caucasian: O45 A40 B11 AB4, Rh+85%; Black American: O49 A27 B20 AB4, Rh+95%; Native North American: O79 A16 B4 AB<1, Rh+100%; etc.)
 - **Antibodies (agglutinins)**: plasma contains antibodies against foreign (not self) surface antigens — genetically determined, present throughout life regardless of prior exposure (for A/B). Type A $\rightarrow$ anti-B antibodies; Type B $\rightarrow$ anti-A antibodies; Type O $\rightarrow$ both anti-A and anti-B; Type AB $\rightarrow$ neither.
 - Rh: Rh- person's plasma has NO anti-Rh antibodies unless previously sensitized by exposure to Rh+ blood (e.g., transfusion, or Rh- mother carrying Rh+ fetus).
 - **Cross-reaction**: antibody meets its specific surface antigen $\rightarrow$ **agglutination** (RBC clumping) and/or **hemolysis** (RBC rupture); clumps/fragments can plug small vessels in kidneys, lungs, heart, brain, causing tissue damage. Donor/recipient blood types must be compatible to avoid cross-reactions.
- **White blood cells (WBCs/leukocytes)**: circulate in blood vessels and scattered in peripheral tissues (most WBCs are in tissues, not blood); defend against pathogens, remove toxins/wastes/damaged cells; all WBCs $\geq$ RBC size. Two groups: **granulocytes** and **agranulocytes**.
 - Normal count: 5000–10,000 WBC/ μL . **Differential count** = % of each WBC type per 100 WBCs in a stained smear.
 - Suffixes: **-penia** = low count (e.g., lymphopenia); **-osis** = high count (e.g., lymphocytosis).

- **Chemotaxis** = WBC attraction to chemicals released by infected/damaged cells. **Diapedesis (emigration)** = WBCs squeezing between endothelial cells to cross capillary walls into tissue.
- Granular leukocytes (granulocytes):
 - **Neutrophils** — 50–70% of circulating WBCs (avg 4150/ μ L, range 1800–7300); diameter $\sim$ 2x RBC; nucleus lobed ("beads on a string"); pale neutral-staining granules with lysosomal enzymes/bactericidal compounds. Highly mobile, first WBCs to injury site; active phagocytes specializing in bacteria. Short lifespan — survive only $\sim$ 10 hours in bloodstream; dies after engulfing 1–2 dozen bacteria.
 - **Eosinophils** — 2–4% of WBCs (avg 165/ μ L, range 0–700); bilobed nucleus; deep red-orange granules (stain with eosin, acidic dye). Attack antibody-coated objects; number increases dramatically in allergic reactions/parasitic infections; release enzymes that reduce inflammation from mast cells/neutrophils, controlling inflammation spread.
 - **Basophils** — $<1\%$ of WBCs (avg 44/ μ L, range 0–150); numerous granules stain deep purple/blue with basic dyes, obscuring nucleus. Migrate to injury sites, cross capillary endothelium, release granule contents (histamine, heparin) into interstitial fluid. Histamine dilates blood vessels; heparin prevents clotting. Enhance local inflammation initiated by mast cells (distinct cell population from mast cells despite similar function). Also release chemicals attracting eosinophils/other basophils.
- Agranular leukocytes (agranulocytes):
 - **Monocytes** — 2–8% of WBCs (avg 456/ μ L, range 200–950); largest WBCs, 2–3x RBC diameter; large kidney bean-shaped nucleus. Circulate only a few days before entering tissues, becoming **free macrophages** (mobile, phagocytic — distinct from immobile fixed macrophages of connective tissue). Arrive at injury sites shortly after first neutrophils; release chemicals attracting/stimulating other monocytes/phagocytes; also secrete substances attracting fibroblasts (which produce scar tissue to wall off injury).
 - **Lymphocytes** — 20–30% of WBCs per text (differential count table says 20–40%; avg 2185/ μ L, range 1500–4000); very little cytoplasm (thin halo) around large round nucleus; slightly larger than RBC. Primary cells of the lymphatic system; bloodstream lymphocytes are small % of total body lymphocyte population. Responsible for **specific immunity** (attacking pathogens/foreign proteins individually). Three functional classes (indistinguishable by light microscopy):

1. **T cells (T lymphocytes)** — enter peripheral tissues, attack foreign cells directly.

2. **B cells (B lymphocytes)** — differentiate into plasma cells (plasmocytes), which secrete antibodies against foreign cells/proteins.

3. **Natural killer (NK) cells** — immune surveillance, destroy abnormal cells; also called large granular lymphocytes; important in cancer prevention.

- Mnemonic for WBC abundance order (most $\rightarrow$ least): "Never Let Monkeys Eat Bananas" = Neutrophils, Lymphocytes, Monocytes, Eosinophils, Basophils.

Platelets

- Flattened, membrane-enclosed cytoplasmic packets (not true cells) circulating in blood; avg 350,000/ μ L (range 150,000–500,000).
- Formed from **megakaryocytes** — enormous bone marrow cells (up to 160 μ m diameter), lobed/ring-shaped nucleus; cytoplasm has Golgi apparatus, abundant ribosomes, mitochondria; plasma membrane connects to extensive membrane network radiating through peripheral cytoplasm.
- Megakaryocytes shed cytoplasm as small membrane-enclosed packets (platelets) entering circulation; one megakaryocyte produces $\sim$ 4000 platelets, then its nucleus is phagocytized/recycled.
- Platelet lifespan: 10–12 days before phagocytosis. At any time, $\sim$ 1/3 of body's platelets are stored in the spleen/other vascular organs (mobilized during circulatory crises like severe bleeding).
- Functions:
 1. Initiate/control clotting process by releasing enzymes/factors.
 2. Clump together at injury site forming a **platelet plug**, slowing blood loss during clotting.

3. Contain actin/myosin filaments that contract during clotting, shrinking the clot and pulling cut vessel edges together.

- **Hemostasis** = prevention of blood loss through damaged vessel walls; restricts blood loss, establishes framework for tissue repair. Complex multistep process; disorder at any step can disrupt whole process. General requirements: adequate calcium ions and vitamin K needed for virtually all hemostasis steps.
- Clot framework = fibrin network; trapped RBCs add mass/red color to clot.
- Clinical: **thrombocytopenia** (low platelet count) — from inadequate production or excessive destruction; symptoms: digestive tract/skin bleeding, occasional CNS bleeding. **Thrombocytosis** (high platelet count, often $>1,000,000/\mu\text{L}$) — indicates accelerated platelet formation from infection, inflammation, or cancer.

20.3 Hemopoiesis

- **Hemopoiesis** = process of blood cell formation; **hematopoietic stem cells** give rise to all blood cells (differentiate into myeloid and lymphoid stem cell lines; myeloid stem cells → progenitor cells → various blood cell classes).

Erythropoiesis

- Formation of erythrocytes; begins early in embryonic development. In adults, **red bone marrow** = primary site — found in vertebrae, sternum, ribs, skull, scapulae, pelvis, proximal limb bones. Under extreme conditions (e.g., severe blood loss), yellow (fatty) bone marrow elsewhere can convert to red marrow to increase RBC production.
- Requires adequate amino acids, iron, and **vitamin B12** (from dairy/meat).
- Regulated by hormone **erythropoietin (EPO)** — glycoprotein formed by kidneys and liver under hypoxic (low O_2) conditions. Two major effects:

1. Stimulates cell division rates in erythroblasts and their stem cells.

2. Speeds RBC maturation, mainly by accelerating Hb synthesis.

- Under maximum EPO stimulation, red bone marrow can increase RBC formation tenfold, to ~30 million cells/second.
- Developmental sequence (from diagram): Proerythroblast → Erythroblast → (ejection of nucleus) → Reticulocyte → Erythrocyte.

Leukopoiesis

- Formation of WBCs; begins early in embryonic development; stem cells originate in red bone marrow, producing granular leukocytes and monocytes.
- **Lymphopoiesis** (lymphocyte production) also originates in red bone marrow; some lymphoid stem cells remain in marrow (differentiate into B cells or NK cells); many migrate to peripheral lymphatic tissues (thymus, spleen, lymph nodes) — lymphoid stem cells migrating to thymus mature into T cells.
- Factors regulating lymphocyte maturation not completely understood; thymic hormones promote T cell differentiation/maintenance prior to maturity.
- **Colony-stimulating factors (CSFs)** regulate other WBC populations; commercially available CSFs used clinically to stimulate WBC production in cancer chemotherapy patients.

Clinical Correlations Summary (Ch 20)

- **Sickle cell disease:** inherited mutation in beta-chain amino acid sequence of Hb; defective Hb releasing O_2 clusters into rods → RBCs become stiff/curved ("sickled"), trapped in capillaries, reducing tissue O_2 and possibly damaging organs (**sickle cell crisis**). Affects ~0.2% of Black Americans, 0.07–0.1% of Hispanic Americans. Treated with transfusions (temporary) and antibiotics (reduce infection deaths). Likely evolved in malaria-endemic areas — sickled cells infected with malaria parasites collapse, preventing parasite multiplication, giving carriers protective advantage against malaria.
- **Hemophilia:** inherited disorder of inadequate clotting factor production; affects ~1 in 10,000 (mostly males). Severity varies — severe cases: excessive bleeding from slight stress, spontaneous internal bleeding in

joints/around muscles. Treated with clotting factor transfusions (expensive, infection risk since pooled from many donors) or gene-spliced factor VIII (limited supply, safer).

- **Blood doping:** RBC transfusions or EPO-type drugs to boost athletic performance. Autologous procedure: remove whole blood weeks before event, separate/store packed RBCs, body replaces lost cells, reinfuse packed cells immediately before event to raise hematocrit/O₂-carrying capacity. Risks: cardiac strain, stroke, heart attack. Training at high altitude = safer alternative (natural EPO stimulation).
- **Anemia:** any condition reducing blood O₂-carrying capacity (low hematocrit or low Hb); symptoms: lethargy, weakness, premature muscle fatigue; many causes/types differentiated by RBC number/size/shape/Hb content; treatment depends on type.
- **Polycythemia:** abnormally high RBC count. **Polycythemia vera** = chronic form, unknown cause; bone marrow hyperplasia, increased blood volume, splenomegaly; hematocrit may reach 80–90, causing tissue O₂ starvation as RBCs block small vessels. Most cases in ages 60–80; no cure; simplest treatment = periodic **phlebotomy** (drawing off blood).
- **Transfusion practices:** whole blood transfusion restores volume after massive hemorrhage. **Exchange transfusion** = removing most of a person's blood, replacing with donor blood (for acute drug poisoning or hemolytic disease of the newborn/maternal-fetal Rh incompatibility). Donor blood screened for pathogens; whole blood stored refrigerated, usable ~3.5 weeks; longer storage requires fractionation (RBCs frozen with antifreeze solution; plasma chilled/frozen/freeze-dried).
- **Packed red blood cells (PRBCs)** = whole blood minus plasma; used for anemia (near-normal volume, low O₂ capacity). Plasma administered for massive fluid loss (e.g., severe burns).
- **Autologous transfusion:** patient's own blood removed/stored/retransfused later (e.g., during surgery); reduces infection risk.
- **Plasma expanders:** contain large carbohydrate molecules (not proteins) to maintain osmolarity; temporary hypovolemia treatment; do not increase O₂ delivery to tissues; easily stored/transported, sterile, no viral/bacterial contamination risk.
- Related clinical terms: **blood bank** (collects, types, separates, stores, prepares blood for transfusion); **dyscrasia** (abnormal blood condition); **phlebotomist** (technician who draws blood); **hematology** (medical study of blood/blood-producing organs); **septicemia-type description** — leukemia (e.g., chronic myelogenous leukemia); large numbers of abnormal RBCs/WBCs/platelets in marrow and peripheral blood; **thrombolytic** (agent causing breakup of a thrombus/clot).
- Clinical Case (Danny): **acute lymphocytic (lymphoblastic) leukemia (ALL)** — cancer of lymphocytes; cancerous lymphoid stem cells take over red bone marrow; most common childhood cancer. Immature lymphoblasts released into blood instead of mature lymphocytes, crowding out healthy RBCs/platelets → pallor, easy bleeding; can invade spinal cord, brain, thymus, liver, spleen. With aggressive treatment (oral/IV/intrathecal chemotherapy), ~95% remission chance; bone marrow transplant from siblings as backup option.

Heart (Ch 21)

21.1 An Overview of the Cardiovascular System

- Heart = fist-sized muscular organ with 4 chambers: right/left **atria** ("chamber") and right/left **ventricles** ("little belly").
- Beats ~100,000 times/day; pumps >1.5 million gallons/year (≈200 train tank cars); would take a fully open kitchen faucet 45 years to match an average lifetime's pumped blood volume.
- Cardiac output varies 5–30 L/min depending on nervous system regulation of body needs.
- Two circuits (in series with each other, functioning like separate loops):
 - **Pulmonary circuit** — carries CO₂-rich blood from heart to lung gas-exchange surfaces, returns O₂-rich blood to heart.
 - **Systemic circuit** — transports O₂-rich blood from heart to body cells, returns CO₂-rich blood to heart.

- Right atrium receives systemic venous blood; right ventricle pumps into pulmonary circuit. Left atrium receives pulmonary venous blood; left ventricle pumps into systemic circuit.
- Each heartbeat: atria contract first, then ventricles; both ventricles contract simultaneously, ejecting equal blood volumes into pulmonary and systemic circuits.
- Blood must complete one circuit before re-entering the heart to start the other (circuits in series).
- Vessel types: **arteries** (carry blood away from heart), **veins** (return blood to heart), **capillaries** (small thin-walled vessels connecting smallest arteries/veins, site of exchange).

21.2 The Pericardium

- Heart lies near anterior chest wall, directly posterior to sternum, within the **mediastinum** (also contains great vessels, thymus, esophagus, trachea).
- **Pericardium** surrounds heart; two parts:
 - **Fibrous pericardium** — outer; dense collagen fiber network; stabilizes heart/vessel position in mediastinum.
 - **Serous pericardium** — inner, two-layered: outer **parietal layer** and inner **visceral layer** (= **epicardium**).
- **Pericardial cavity** = potential, fluid-filled space between parietal and visceral serous layers; contains up to **50 mL of pericardial fluid** (lubricant, reduces friction during heartbeat).
- Analogy: pericardium = partially inflated balloon; heart = fist pushed into it.
- Clinical: **Pericarditis** = infection/inflammation of pericardium; most common cause = viral infection; also bacterial (e.g., TB complication), cancer invasion, kidney failure (uremic pericarditis), severe chest trauma with bleeding into pericardial sac.

21.3 Structure of the Heart Wall

- Three layers of heart wall:

1. **Epicardium** (visceral layer of serous pericardium) — covers heart surface; mesothelium + underlying areolar connective tissue.

2. **Myocardium** — cardiac muscle tissue forming atria/ventricles; includes cardiac muscle cells, connective tissue, blood vessels, nerves. Atrial myocardium thin, organized in figure-eight layers between atria. Ventricular myocardium much thicker; superficial muscles wrap both ventricles, deeper layers spiral around/between individual ventricles.

3. **Endocardium** — simple squamous epithelium lining inner heart surfaces (incl. valves); continuous with endothelium of attached great vessels.

- **Cardiac muscle cells:** small (10–20 μm diameter, 50–100 μm length), single centrally-placed nucleus; resemble skeletal muscle (organized myofibrils, striations) but differ:
 - Almost totally dependent on **aerobic respiration**: sarcoplasm has hundreds of mitochondria, large myoglobin reserves (O_2 storage), abundant glycogen and lipid inclusions (energy reserves).
 - T tubules shorter than skeletal muscle; do NOT form triads with sarcoplasmic reticulum.
 - More blood vessels than even red (slow-twitch) skeletal muscle.
 - Contract without nervous system stimulation (autorhythmicity).
 - Interconnected by **intercalated discs**.
- **Intercalated discs:** unique cell junctions connecting cardiac muscle cells, giving jagged membrane appearance. Components:
 - **Desmosomes** — bind plasma membranes together, lock cells to prevent separation during contraction.
 - **Fascia adherens** — actin filaments anchor to plasma membrane here, tying together actin of adjacent cells for maximal contraction efficiency.
 - **Gap junctions (communicating junctions)** — allow ions/small molecules to pass between cells, creating direct electrical connection; action potential moves cell-to-cell as if sarcolemmae were continuous.

- Because cells are mechanically, chemically, electrically connected, cardiac tissue acts as a single enormous cell — a **functional syncytium**; contraction of one cell triggers spreading contraction through the myocardium.
- Connective tissue of heart: reticular, collagen, elastic fibers. Each cardiac cell wrapped in elastic sheath; adjacent cells linked by fibrous "struts"; muscle layers wrapped in fibrous sheets. These connect to bands of dense connective tissue forming the **cardiac skeleton**, which:
 1. Encircles bases of pulmonary trunk and aorta.
 2. Encircles heart valves.
 3. Connects fibrous rings around valve openings.
 4. Extends into muscle separating atria/ventricles.
 - Functions of cardiac skeleton: stabilizes muscle cell/valve positions; physically supports muscle cells, vessels, nerves; distributes contraction forces; reinforces valves, prevents overexpansion; provides elasticity for post-contraction shape recovery; physically isolates atrial from ventricular muscle cells (vital for coordinating contractions).

21.4 Orientation and Superficial Anatomy of the Heart

- Heart is NOT centered — (1) lies slightly left of midline, (2) sits at an angle to body's longitudinal axis, (3) rotated to the left. Located within mediastinum between the two lungs; cardiac notch of left lung is deeper than right lung's due to leftward heart position.
- **Base** = broad superior portion, attachment point of major systemic/pulmonary vessels and superior surfaces of both atria; posterior to sternum at ~3rd costal cartilage.
- **Apex** = inferior, pointed tip, formed mainly by left ventricle; points laterally, reaches 5th intercostal space, extends left of midline.
- Borders: base = superior border; right atrium = right border; left ventricle + small left atrium portion = left border (extends to apex); inferior wall of right ventricle = inferior border.
- Surfaces:
 - **Anterior (sternocostal) surface** — faces anterior thoracic wall; mostly right ventricle wall, some left ventricle.
 - **Posterior surface** (at base) — formed by left atrium and small right atrium portion.
 - **Diaphragmatic surface** — mainly posterior/inferior surfaces of right and left ventricles.
- External **sulci** (grooves) mark chamber borders, contain adipose tissue (removed to view in dissection) plus coronary vessels: **coronary sulcus** (between atria/ventricles), **anterior/posterior interventricular sulci** (between ventricles).
- Atria vs ventricles: atria receive venous blood, thin muscular walls; when not filled, anterior portion deflates into wrinkled flap called the **auricle**. Ventricles propel blood to periphery/lungs.
- Right atrium: anterior, inferior, right of left atrium (due to angled heart position). Left atrium curves posteriorly, forms most of posterior heart surface superior to coronary sulcus.
- Clinical: **Myocarditis** = myocardium inflammation, commonly viral, but also autoimmune, toxins, alcohol, medications, chemo/radiation (e.g., breast cancer therapy); severe cases → chronic cardiomyopathy, impaired pumping, blood clots (risk of heart attack/stroke), possible heart transplant. **Endocarditis** = endocardium inflammation, almost always bacterial (sometimes fungal); risk factors: cardiac birth defects, damaged/abnormal valves; bacteria enter bloodstream (dental surgery, central venous lines, unsterile drug injection) and settle on endocardium; can destroy valves requiring surgical replacement.

21.5 Internal Anatomy and Organization of the Heart

- **Interatrial septum** separates atria; **interventricular septum** separates ventricles. Blood flows from atrium → ventricle of same side through AV valves (folds of endocardium) that prevent backflow, maintaining one-way flow.
- Atrial demands are similar (chambers look nearly identical); ventricular demands differ greatly (anatomically very different).

Right Atrium

- Receives deoxygenated systemic venous blood via **superior vena cava** (opens into posterior-superior right atrium; drains head, neck, upper limbs, chest) and **inferior vena cava** (posterior-inferior right atrium).
- **Coronary sinus** (collects blood from **coronary veins** draining the heart wall itself) opens into posterior wall of right atrium, inferior to IVC opening.
- **Pectinate muscles** ("comb") line inner surface of right auricle and anterior right atrial wall.
- **Interatrial septum**: contains **foramen ovale** from 5th embryonic week until birth — allows blood to bypass developing lungs (right atrium → left atrium directly). Closes at birth when lungs expand/function; permanently sealed within 3 months, leaving the fossa ovalis. If it remains patent (open), blood recirculates into pulmonary circuit, reducing systemic efficiency, raising pulmonary vessel pressure — untreated leads to cardiac enlargement, pulmonary fluid buildup, heart failure.

Right Ventricle

- Receives blood from right atrium through the **right AV valve (tricuspid valve)** — 3 fibrous cusps/flaps. Cusps attach to cardiac skeleton on one side; free edges attach via **chordae tendineae** ("tendinous cords") to **papillary muscles** (cone-shaped projections of ventricle wall). Chordae/papillary muscles limit cusp movement, preventing regurgitation into right atrium during contraction.
- Internal surface has irregular muscular ridges: **trabeculae carneae** ("fleshy"). **Moderator band** = muscular ridge from inferior interventricular septum to anterior papillary muscle (conducts stimulus to papillary muscles).
- Superior end tapers to **conus arteriosus** (smooth, cone-shaped pouch), ending at the **pulmonary valve (pulmonary semilunar valve)** — 3 thick semilunar cusps; blood passes into **pulmonary trunk** (first vessel of pulmonary circuit); valve prevents backflow into right ventricle during relaxation. Pulmonary trunk branches into pulmonary arteries → lungs → pulmonary capillaries for gas exchange.

Left Atrium

- Receives oxygenated blood from pulmonary capillaries via small veins uniting into 4 **pulmonary veins** (usually 2 per lung) emptying into posterior left atrium.
- Differs from right atrium: (1) more cuboidal shape, (2) left auricle longer/narrower/more hook-shaped, (3) all pectinate muscles confined to the left auricle.
- Blood passes to left ventricle via **left AV valve (mitral/bicuspid valve)** — 2 cusps (vs. 3 in right AV valve); permits one-way flow left atrium → left ventricle.

Left Ventricle

- Wall ~3x thicker than right ventricle wall — must generate enough pressure to push blood through entire systemic circuit (vs. right ventricle's short ~30 cm trip to lungs and back).
- Internal organization resembles right ventricle but: (1) more prominent trabeculae carneae, (2) no moderator band, (3) only 2 large papillary muscles (matching 2-cusp mitral valve, vs 3 in right).
- Blood exits via **aortic valve (aortic semilunar valve)** into ascending aorta. **Aortic sinuses** = small saclike dilations at base of ascending aorta next to each cusp — prevent cusps from sticking to aortic wall when valve opens; right and left coronary arteries originate at the aortic sinuses. Aortic valve prevents backflow into left ventricle.
- Blood path: ascending aorta → aortic arch → descending aorta. **Ligamentum arteriosum** = fibrous band connecting pulmonary trunk to aortic arch — remnant of fetal vessel (ductus arteriosus) that once linked pulmonary/systemic circuits (fetal circulation changes at birth, detailed Ch 22).

Structural Differences Between Right and Left Ventricles

- Right ventricle: thin wall (pouch-like, attached to massive left ventricle wall) — lungs are close, pulmonary vessels short/wide, so low pressure suffices; higher pressure would be dangerous (delicate pulmonary capillaries; excess pressure would force fluid into alveoli). Right ventricle contraction moves toward left ventricle wall, compressing blood, pushing it through pulmonary valve efficiently with minimal effort.
- Left ventricle: needs 4–6x more force to push blood through systemic circuit; extremely thick wall, round in cross-section. On contraction: shortens/narrows, base-apex distance decreases, chamber diameter decreases (like squeezing/rolling a toothpaste tube) — generates enough pressure to open aortic valve. Also bulges into

right ventricular cavity during contraction, improving right ventricle's pumping efficiency — this is why individuals with severe right ventricle damage may survive (left ventricle helps push blood into pulmonary circuit).

Structure and Function of Heart Valves

- **AV valves** (tricuspid, mitral) have 4 components: (1) ring of connective tissue attached to cardiac skeleton, (2) connective tissue cusps, (3) chordae tendineae, (4) papillary muscles (tense chordae tendineae). Papillary muscle/chordae tension prevents cusps from swinging into atria (prevents regurgitation) during ventricular contraction.
- **Semilunar valves** (pulmonary, aortic) — lack chordae tendineae; three symmetrical cusps support one another like tripod legs. Pulmonary valve at right ventricle/pulmonary artery junction; aortic valve at left ventricle/ascending aorta junction.
- Clinical: **Heart murmur** = soft whooshing/swishing sound from turbulent blood flow; most are "innocent" (no clinical significance, e.g., in pregnancy from increased blood volume/cardiac output with unchanged valve size). Congenital defects (atrial/ventricular septal defects = holes in septa) cause abnormal flow during systole, generating murmurs. **Valvular stenosis** = narrowed valve opening, reduced flow. **Valvular insufficiency** = regurgitation from incomplete valve closure. Stethoscope used to assess heart sounds (muffled by pericardium/tissue/chest wall — placement doesn't always match valve location).

21.6 Coronary Blood Vessels

- **Coronary circulation** supplies heart muscle; during max exertion, blood flow to heart can increase up to 9x resting levels.
- **Right and left coronary arteries** originate at base of ascending aorta within the aortic sinus — first branches off the aorta; highest blood pressure in systemic circuit occurs here, ensuring continuous flow to active cardiac muscle.
- **Right coronary artery (RCA)**: branches off ascending aorta, runs right within coronary sulcus, passes between right auricle and pulmonary trunk. In 60% of people, RCA is the "dominant" coronary artery (gives off posterior interventricular branch). Typically supplies: right atrium, portion of left atrium, interatrial septum, entire right ventricle, variable portion of left ventricle, postero-inferior 1/3 of interventricular septum, portions of conducting system (SA node).
 - Gives off atrial branches (supply right atrium, part of left atrium) and the **right marginal branch** (near right heart border, toward apex along right ventricle anterior surface; supplies right atrium, interatrial septum, right ventricle).
 - Continues posteriorly, gives off **posterior interventricular branch (posterior descending artery)** — supplies interventricular septum and adjacent ventricle portions.
 - Small branch reaches the **SA node** ("cardiac pacemaker"); another small branch (near posterior interventricular branch) supplies the **AV node**.
- **Left coronary artery (LCA)**: typically larger lumen than RCA. Supplies most of left ventricle, small right ventricle segment, most of left atrium, anterior 2/3 of interventricular septum.
 - Forms **anterior interventricular branch** (left anterior descending) — supplies left ventricular myocardium and anterior 2/3 of interventricular septum; often continuous (via small branches) with posterior interventricular branch of RCA.
 - Forms **circumflex branch** — curves left within coronary sulcus toward posterior surface, gives rise to diagonal branches supplying left ventricle portions.
 - LCA usually forms **left marginal branch**, reaching apex, supplying much of left ventricle.
 - RCA's **posterior left ventricular branch** (small, variable, sometimes absent) may be replaced by circumflex artery's posterior interventricular branch.
- **Anastomoses** = interconnections between coronary arteries; keep ventricular muscle blood supply relatively constant despite pressure fluctuations in either coronary artery.

- **Coronary veins:** **great cardiac vein** and **middle cardiac vein** collect blood from myocardial capillaries, deliver to **coronary sinus** (large thin-walled vein in posterior coronary sulcus); coronary sinus drains into right atrium inferior to IVC opening.
 - Veins draining into great cardiac vein/coronary sinus: (1) **posterior vein of left ventricle** (drains area served by circumflex branch of LCA), (2) **middle cardiac vein** (drains area from posterior interventricular branch of LCA — note: some ambiguity but per text association), (3) **small cardiac vein** (posterior right atrium/ventricle).
 - **Anterior cardiac veins** drain anterior right ventricle surface, empty directly into right atrium, bypassing coronary sinus.
- Clinical: **Coronary artery disease (CAD)** = leading cause of death for men and women in US; cardiac arteries hardened/narrowed from cholesterol/calcium/plaque buildup = **atherosclerosis**; decreases blood supply → **cardiac ischemia** → **angina** (crushing/smothering chest pain, may radiate to back/neck/jaw/left arm). Blood clot at plaque site = **coronary thrombosis**, can cause heart attack, impairing conduction system → arrhythmias, heart failure. Preventable causes: smoking, obesity, hypertension, hypercholesterolemia, diabetes, sedentary lifestyle.
 - Treatment: thrombolytic (clot dissolver), coronary angioplasty; **percutaneous balloon angioplasty** — catheter threaded to atherosclerosis site, balloon inflated to compress plaque, metal stent inserted to keep vessel open. **Coronary artery bypass graft (CABG)** — replace damaged coronary artery sections with venous graft, typically using great saphenous vein (leg) or internal thoracic artery (chest wall).
 - ECG can demonstrate cardiac ischemia.

21.7 The Coordination of Cardiac Contractions

- Pump function: (1) develop pressure, (2) move fluid volume in a direction at acceptable speed. Heart cycles through **systole** (contraction) and **diastole** (relaxation), with pressure rising/falling in each chamber per cycle.
- AV, aortic, pulmonary valves ensure one-way flow: atrium → ventricle only when AV valve open and atrial pressure > ventricular pressure; ventricle → artery only when semilunar valve open and ventricular pressure > arterial pressure.
- Cardiac muscle contracts on its own without neural/hormonal stimulation (**autorhythmicity/automaticity**) — even a transplant heart continues beating unless chilled in preservation solution. Neural/hormonal stimuli modify the basic rhythm.
- Two distinct populations of specialized (non-contractile) cardiac cells coordinate contraction:

1. **Nodal cells** — establish rate of cardiac contraction.

2. **Conducting cells** — distribute the contractile stimulus to the myocardium.

- **Conducting system components:**
 - **Sinoatrial (SA) node** — contains pacemaker cells that initiate the electrical impulse triggering a heartbeat; located in posterior wall of right atrium near SVC entrance.
 - **Internodal pathways** — conducting fibers in atrial wall carrying impulse to AV node while stimulating atrial contraction.
 - **Atrioventricular (AV) node** — slows/delays electrical impulse arriving from internodal pathways (~100 msec delay).
 - **AV bundle** — conducts impulse from AV node to bundle branches.
 - **Left/right bundle branches** — extend toward apex, radiate across inner ventricular surfaces.
 - **Moderator band** — relays stimulus through right ventricle to papillary muscles, tensing chordae tendineae before contraction.
 - **Purkinje fibers** — convey impulses very rapidly to contractile cells throughout ventricular myocardium.
- Sequence with approximate timing: SA node depolarizes (time 0) → depolarization spreads across atria, reaches AV node (50 msec) → AV node delays spread to AV bundle (by 100 msec total; atrial contraction begins) → impulses travel AV bundle within interventricular septum to apex, also to papillary muscles via moderator band

(150 msec; atrial contraction completing, ventricular contraction beginning) → impulse distributed via Purkinje fibers throughout ventricular myocardium (175–225 msec).

21.8 The Cardiac Cycle

- **Cardiac cycle** = period between start of one heartbeat and start of the next; alternating contraction (systole) and relaxation (diastole) for each chamber.
- Nodal cell plasma membranes spontaneously depolarize to threshold; intercellular junctions electrically couple nodal cells to each other, conducting fibers, and cardiac muscle cells. Depolarization generates action potential traveling through conducting system → contraction of all cardiac muscle = one cardiac cycle/heartbeat.
- Not all nodal cells depolarize at the same rate — the fastest-depolarizing cells set the pace and bring others to threshold; these are **pacemaker cells**, located in the **SA node (cardiac pacemaker)**.
- SA node pacemaker cells depolarize spontaneously/rapidly: generate **80–100 action potentials/minute** intrinsically. Each impulse = one heartbeat, so intrinsic resting heart rate = 80–100 bpm; however, ANS activity modifies this. Under normal resting conditions, parasympathetic activity reduces rate to a more typical **70–80 bpm**.
- SA node cells electrically connected to AV node via conducting fibers in atrial walls.
- Clinical: **Bradycardia** = slower-than-normal heart rate; **Tachycardia** = faster-than-normal heart rate (definitions vary by individual baseline/conditioning).
- Cardiac cycle phase sequence (per illustration): ventricles relaxed (AV valves open, semilunar valves closed, chordae loose, papillary muscles relaxed) → atrial systole ends/atrial diastole begins → ventricular diastole late (pressure rises to open semilunar valves) → ventricular systole (2nd phase: pressure exceeds arterial pressure, semilunar valves open, blood ejected) → ventricular diastole early (ventricles relax, pressure drops until reverse flow closes semilunar valve cusps; blood flows into relaxed atria).

21.9 Autonomic Control of Heart Rate

- Sympathetic and parasympathetic ANS divisions innervate the heart via the **cardiac plexus**; both innervate SA and AV nodes, atrial/ventricular cardiac muscle, and smooth muscle of cardiac blood vessel walls.
- **Norepinephrine (NE)** release (sympathetic) — increases heart rate and contraction force by stimulating **beta receptors** on nodal and contractile cells.
- **Acetylcholine (ACh)** release (parasympathetic) — decreases heart rate and contraction force by stimulating **muscarinic receptors** on nodal and contractile cells.
- **Cardiac centers** in the medulla oblongata contain ANS cardiac control centers:
 - **Cardioacceleratory center** — activates sympathetic neurons.
 - **Cardioinhibitory center** — activates parasympathetic neurons.
 - Both receive input from higher centers, especially hypothalamic sympathetic/parasympathetic headquarters.
- Autonomic innervation pathway: cardiac centers modify heart rate/output via **vagus nerves (X)** (parasympathetic) and **cardiac nerves** (sympathetic, from cervical ganglia and postganglionic ganglia T1–T4).
- Sensory input to cardiac centers arrives via visceral sensory fibers of **glossopharyngeal (IX)** and **vagus (X)** nerves, from **baroreceptors** (blood pressure) and **chemoreceptors** (dissolved gas concentration). Cardiac centers respond quickly to maintain circulation to vital organs (e.g., brain) — e.g., drop in blood pressure or rise in CO₂ → increased heart rate/contraction force via sympathetic activation.
- Mnemonic: "SSS" — Sympathetic input Speeds and Strengthens the heartbeat.

Clinical Correlations Summary (Ch 21)

- **Pericarditis**: infection/inflammation of pericardium (viral most common; also bacterial/TB, cancer, kidney failure/uremic, trauma).
- **Myocarditis**: inflammation of myocardium (viral, autoimmune, toxins, alcohol, meds, chemo/radiation); severe cases → chronic cardiomyopathy, clotting risk, possible transplant.

- **Endocarditis:** bacterial (or fungal) infection of endocardium; risk with congenital defects/damaged valves; can destroy valves requiring replacement.
- **Heart murmur:** turbulent flow sound; innocent (e.g., pregnancy) vs pathologic (septal defects).
- **Valvular stenosis / insufficiency:** narrowed opening / incomplete closure with regurgitation.
- **Coronary artery disease (CAD):** atherosclerosis → cardiac ischemia → angina; risk factors smoking, obesity, hypertension, hypercholesterolemia, diabetes, sedentary lifestyle; coronary thrombosis → heart attack/arrhythmia/heart failure; treated via thrombolytics, balloon angioplasty + stent, or CABG (saphenous vein or internal thoracic artery graft); ECG detects ischemia.
- **Bradycardia / Tachycardia:** abnormal pacemaker rates.
- Clinical Case (Ellen): **Takotsubo cardiomyopathy** ("broken heart syndrome") — triggered by severe emotional stress; sympathetic NE release "overdrive" causes left ventricular apical wall hypokinesia (balloon-shaped ventricle resembling a Japanese octopus trap/tako-tsubo at end systole), mimicking a heart attack (ECG changes resemble MI) but coronary vessels normal; typically resolves with supportive care.
- Related clinical terms: **artificial pacemaker** (battery device maintaining regular heart rhythm, permanent or temporary); **asystole** (absence of cardiac activity/no contraction/no output); **cardiomyoplasty** (surgical use of stimulated latissimus dorsi wrapped around ventricles to assist cardiac function); **cardiomegaly** (enlarged heart, sign of underlying condition e.g. stress, muscle weakening, CAD); **fibrillation** (fast twitching of muscle fibers with little/no coordinated movement; atrial fibrillation = atria beating out of sync with ventricles); **heart block** (delay in normal electrical impulses); **palpitation** (irregular/rapid heartbeat sensation); **sick sinus syndrome** (group of rhythm disorders where SA node fails to properly regulate heart rhythm); implantable defibrillator (detects shockable rhythm, delivers shock to restore normal rhythm — described in text without explicit name); balloon-tipped catheter used to enlarge narrowed artery (angioplasty).

Vessels and Circulation (Ch 22)

22.1 Histological Organization of Blood Vessels

- Cardiovascular system is closed; pulmonary circuit supplies lungs, systemic circuit supplies rest of body; heart pumps into both simultaneously.
- Pulmonary circuit: pulmonary valve → ends at entrance to left atrium. Systemic circuit: aortic valve → ends at entrance to right atrium.
- After entering an organ, vessels branch progressively smaller; hundreds of millions of tiny arteries feed >10 billion capillaries (each ~RBC diameter). Combined capillary length estimated 5,000–25,000 miles (could cross the continental US or circle the globe).
- Exchange between blood and interstitial fluid occurs across capillary walls only.
- Vessel wall = 3 layers, superficial to deep:

1. **Adventitia (tunica externa)** — outer connective tissue sheath, mostly collagen fibers with scattered elastic fibers; thicker than media in veins; blends into surrounding tissue, anchoring the vessel.

2. **Media (tunica media)** — middle layer; concentric smooth muscle layers supported by loose connective tissue; smooth muscle encircles lumen. Sympathetic ANS stimulation → contraction → **vasoconstriction** (reduced lumen diameter); relaxation → **vasodilation** (increased diameter). Smooth muscle also responds to local stimuli (pH, pCO₂, pO₂). Collagen fibers bind media to intima/adventitia. Arteries have **external elastic membrane** (thin elastic fiber band) between media and adventitia.

3. **Intima (tunica intima)** — innermost; endothelial lining + underlying connective tissue with variable elastic fibers. In arteries, outer margin has **internal elastic membrane** (thick elastic layer); largest arteries have thickest intima.

- **Vasa vasorum** ("vessels of vessels") — small arteries/veins within the walls of large vessels supplying the smooth muscle/fibroblasts of media/adventitia (since vessel walls too thick for direct diffusion).
- **Distinguishing arteries from veins:** (1) artery walls thicker (more smooth muscle/elastic fibers in media) than veins; (2) artery lumen appears smaller/circular (retains shape) vs vein lumen larger/collapsed-flattened in

section; (3) constricted artery endothelium folds → pleated appearance (veins lack this); (4) veins have **valves** (prevent backflow), arteries do not.

- **Arteries** (elastic → muscular → arterioles, decreasing diameter toward periphery):
 - **Elastic arteries:** large vessels, luminal diameter up to ~2.5 cm; transport large blood volumes away from heart (e.g., pulmonary artery, aorta, brachiocephalic trunk, carotid/subclavian/common iliac arteries). Thick intima (endothelium + basal lamina of collagen/elastic fibers); internal elastic membrane separates intima from media. Media has high density of elastic fibers, relatively few smooth muscle cells (which don't contract in response to sympathetic/local stimulation) → very resilient, tolerates cardiac cycle pressure changes. During systole, arteries stretch (cushions pressure rise); during diastole, elastic recoil maintains forward flow (slows pressure drop, forces blood onward).
 - **Muscular (medium-sized) arteries:** transport blood to skeletal muscles/internal organs; more smooth muscle in media than elastic arteries; prominent internal elastic membrane. Diameter/wall controlled by sympathetic ANS + local stimulation — regulates blood flow per organ independently. Average luminal diameter ~4 mm (e.g. radial, ulnar arteries), wall thickness ~1 mm.
 - **Arterioles:** much smaller than muscular arteries; avg luminal diameter ~30 μm , wall thickness ~6 μm ; thin adventitia; media = 1-2 layers of smooth muscle cells that may not form a complete layer; visible only by microscope. Control blood flow between arteries and capillaries; diameter changes with local conditions or sympathetic/endocrine stimulation.
 - Vessel characteristics change gradually along the arterial tree (largest muscular arteries have much elastic tissue; smallest resemble heavily muscled arterioles).
 - Clinical: **Arteriosclerosis** = hardening of arteries (types include atherosclerosis). **Atherosclerosis** = fatty plaque deposition on arterial walls, beginning with endothelial damage; most common arteriosclerosis type; develops gradually, may begin early in life. Effects by location: coronary vessels → CAD → angina/MI; carotid/cerebral arteries → transient ischemic attack (TIA/mini-stroke), untreated can progress to stroke; extremity arteries (esp. lower legs) → peripheral artery disease (PAD) — intermittent claudication (limping/calf pain with walking), ischemia, possible gangrene if blood supply lost; renal arteries → hypertension → kidney failure. Non-preventable risk factors: aging, family history, inflammatory diseases. Preventable: smoking (worst factor), hypertension, diabetes, obesity, sedentary lifestyle, high cholesterol/triglycerides.
- **Capillaries:** smallest, most delicate vessels; only vessels permitting nutrient/waste exchange with interstitial fluid (thin walls = short diffusion distance). Avg luminal diameter ~8 μm (close to single RBC). Blood flows slowly through capillaries, allowing time for diffusion/active transport.
 - **Continuous capillaries:** found in all tissues except epithelia/cartilage; complete endothelial lining, cells connected by tight junctions and desmosomes; single endothelial cell wraps around lumen.
 - **Fenestrated capillaries:** "windows"/pores from incomplete/perforated endothelial lining ("Swiss cheese" appearance); allow peptides/small proteins to pass; permit rapid fluid/solute exchange. Found in: choroid plexus (brain), endocrine glands, gallbladder, kidney, pancreas, some intestinal locations.
 - **Sinusoids (discontinuous capillaries):** flattened, irregularly shaped, fenestrated + gaps between endothelial cells; basal lamina thin or absent (absent in liver). Blood moves slowly, maximizing absorption/secretion time. Found in liver, bone marrow, spleen, adrenal glands.
 - 4 mechanisms of capillary exchange: (1) diffusion across endothelial cells (lipid-soluble materials, gases, water via osmosis); (2) diffusion through gaps between endothelial cells (water, small solutes; larger solutes in sinusoids); (3) diffusion through fenestrated/sinusoid pores (water, solutes); (4) vesicular transport (endocytosis at luminal side, exocytosis at basal side).
- **Capillary beds:** collection of interconnected capillaries; single arteriole gives rise to dozens of capillaries emptying into several venules. Smooth muscle in arterioles cycles contraction/relaxation ~a dozen times/minute → pulsatile (not steady) capillary flow; blood routing through the bed shifts moment to moment = **autoregulation** (tissue-level control).
 - **Collaterals** = multiple muscular arteries supplying one capillary bed, fusing together (**arterial anastomosis**) rather than forming a series — found in brain, heart, stomach, high-demand areas; ensures reliable supply if one artery blocked.

- **Arteriovenous anastomoses** = direct arteriole-venule connections, common in visceral organs/joints; smooth muscle regulates flow; when dilated, blood bypasses the capillary bed entirely.
- **Thoroughfare channels** = smooth-muscle-lined passageways connecting arterioles to venules in some connective tissues, regulating flow by constricting/dilating.
- **Metarterioles (precapillary sphincters)** = guard capillary entrances in mesenteries; represent thoroughfare channels within mesenteric circulation (not well-supported outside mesenteries).
- **Veins:** collect blood, return to heart; thinner walls than corresponding arteries (lower venous blood pressure); larger luminal diameter than corresponding arteries despite thinner walls.
 - **Venules** (smallest veins): collect blood from capillaries; smallest ("postcapillary venules") resemble expanded capillaries; venules $<50\ \mu\text{m}$ lack complete media (only isolated smooth muscle cells deep to intima); venules $>50\ \mu\text{m}$ have 1-2 smooth muscle layers, thin/connective-tissue-dominated.
 - **Medium-sized veins:** luminal diameter 2–10 mm; most deep veins (radial, tibial, popliteal) in this category; typically paired with a muscular artery in same connective tissue sheath. Best-visualized 3-layer wall organization; intima = endothelium + basal lamina; media has several smooth muscle layers + collagen/elastic tissue; adventitia typically thicker than media, elastic/collagenous fibers. **Valves** characteristic of medium-sized veins — infoldings of intima, one-way, prevent backflow (like heart valves). Especially important in limbs where venous pressure can't overcome gravity alone. **Skeletal muscle pump** = surrounding muscle contraction squeezes blood toward heart (valves prevent backflow to compartmentalize venous blood weight).
 - **Large veins:** superior/inferior venae cavae + tributaries (subclavian, renal, mesenteric, portal veins) in abdominopelvic/thoracic cavities. Intima/media small, hard to distinguish; adventitia = thickest layer (collagen/elastic fibers, isolated longitudinal smooth muscle cells). No valves in large veins; body cavity pressure changes help move blood toward heart.
 - **Venoconstriction:** sympathetic-triggered constriction of medium-sized vein smooth muscle during hemorrhage — reduces venous blood volume, increases arterial/capillary volume, helping maintain arterial blood volume/flow to vital organs (brain, active muscle) despite blood loss.
 - Veins = **capacitance vessels** (high capacitance = large volume change accommodated per pressure change); stretch $\sim 8\times$ more than corresponding artery per given pressure rise; act as blood reservoirs.

22.2 The Distribution of Blood

- Blood volume distribution: heart + arteries + capillaries $\approx 30\text{--}35\%$ of blood volume ($\sim 1.5\ \text{L}$); venous system $\approx 65\text{--}70\%$ ($\sim 3.5\ \text{L}$).
- Veins more distensible than arteries (thinner walls, more elastic tissue, less smooth muscle).

22.3 Blood Vessel Distribution (Functional Patterns)

- Divided into pulmonary circuit (heart $\leftrightarrow$ lungs, short distance, lower pressure, thinner pulmonary artery walls) and systemic circuit (heart $\leftrightarrow$ all other tissues, longer distance).
- Three important functional patterns:

1. Peripheral arterial/venous distribution is usually mirror-identical on left/right sides of body, except near the heart (largest vessels connect asymmetrically to atria/ventricles).

2. A single vessel may have multiple names as it crosses anatomical boundaries (enables precise anatomical description over long peripheral course).

3. Anastomoses commonly found between arteries and veins — reduce impact of vessel occlusion/blockage.

22.4 The Pulmonary Circuit

- Blood enters right atrium from peripheral/myocardial capillary beds (O_2 released, CO_2 absorbed) $\rightarrow$ right ventricle $\rightarrow$ pulmonary valve $\rightarrow$ pulmonary trunk (start of pulmonary circuit).
- Pulmonary circuit holds $\sim 9\%$ of total blood volume at any moment; short distance — pulmonary trunk base to lungs only $\sim 15\ \text{cm}$ (6 in.).

- Pulmonary arteries carry deoxygenated blood (opposite of systemic arteries) — often color-coded blue like systemic veins.
- Path: pulmonary trunk curves over heart's superior border → divides into left/right pulmonary arteries → enter lungs, branch repeatedly → pulmonary arterioles → capillary networks around alveoli (thin walls permit gas exchange) → oxygenated blood collects into venules → merge into 4 pulmonary veins (typically 2/lung) → empty into left atrium (completes pulmonary circuit).
- Systemic circuit begins at aortic valve, ends at right atrium entrance; holds ~84% of total blood volume at any moment; supplies capillary beds everywhere except lungs.

22.5 Systemic Arteries

The Aorta

- Begins at aortic valve of left ventricle. Segments: ascending aorta → aortic arch → descending aorta (thoracic then abdominal).
- **Ascending aorta:** left/right coronary arteries originate here, just superior to aortic valve.
- **Aortic arch:** curves like a cane handle over the heart, connects ascending to descending aorta. Three elastic arteries branch off: **brachiocephalic trunk**, **left common carotid artery**, **left subclavian artery** — supply head, neck, shoulders, upper limbs. Brachiocephalic trunk (only one; ascends briefly, then splits posterior to right sternoclavicular joint) → **right subclavian artery** + **right common carotid artery**. Mnemonic: "ABCs" = Aortic arch, Brachiocephalic trunk, Common carotid, Subclavian.

Subclavian Arteries

- Arch over lung apices; supply upper limbs, chest wall, shoulders, back, brain, spinal cord. Three branches before leaving thoracic cavity: (1) **thyrocervical trunk** (neck/shoulder/upper back muscles), (2) **internal thoracic artery** (pericardium, anterior chest wall), (3) **vertebral artery** (brain, spinal cord).
- After crossing first rib → becomes **axillary artery** (supplies pectoral region/axilla muscles; gives off humeral circumflex arteries near humeral head) → becomes **brachial artery** (supplies upper limb). Brachial artery's first branch = **deep brachial artery** (posterior arm structures); also gives **ulnar collateral** and **ulnar recurrent arteries** (elbow region). Near coronoid fossa, divides into **radial** and **ulnar arteries** (forearm, following radius/ulna) → anastomose at wrist forming **superficial** and **deep palmar arches** → supply palm and digital arteries of thumb/fingers.

Carotid Arteries and Brain Blood Supply

- Common carotid arteries ascend lateral to trachea; at larynx level, each divides into **external carotid artery** and **internal carotid artery**. **Carotid sinus** at internal carotid base (extends along common carotid) contains baroreceptors and chemoreceptors for cardiovascular regulation.
- External carotid arteries → neck, pharynx, esophagus, larynx, lower jaw, face.
- Internal carotid arteries → enter skull via carotid canals → brain; each ascends to optic nerve level, divides into 3 branches: (1) **ophthalmic artery** (eye), (2) **anterior cerebral artery** (frontal/parietal lobes), (3) **middle cerebral artery** (midbrain, lateral cerebral hemisphere surfaces).
- Brain circulatory interruption: unconsciousness within seconds; permanent neural damage after ~4 minutes. Redundant supply from 2 sources: vertebral arteries + internal carotid arteries.
- Vertebral arteries branch from subclavian arteries, ascend through cervical vertebrae transverse foramina, enter cranium at foramen magnum, fuse along medulla oblongata ventral surface forming the **basilar artery** → continues along pons ventral surface → branches, divides into **posterior cerebral arteries** → **posterior communicating arteries**.
- Internal carotids supply anterior cerebrum; vertebral/basilar supply the rest. Internal carotids and basilar artery interconnect forming the **cerebral arterial circle (circle of Willis)** — ring anastomosis encircling the pituitary infundibulum — ensures brain blood supply redundancy from carotid and/or vertebral sources.
- Clinical: **Aneurysm** = arterial wall bulge/weakness, can occur anywhere (focus: aorta). Aortic aneurysm types: thoracic (ascending/descending) or abdominal. Often asymptomatic early, found incidentally via imaging; screening recommended for people >65 with family history, and men >65 who ever smoked. Rupture mortality

>90%. Repair options: open surgical graft replacement (cross-clamp aorta, insert synthetic tube graft) or less invasive **endovascular stent graft** (via femoral artery catheter, guide wire, expandable stent).

Descending Aorta

- Diaphragm divides it into **thoracic aorta** (superior) and **abdominal aorta** (inferior).
- Thoracic aorta begins at vertebra T5 (slightly left of midline), shifts toward midline, passes through diaphragm at vertebra T12; supplies thoracic organs (except heart), chest/diaphragm muscles, thoracic spinal cord.
 - Visceral branches: **bronchial arteries** (lung airways), **pericardial arteries** (pericardium), **mediastinal arteries** (mediastinal structures), **esophageal arteries** (esophagus).
 - Parietal branches: **intercostal arteries** (chest muscles, vertebral column area), superior [phrenic] arteries (diaphragm — text truncated but implied).
- Abdominal aorta: begins immediately inferior to diaphragm, descends slightly left of vertebral column, posterior to peritoneal cavity, cushioned by adipose tissue; splits at vertebra L4 into right/left **common iliac arteries** (terminal segment of aorta).
 - 3 unpaired visceral branches:
 1. **Celiac trunk** — liver, stomach, esophagus, gallbladder, duodenum, pancreas, spleen. Divides into: **left gastric artery** (stomach, inferior esophagus), **splenic artery** (spleen; + left gastro-epiploic artery to stomach, pancreatic arteries to pancreas), **common hepatic artery** (liver via hepatic artery proper; + right gastric artery to stomach, cystic artery to gallbladder, gastroduodenal/right gastro-epiploic/superior pancreaticoduodenal arteries to duodenal area).
 2. **Superior mesenteric artery** (~2.5 cm inferior to celiac trunk) — pancreas & duodenum (inferior pancreaticoduodenal artery), small intestine (intestinal arteries), most of large intestine (right colic, middle colic, ileocolic arteries).
 3. **Inferior mesenteric artery** (~5 cm superior to aortic terminal segment) — terminal colon (left colic, sigmoid arteries), rectum (rectal arteries).
 - 5 paired branches: (1) **inferior phrenic arteries** (inferior diaphragm, inferior esophagus), (2) **adrenal (suprarenal) arteries** (adrenal glands, near superior mesenteric artery origin), (3) **renal arteries** (~7.5 cm long, ~2.5 cm inferior to SMA, kidneys), (4) **gonadal arteries** (testicular arteries in males → testes/scrotum; ovarian arteries in females → ovaries/uterine tubes/uterus; distribution differs by sex), (5) **lumbar arteries** (vertebrae, spinal cord, abdominal wall).

Arteries of Pelvis and Lower Limbs

- At L4, aorta splits into right/left **common iliac arteries** + small **median sacral artery**; common iliacs travel along inner ilium surface, posterior to cecum/sigmoid colon.
- At lumbosacral joint, each common iliac divides into **internal iliac artery** (enters pelvic cavity; supplies urinary bladder, internal/external pelvic walls, external genitalia, medial thigh; branches: superior gluteal, internal pudendal, obturator, lateral sacral arteries; in females also supplies uterus/vagina) and **external iliac artery** (larger diameter; supplies lower limbs).
- External iliac artery crosses iliopsoas, penetrates abdominal wall between ASIS and pubic symphysis → becomes **femoral artery** on anteromedial thigh. ~5 cm distal, **deep femoral artery** branches off laterally, giving **medial** and **lateral circumflex arteries** (thigh muscles/skin).
- Femoral artery continues posterior to femur; at popliteal fossa gives off **descending genicular artery** (medial knee) → becomes **popliteal artery** through adductor magnus → crosses popliteal fossa, branches into **posterior tibial artery** (deep to soleus, posterior leg skin/muscles) and **anterior tibial artery** (between tibia/fibula, anterior leg). Posterior tibial gives rise to **fibular (peroneal) artery** (~4 cm distal, lateral leg).
- At ankle: anterior tibial artery → **dorsalis pedis artery** (ankle, dorsal foot). Posterior tibial artery divides into **medial** and **lateral plantar arteries** (plantar foot surface); connects to dorsalis pedis via anastomoses linking **dorsal (arcuate) arch** to **plantar arch**, supplying toes/distal foot.

22.6 Systemic Veins

- Veins drain into right atrium via superior/inferior venae cavae (cardiac veins excepted — drain via coronary sinus).
- Two key arterial/venous system differences in neck/limbs: (1) venous drainage is variable (number/location of veins vary significantly between individuals); (2) neck/limb arteries are deep (protected by bone/tissue), while veins have both superficial (visible, used for blood draws, e.g. antecubital) and deep sets — dual drainage aids temperature control (cold: blood shunted to deep veins, superficial bypassed; heat: superficial veins dilate, become prominent during exercise).
- Venous branching is more variable than arterial (arteries grow directly toward active tissue during development; venous pressures are low by the time blood reaches veins, so routing variation matters less functionally).

Venous Return from the Cranium

- Superficial cerebral veins → dural sinus network: superior/inferior sagittal sinuses, petrosal sinuses, occipital sinus, left/right transverse sinuses, straight sinus. Superior sagittal sinus lies within the falx cerebri.
- Internal cerebral veins → **great cerebral vein** (collects from interior cerebral hemispheres, choroid plexus) → **straight sinus**.
- Other cerebral veins + orbital veins → **cavernous sinus** → **petrosal sinuses** → internal jugular vein.
- Venous sinuses converge near lambdoid suture; left/right **transverse sinuses** form at confluence of sinuses (near petrous temporal bone) → each drains to a **sigmoid sinus** → continues as **internal jugular vein** (exits skull at jugular foramen, descends parallel to common carotid artery).
- **Vertebral veins** drain cervical spinal cord/posterior skull, descend in cervical transverse foramina next to vertebral arteries → empty into brachiocephalic veins.
- Superficial head veins: **temporal, facial, maxillary veins**. Temporal + maxillary → **external jugular vein**; facial → **internal jugular vein**. Anastomosis at mandible angle gives dual facial/scalp/cranial drainage. External jugular descends superficial to sternocleidomastoid, empties into subclavian vein posterior to clavicle (easily palpable — jugular venous pulse/JVP).

Venous Return from the Upper Limb

- Digital veins → superficial/deep palmar veins → **palmar venous arches**.
- Superficial arch → **cephalic vein** (radial side forearm), **median antebrachial vein** and **basilic vein** (ulnar side forearm). **Median cubital vein** (anterior elbow) interconnects cephalic/basilic — standard site for venous blood draws.
- Deep palmar veins → **radial** and **ulnar veins** → fuse with **anterior interosseous vein** forming the **brachial vein** (parallels brachial artery); receives basilic vein → enters axilla as **axillary vein**.
- Cephalic vein joins axillary vein forming **subclavian vein** → passes over first rib deep to clavicle into thoracic cavity → merges with external/internal jugular veins forming **brachiocephalic vein** (also receives vertebral vein). Left/right brachiocephalic veins merge at 1st/2nd rib level → **superior vena cava**. **Internal thoracic vein** empties into brachiocephalic vein near this junction.
- **Azygos vein**: ascends from lumbar region over right vertebral column side, enters thorax through diaphragm, joins SVC at vertebra T2; receives blood from intercostal veins, esophageal veins, other mediastinal veins, plus the **hemi-azygos vein** (left side; may also drain via highest intercostal vein into left brachiocephalic vein). **Accessory hemi-azygos vein** starts at 4th/5th intercostal space, travels inferiorly on left, crosses to right at T8, empties into azygos vein. Azygos/hemi-azygos = chief thoracic collecting vessels.

Inferior Vena Cava

- Collects most venous blood from below diaphragm (small amount reaches SVC via azygos/hemi-azygos instead).
- Veins draining lower limb: plantar veins (sole) → plantar venous arch → deep leg veins: **anterior tibial, posterior tibial, fibular (peroneal) veins**. **Dorsal venous arch** (dorsal foot/digital veins) drains via extensive anastomoses with plantar arch to either superficial or deep veins.

- Dorsal venous arch → 2 superficial veins: **great saphenous vein** (longest vein in body; medial leg/thigh → femoral vein near hip; often harvested for CABG) and **small saphenous vein** (posterior/lateral calf → popliteal fossa → popliteal vein).
- **Popliteal vein** = union of anterior tibial, posterior tibial, fibular veins; palpable in popliteal fossa; becomes **femoral vein** proximally (parallels femoral artery).
- Before entering abdominal wall, femoral vein receives: (1) great saphenous vein, (2) **deep femoral vein** (deep thigh structures), (3) **femoral circumflex vein** (femoral neck/head area) → femoral vein penetrates body wall, becomes **external iliac vein**.
- Pelvic veins: external iliac vein (lower limbs/pelvis/lower abdomen) merges with **internal iliac vein** (pelvic organs; formed from gluteal, internal pudendal, obturator, lateral sacral veins) → **common iliac vein**. **Median sacral vein** empties into left common iliac. Left/right common iliac veins ascend, unite anterior to vertebra L5 → **inferior vena cava** (retroperitoneal, ascends right of and parallel to aorta) → right atrium (mixes with SVC blood) → right ventricle → pulmonary circuit.
- Abdominal IVC collects from 6 major veins: (1) **lumbar veins** (connected to azygos/hemi-azygos), (2) **gonadal veins** (right → IVC directly; left → left renal vein), (3) **hepatic veins** (from liver, at vertebra T8 level), (4) **renal veins** (largest tributaries of IVC), (5) **adrenal veins** (right → IVC; left → left renal vein), (6) **phrenic veins** (right → IVC; left → left renal vein).

Hepatic Portal System

- Begins in digestive organ capillaries, ends in liver sinusoids; liver is the only digestive organ draining directly into IVC. Blood from celiac, superior mesenteric, inferior mesenteric artery-supplied capillary beds flows into hepatic portal system instead of directly to IVC.
- Portal vessel = vessel connecting two capillary beds; portal system = the network.
- Hepatic portal blood differs from other systemic venous blood — contains substances absorbed from stomach/intestines (glucose, amino acid levels can exceed anywhere else in cardiovascular system); delivered directly to liver for storage/metabolic conversion/excretion.
- **Hepatic portal vein** = largest vessel; delivers blood to liver; receives blood from 3 large veins:
 1. **Inferior mesenteric vein** — inferior large intestine capillaries; receives **left colic vein** and **superior rectal veins** (descending colon, sigmoid colon, rectum).
 2. **Splenic vein** — receives inferior mesenteric vein + veins from spleen, stomach lateral border (left gastro-epiploic vein), pancreas (pancreatic veins).
 3. **Superior mesenteric vein** — stomach (right gastro-epiploic vein), small intestine (intestinal, pancreaticoduodenal veins), 2/3 of large intestine (ileocolic, right colic, middle colic veins).
- Hepatic portal vein forms via fusion of superior mesenteric + splenic veins (superior mesenteric contributes greater blood volume/nutrients). Also receives **gastric veins** (medial stomach border) and **cystic veins** (gallbladder) en route to liver.
- Because intestinal blood goes to liver first, systemic circuit blood composition stays relatively stable regardless of ongoing digestive activity.

22.7 Cardiovascular Changes at Birth

- Fetal circulation differs from adult due to different respiratory/nutritional sources: fetal lungs collapsed/nonfunctional, digestive tract has nothing to digest; the **placenta** handles fetal respiratory/nutritional exchange with maternal blood.
- **Two umbilical arteries** branch off fetal internal iliac arteries, enter umbilical cord, deliver blood to placenta.
- **Umbilical vein** (single) returns oxygen/nutrient-rich blood from placenta to fetus → drains into **ductus venosus** (connects to fetal liver venous network) → returns blood to IVC.
- At birth, placental connection breaks; umbilical vessels cease flow and degenerate.
- **Foramen ovale**: interatrial opening with a valve-like flap; allows right-to-left atrial blood flow (bypassing pulmonary circuit) but backflow closes the valve, isolating chambers.
- **Ductus arteriosus**: short muscular vessel connecting pulmonary trunk and aortic arch (2nd bypass route).

- Fetal lungs collapsed → pulmonary capillaries compressed, little blood flows through lungs. During diastole: blood enters right atrium → right ventricle, but also passes right atrium → left atrium via foramen ovale (~25% of right atrial blood bypasses pulmonary circuit this way). >90% of right ventricle output passes through ductus arteriosus into systemic circuit rather than continuing to lungs.
- At birth: first breath → lungs/pulmonary vessels expand → ductus arteriosus smooth muscle contracts, isolating pulmonary/aortic trunks, blood begins flowing through pulmonary circuit → rising left atrial pressure closes the foramen ovale's valvular flap, completing cardiovascular remodeling.
- Postnatal remnants: foramen ovale → **fossa ovalis** (shallow depression in interatrial septum); ductus arteriosus → **ligamentum arteriosum** (fibrous cord).
- Clinical: if these changes fail to occur properly, congenital heart defects result (severity depends on which connection remains open/opening size); treatment may include surgical closure of foramen ovale, ductus arteriosus, or both. Other defects arise from abnormal cardiac development or inappropriate heart-vessel connections. Congenital cardiovascular problems serious enough to threaten homeostasis are relatively rare; usually surgically correctable (may need multiple surgeries; severe defects can shorten life expectancy).

22.8 Aging and the Cardiovascular System

- Cardiovascular capabilities gradually decline with age.
- Blood changes: (1) decreased hematocrit, (2) peripheral vein constriction/blockage by thrombus (stationary clot) — can detach, travel through heart, lodge in a small artery (commonly pulmonary) causing **pulmonary embolism**, (3) blood pooling in leg veins from less efficient valve function.
- Heart changes: (1) reduced maximum cardiac output, (2) decreased cardiac skeleton elasticity, (3) progressive atherosclerosis restricting coronary circulation, (4) damaged cardiac muscle replaced by scar tissue, (5) changes in pacemaker/conducting cell activity.
- Blood vessel changes (often arteriosclerosis-related): (1) decreased arterial/venous elasticity, (2) arterial walls less tolerant of sudden pressure increases → risk of **aneurysm** (enlarged vessel — may cause stroke, heart attack, massive blood loss depending on vessel), (3) calcium salt deposition on weakened vascular walls → increased stroke/heart attack risk, (4) thrombi forming at atherosclerotic plaques.

Clinical Correlations Summary (Ch 22)

- **Arteriosclerosis / Atherosclerosis**: plaque buildup, hardened/narrowed arteries; leads to CAD, TIA/stroke, PAD, renal hypertension/failure depending on location.
- **Aortic aneurysm**: thoracic or abdominal; often asymptomatic, found incidentally; screening recommended for at-risk groups; >90% mortality if ruptured; treated via open graft surgery or endovascular stent.
- **Congenital cardiovascular problems**: rare but serious; abnormal heart formation or heart-vessel interconnections; usually surgically correctable.
- Clinical Case (Rodrigo): **Arteriovenous malformation (AVM)** — tangle of abnormal vessels connecting arteries directly to veins with no intervening capillary bed; blood flows quickly/directly from artery to thin-walled vein without perfusing surrounding tissue with O₂. In Rodrigo's case, a branch of the left middle cerebral artery connects directly to a thin-walled vein, depriving the left primary motor cortex (precentral gyrus) and primary somatosensory cortex (postcentral gyrus) of blood → right hand numbness/weakness; higher arterial pressure forced into low-pressure thin-walled veins → rupture → hemorrhagic stroke (brain bleed) → seizure. Likely congenital (formed during fetal development), grown into a large tangle pressing on brain tissue. Treatment options: open surgery (aneurysm/AVM repair), endovascular embolization (cuts off blood supply to the region), or radiation therapy.
- Related clinical terms: **angiogram** (x-ray of a vessel visualized via injected dye); **carotid sinus massage** (rubbing carotid bifurcation area, a diagnostic/therapeutic procedure); **deep vein thrombosis (DVT)** (clot in major vein, usually legs, often after prolonged inactivity e.g. long flights; can embolize to lungs causing respiratory distress/failure); **intermittent claudication** (cramping leg pain limp from arterial insufficiency); **orthostatic hypotension** (low BP on standing, causes dizziness/lightheadedness); **phlebitis** (vein inflammation); **Raynaud's phenomenon** (finger/toe discoloration from temperature changes or emotional stress); **sclerotherapy** (varicose vein treatment via irritant injection causing

inflammation/coagulation/narrowing); **sounds of Korotkoff** (turbulent arterial flow sounds heard via stethoscope during BP measurement); **syncope** (temporary loss of consciousness from sudden BP drop); **thrill** (vibration felt in a vessel from abnormal flow, e.g. at hemodialysis fistula); **thrombophlebitis** (vein inflammation with thrombus formation); **vascular murmur** (abnormal sounds from turbulent blood flow on auscultation); **white coat hypertension** (short-term BP rise triggered by seeing medical personnel); **normotensive** (normal blood pressure); sphygmomanometer (inflatable cuff BP measurement device, described without being named directly in extracted text).

Lymphatic (Ch 23)

23.1 An Overview of the Lymphatic System

- Lymphatic (lymphoid) system = functionally part of circulatory and immune systems; monitors/transport **lymph** (fluid connective tissue).
- **Lymphatic vessels** carry lymph; **lymphocytes** = cells suspended in lymph/blood; specialized lymphatic tissues/organs adjust lymph composition and produce lymphocytes.
- Lymphatic vessels originate in peripheral tissues, deliver lymph to venous system. Lymph = (1) interstitial fluid resembling plasma but lower protein concentration, (2) lymphocytes (immune response cells), (3) macrophages (phagocytic cells).
- Functions of the lymphatic system:

1. **Produces, maintains, distributes lymphocytes** — produced/stored in spleen, thymus, red bone marrow.

Lymphatic tissues/organs classified as:

- **Primary lymphatic structures** — sites where lymphocytes are produced and mature (red bone marrow, adult thymus).
- **Secondary lymphatic structures** — sites where lymphocytes are activated to produce more of the same type (e.g., activated B cells dividing to fight infection); located at "front lines" (tonsils, lymph nodes).

2. **Maintains normal blood volume, eliminates local interstitial fluid composition variations** — proximal systemic capillary blood pressure ≈ 35 mm Hg forces water/solutes out of plasma into interstitial fluid; ~ 3.6 L (72% of total blood volume) enters interstitial fluid daily; lymphatic vessels return equal volume back to bloodstream, eliminating regional interstitial fluid composition differences. A break in a major lymphatic vessel can cause rapid, potentially fatal blood volume decrease (since so much fluid cycles through daily).

3. **Provides alternate route for transporting hormones, nutrients, wastes** — e.g., lipids absorbed by digestive tract are carried to bloodstream via lymphatic vessels rather than direct capillary absorption.

23.2 Structure of Lymphatic Vessels

- Lymphatic vessels carry lymph one-way (peripheral tissues $\rightarrow$ venous system), unlike blood vessels' complete circuit.
- **Lymphatic capillaries**: begin the lymphatic network in peripheral tissues; closed tubes. Differ from vascular capillaries:

1. Larger luminal diameters.
2. Thinner walls; endothelial cells lack continuous basal lamina.
3. Flat or irregular outline.
4. Endothelial cells anchored by fine filaments to surrounding connective tissue (keep passageways open under increased interstitial pressure).
5. Greater permeability — endothelial cells overlap rather than tightly bind, forming one-way valves permitting fluids/solutes/proteins (and viruses, bacteria, cell debris) to enter but not exit back to intercellular spaces.

- Lymphatic capillaries present in most tissues; especially numerous in connective tissue deep to skin/mucous membranes and digestive tract mucosa/submucosa. **Lacteals** = lymphatic capillaries in small intestine, transport absorbed lipids. Absent in avascular areas (e.g., cornea).
- Clinical: blocked/compressed/valve-damaged lymphatic vessel → lymph drainage slows/stops in area → interstitial fluid volume/pressure rises → tissue swelling = **lymphedema**.
- Visual: arteries bright red, veins dark red, lymphatics pale golden color.
- **Larger lymphatic vessels**: differ from veins — (1) thinner walls, (2) wider lumens, (3) no clear boundaries between the three wall layers. Small/medium lymphatic vessels have internal valves (like veins) preventing lymph backflow, especially in limbs — essential since lymphatic pressure is very low. Skeletal muscle contractions help move lymph.

23.3 Major Lymph-Collecting Vessels

- **Superficial lymphatics**: anastomose freely, more numerous than superficial veins; travel adjacent to superficial veins. Locations: subcutaneous layer deep to skin; loose connective tissue of mucous membranes (digestive, respiratory, urinary, reproductive tract linings); loose connective tissue of serous membranes (pleural, pericardial, peritoneal cavity linings).
- **Deep lymphatics**: large vessels accompanying deep arteries/veins; collect lymph from skeletal muscles and other organs of neck, limbs, trunk, plus thoracic/abdominopelvic visceral organs.
- Superficial + deep lymphatics converge into larger **lymphatic trunks**: (1) lumbar trunks, (2) intestinal trunks, (3) bronchomediastinal trunks, (4) subclavian trunks, (5) jugular trunks.
- **Thoracic duct**: collects lymph from both sides of body inferior to diaphragm AND left side of body superior to diaphragm. Begins as the **cisterna chyli** — expanded saclike chamber inferior to diaphragm at vertebra L2, where deep lumbar and intestinal trunks join; lies anterior to vertebral column, slightly right of midline; receives lymph from abdomen, pelvis, lower limbs via right/left lumbar trunks and intestinal trunk. Passes through diaphragm at the **aortic hiatus** (with the aorta), becoming the thoracic duct. At vertebra T5, shifts left, lying along left side of esophagus; ascends to left clavicle level; collects from left bronchomediastinal trunk, left subclavian trunk, left jugular trunk; empties into the **left subclavian vein** near the left internal jugular vein.
- **Right lymphatic duct**: collects lymph from the right side of the body superior to the diaphragm (right side of head, neck, thorax); forms near the right clavicle; empties into venous system at/near junction of right internal jugular and right subclavian veins.

23.4 Lymphocytes

- Lymphocytes = responsible for specific immunity; combat (1) invading organisms (bacteria, viruses), (2) abnormal body cells (virus-infected, cancer cells), (3) foreign proteins (e.g., bacterial toxins).
- Circulate through bloodstream and lymph, exit into interstitial space/peripheral tissues, return via lymphatic system; time spent within lymphatic system varies (hours to years). Encountering a pathogen/foreign protein triggers an immune response.
- Three classes of lymphocytes (biochemically/functionally distinct, few structural differences):
 - **T cells (thymus-dependent)**: ~80% of circulating lymphocytes. Originate in red bone marrow, migrate to thymus to become **immunocompetent** (activated). Types:
 - **Cytotoxic T cells** — direct cellular attack, **cell-mediated immunity**; enter peripheral tissues, physically/chemically attack antigens.
 - **Helper T cells** — stimulate responses of both T cells and B cells; must activate B cells before antibody production.
 - **Regulatory T cells** — moderate the immune response (subset of T cells); includes **suppressor/inducer T cells** (suppress B cell activity, stimulate other T cells).
 - **Memory T cells** — respond to previously-encountered antigens by cloning more lymphocytes; enable rapid secondary response if antigen reappears (initiated by memory cells).
 - **B cells (bone marrow-derived)**: originate/become immunocompetent in red bone marrow; ~10–15% of circulating lymphocytes. Activated B cells divide → **plasma cells (plasmocytes)** (produce/secrete

antibodies) and **memory B cells** (activated only if antigen reappears later). Antibodies = **immunoglobulins**; antibody-antigen binding triggers antigen destruction/neutralization/elimination. B cells provide **antibody-mediated (humoral) immunity** — defense within body fluids.

- **NK (natural killer) cells**: ~5–10% of circulating lymphocytes; attack foreign cells, virus-infected normal cells, cancer cells. Constant monitoring by NK cells + cytotoxic T cells = **immune surveillance**.
- Lymphocytes retain ability to divide as they migrate through tissues; division produces daughter cells of same type/antigen sensitivity (e.g., dividing B cell → more B cells only). This clonal expansion capacity is critical for effective immune response. If compromised (e.g., **AIDS** — virus selectively destroys T cells), individual cannot mount effective defense, succumbing to infections normally easily overcome.
- Immune system prepares for any possible antigen in advance: trillion+ lymphocytes in body, millions of different populations, each with thousands of cells recognizing a specific antigen (= **immunocompetence** — the ability to recognize antigens).
- Activation sequence: antigen phagocytized by macrophage → macrophage incorporates antigen pieces into its membrane, "presents" antigen to T cells → T cells with matching receptors bind, activate, divide → cell-mediated immunity.
- **Distribution/life span of lymphocytes**: B:T ratio varies by tissue — B cells seldom in thymus; blood T:B ratio 8:1; spleen ratio 1:1; red bone marrow B:T ratio 1:3. Lymphocytes are "visitors" not residents in organs — continually migrate (e.g., wandering T cell: ~30 min in blood, then 15-20 hrs in a lymph node; typical B cell: ~30 hrs in a lymph node before moving on). Long lifespans (far longer than other blood formed elements) — ~80% survive 4 years, some 20+ years.
- **Lymphocytopoiesis** (lymphocyte production): involves red bone marrow, thymus, peripheral lymphatic tissues. Hematopoietic stem cells in red bone marrow → lymphatic stem cells with 2 fates: (1) remain in bone marrow → mature into NK cells and B cells (which then migrate to peripheral tissues); (2) migrate to thymus → mature into T cells (then migrate to bloodstream, red bone marrow, spleen, other lymphatic tissues).
- Immune surveillance (NK cells), antibody-mediated immunity (B cells/plasma cells/antibodies), cell-mediated immunity (cytotoxic T cells) are the three defense mechanisms summarized.

23.5 Lymphatic Tissues

- Two types:
 - **Diffuse lymphatic tissue**: lymphocytes loosely aggregated within connective tissue (e.g., respiratory/urinary tract mucous membranes).
 - **Lymphatic nodules**: aggregations of lymphocytes within a supporting reticular cell/fiber framework; oval-shaped; lack a capsule (indistinct boundaries); found within walls of various digestive tract segments. Pale central **germinal center** contains activated, dividing lymphocytes. Average diameter ~1 mm.
- **Tonsils**: first lymphatic nodules to encounter invading bacteria/viruses entering through the mouth. Types:
 - **Palatine tonsils** — pair, posterior/inferior oral cavity margin, along pharynx-soft palate boundary.
 - **Lingual tonsils** — pair, deep to mucosa at tongue base (not visible).
 - (Pharyngeal tonsil implied elsewhere as chronically-infectable, forming abscess/bacteremia source.)
 - Collectively part of **mucosa-associated lymphoid tissue (MALT)** — lymphatic tissue within epithelia of digestive, respiratory, urinary, reproductive systems.
 - Clinical: **Tonsillitis** — infectious overwhelm, especially in children; chronically infected pharyngeal tonsils can abscess, become infection source, allowing bacteria into bloodstream via lymphatic capillaries/vessels → **bacteremia**; can cause swallowing/breathing difficulty (esp. during sleep). Early tonsillitis controlled with antibiotics; **tonsillectomy** considered if peritonsillar abscess forms, breathing/swallowing difficulty occurs, or child has >5-7 episodes/year. Tonsils most immunologically active before puberty; atrophy after puberty, may disappear by late adulthood.
- **Aggregated lymphoid nodules (Peyer's patches)**: clusters of lymphatic nodules in small intestine mucosal lining.
- **Appendix**: tube-shaped pouch near small/large intestine junction; walls contain fused lymphoid nodule mass.

- Lymphocytes in these nodules aren't always able to destroy bacteria/viruses that cross the digestive epithelium → infection can develop (e.g., tonsillitis, appendicitis).

23.6 Lymphatic Organs

- Lymphatic organs = separated from surrounding tissue by fibrous connective tissue capsule; includes lymph nodes, thymus, spleen.

Lymph Nodes

- ~1–25 mm diameter; dense fibrous connective tissue capsule; widely distributed throughout body. **Trabeculae** = fibrous capsule extensions into node interior.
- Blood vessels/nerves enter/exit at the **hilum**. **Afferent lymphatic vessels** bring lymph to node from peripheral tissues (penetrate capsule opposite the hilum); **efferent lymphatic vessels** carry lymph away (exit at hilum) — note: several afferent vessels but only ONE efferent vessel per node.
- Internal organization: outer **cortex**, inner **medulla**, intervening **paracortex**.
 - Lymph enters **subcapsular space** (meshwork of reticular fibers, macrophages, dendritic cells — dendritic cells start immune responses) → flows through **cortex** (periphery has B cells in germinal centers, like lymphoid nodules) → **paracortex** (dominated by T cells; lymphocytes cross blood vessel walls to enter node here) → **medulla** (B cells and macrophages; **medullary cords** = elongated dense lymphoid tissue masses between sinuses) → efferent lymphatics at hilum.
- Function: filter/purify lymph before it reaches venous system — ≥99% of antigens in arriving lymph are removed. Fixed macrophages in lymphatic sinuses engulf debris/pathogens; antigens processed and "presented" to nearby T cells, or stick to reticular fibers/dendritic cell surfaces, stimulating T cell activity.
- Largest lymph nodes ("lymph glands") found where peripheral lymphatics connect with a lymphatic trunk: base of neck, axillae, groin. "Swollen glands" = inflammation/infection indicator.
- Major named groups and drainage areas:
 - **Cervical lymph nodes** — head and neck, including meningeal lymphatics.
 - **Axillary lymph nodes** — trunk lymph from upper limbs; in women, also drain mammary glands.
 - **Popliteal lymph nodes** — thigh lymph arriving from the leg.
 - **Inguinal lymph nodes** — trunk lymph arriving from lower limbs.
 - **Thoracic lymph nodes** — lungs, respiratory passageways, mediastinal structures.
 - **Abdominal lymph nodes** — urinary and reproductive systems.
 - **Intestinal and mesenteric lymph nodes** — digestive tract.
- Clinical: **HIV** = retrovirus (RNA virus) targeting helper T cells via CD4 surface binding; invades and hijacks host replication machinery; destroys "CD4 cell" families of helper T cells → opportunistic infections. Normal CD4 count = 500–1,600 cells/mm³; **AIDS** diagnosis when CD4 count falls below 200. HIV cannot be fully cleared — lifelong infection once contracted; multiple antiretroviral drug classes target different HIV life cycle stages; good care/early treatment can control HIV, prevent AIDS progression.
- Clinical: **Cancer in lymph nodes: lymphoma** = cancer starting in the lymph nodes; more often cancer starts elsewhere (primary tumor) and spreads to nodes via bloodstream, most escaped cells die/killed by NK cells, but some settle and grow = **metastatic tumors**. Hematogenous-spreading cancers: **hepatomas** (primary liver tumors), **sarcomas** (connective tissue tumors). Lymphatic-spreading cancers: **carcinomas** (epithelial-derived — glandular/squamous, e.g. breast, ovarian cancer), **melanomas** (melanin-forming cell tumors). All lymph eventually enters venous system, so lymphatic-system cancers eventually spread through blood too.

The Thymus

- Located posterior to the sternal manubrium, superior mediastinum. Reaches greatest size relative to body size in first 1–2 years after birth; reaches max absolute size at puberty (30–40 g). After puberty, gradually shrinks, functional tissue replaced by fibrous connective tissue and fat = **involution of the thymus**.
- Capsule divides thymus into 2 **thymic lobes**; fibrous septa divide lobes into **lobules** (avg 2 mm wide), each with dense outer **cortex** and paler central **medulla**.

- Cortex: rapidly dividing lymphatic stem cells → daughter cells mature into T cells, migrate to medulla. During maturation, T cells sensitive to normal tissue antigens are destroyed (self-tolerance). Thymic capillaries form a **blood-thymus barrier** (like CNS capillaries) — prevents free exchange between circulation and thymic interstitial fluid, protecting developing T cells from premature antigen stimulation.
- **Epithelial reticular cells** scattered among lymphocytes produce thymic hormones promoting T cell differentiation. In the medulla, reticular cells cluster in concentric layers forming **thymic corpuscles** (function unknown).

The Spleen

- Largest lymphatic organ in the body; ~12 cm (5 in.) long, up to 160 g (5.6 oz); located left side, along lateral stomach border, between ribs 9-11; attached to stomach's lateral border by the **gastrosplenic ligament** (mesentery).
- Deep red color (large blood content, on gross dissection). Functions parallel lymph nodes but for blood: (1) removes abnormal blood cells/components via phagocytosis, (2) stores iron recycled from RBC breakdown, (3) initiates immune responses (B cells, T cells) to antigens in circulating blood.
- Surfaces: **diaphragmatic surface** — smooth, convex, conforms to diaphragm/body wall shape. **Visceral surface** (inferior) — indentations matching stomach (**gastric area**) and left kidney (**renal area**). Splenic artery, splenic vein, lymphatic vessels enter/leave at the **hilum**.
- Histology: capsule (collagen + elastic fibers); cellular components = **pulp**.
 - **Red pulp**: forms **splenic cords**, contains large RBC quantities; cell population includes all normal circulating blood components plus fixed/free macrophages; reticular fiber network is structural framework; blood passes through this network into large sinusoids lined by fixed macrophages; sinusoids empty into small veins → merge into **trabecular veins** → toward hilum.
 - **White pulp**: forms lymphatic nodules (lymphocyte-dominated).
 - Splenic artery enters at hilum, branches into **trabecular arteries** radiating to capsule; arteriolar branches surrounded by white pulp areas; capillaries discharge blood into venous circulation.
 - This arrangement lets splenic phagocytes identify/engulf damaged or infected circulating blood cells; macrophages scattered through red pulp; area around white pulp has high lymphocyte/dendritic cell concentration — microorganisms/antigens in blood quickly trigger immune response.
- Clinical: **Lymphoma** = blood cancer group developing in lymphatic system; begins with a mutated lymphocyte in a lymph node/lymphoid tissue growing uncontrolled, producing abnormal daughter cells that accumulate, crowd out normal lymphocytes, impair lymphatic function. Two main types:
 - **Hodgkin lymphoma** — ~10% of lymphomas; cancer of blood/bone marrow; characterized by abnormal **Reed-Sternberg cells** (B cell origin); associated with Epstein-Barr virus infection; affects people in 20s and >55, most common in US/Canada/northern Europe, rare in Asia; strong genetic link (identical twin of affected person has very high risk); very treatable, most cases curable.
 - **Non-Hodgkin lymphoma (NHL)** — ~90% of lymphomas; diverse group, may involve bone marrow/blood; ~85% affect B cells, remainder T cells/NK cells; indolent (slow) or aggressive (fast); can occur any age but most common >60; cause unknown, implicated factors: certain chemicals (benzene), nuclear radiation, immunocompromised state.
 - Lymphoma symptoms: persistent enlarged lymph nodes, unexplained weight loss, fever, night sweats, unusual infections.
 - Treatment: chemotherapy, radiation therapy, immunotherapy; stem cell transplant if these fail.

23.7 Aging and the Lymphatic System

- With advancing age, T cells become less responsive to antigens → fewer cytotoxic T cells respond to infection. Helper T cell numbers decline → B cells less responsive, antibody levels rise more slowly after antigen exposure.
- Result: increased susceptibility to viral/bacterial infections — vaccinations (e.g., flu/influenza) strongly recommended for older adults.

- Increased cancer incidence in elderly reflects declining immune surveillance — tumor cells not eliminated as effectively.

Clinical Correlations Summary (Ch 23)

- **Lymphedema**: tissue swelling from blocked/damaged lymphatic drainage.
- **Tonsillitis**: tonsil infection/inflammation; can lead to bacteremia if chronic/abscessed; tonsillectomy criteria: peritonsillar abscess, breathing/swallowing difficulty, ≥ 5 -7 episodes/year.
- **HIV/AIDS**: retrovirus destroying CD4+ helper T cells; AIDS diagnosis at CD4 < 200 cells/mm³; lifelong infection, managed with antiretrovirals.
- **Lymphoma** (Hodgkin and non-Hodgkin): described above.
- **Sentinel node biopsy**: concept referenced in related terms (first node draining a tumor, used to check for metastasis).
- Clinical Case (Grace): **Ebola virus disease** — a **zoonotic disease** (transmitted between humans and animals; reservoir hosts include bats); zoonotic diseases cause 3 of 5 new human illnesses annually in the US per CDC. No vaccine available in the scenario; supportive care (rehydration) is the mainstay; recovery depends on the immune response (activated cytotoxic T cells providing cell-mediated immunity; helper T cell-regulated B cell differentiation into plasma cells providing antibody-mediated immunity) outpacing viral progression.
- Related clinical terms: **adenitis** (pharyngeal tonsil/adenoid inflammation); **allograft** (transplant between compatible same-species individuals); **anamnestic response** (immune response initiated by memory cells); **autograft** (transplant from the same person); **Burkitt's lymphoma** (malignant B lymphocyte cancer); **Coombs test** (detects antibodies/complement in blood); **dermatomyositis** (autoimmune inflammation of skin and muscles); **eczema** (genetic inflammatory skin disorder); **host versus graft disease** (donor tissue immune cells attack recipient); **hybridoma** (cancer cell + lymphocyte fusion, mass-produces specific antibody); **infectious mononucleosis** (acute Epstein-Barr virus disease — fever, lymph node swelling, sore throat, increased blood lymphocytes); **latex allergy** (hypersensitivity to rubber sap products); **polymyositis** (autoimmune muscle inflammation/atrophy); **sentinel node** (first node draining a tumor, checked for metastasis); **splenomegaly** (enlarged spleen); **xenograft** (transplant between different species); autoimmune disease (general concept — immune system attacks own tissues, referenced in context of dermatomyositis/polymyositis definitions); DiGeorge syndrome-type description (congenital absence of thymus and parathyroid glands with immunity deficiency, referenced without being fully named in extracted text); immunology (study of immune system structure/function, referenced in text).

Digestive (Ch 25)

25.1 An Overview of the Digestive System

- Digestive system = digestive tract (alimentary canal: oral cavity, pharynx, esophagus, stomach, small intestine, large intestine) + accessory organs (teeth, tongue, salivary glands, liver, gallbladder, pancreas).
- Accessory glandular organs secrete water, enzymes, buffers, other components into ducts emptying into the digestive tract, preparing nutrients for absorption.
- Six major functions: **Ingestion** (food/liquid entering via mouth); **Mechanical processing** (tongue/teeth before swallowing; swirling/mixing/churning/propulsive motions after); **Digestion** (chemical/enzymatic breakdown of sugars, lipids, proteins into small absorbable molecules); **Secretion** (acids, enzymes, buffers — mostly from accessory organs like pancreas); **Absorption** (movement of organic molecules, electrolytes, vitamins, water across epithelium into interstitial fluid); **Excretion** (elimination of undigested residue and metabolic wastes, mostly via liver); **Compaction** (progressive dehydration of undigested material/wastes prior to excretion — compacted material = feces; **defecation** = elimination of feces via anus).
- Digestive tract lining protects surrounding tissue against (1) corrosive digestive acids/enzymes, (2) mechanical stress/abrasion, (3) swallowed pathogens.

Histological Organization of the Digestive Tract — 4 major layers, superficial to deep... actually inner to outer:

- **Mucosa:** inner lining; mucous membrane = loose connective tissue covered by epithelium moistened by glandular secretions. Epithelium type varies by region/stress: oral cavity/esophagus = nonkeratinized stratified squamous (abrasion resistance); stomach/small intestine/most large intestine = simple columnar (secretion/absorption). **Circular folds (plicae circulares)** = transverse/longitudinal mucosal folds increasing surface area; some permanent (mucosa+submucosa), others temporary (disappear as lumen fills, allowing post-meal expansion). Deep to epithelium = **lamina propria** (areolar connective tissue with blood vessels, sensory nerve endings, lymphatic vessels, smooth muscle fibers, scattered lymphatic tissue). Mucosal border = **muscularis mucosae** (narrow smooth muscle/elastic fiber band, 2 concentric layers: internal circular, external longitudinal) — contraction alters lumen shape, moves epithelial folds.
- **Submucosa:** areolar connective tissue deep to muscularis mucosae; contains large blood/lymphatic vessels; in some regions has exocrine glands secreting buffers/enzymes into lumen. **Submucous neural plexuses** innervate the mucosa — contain sensory neurons, parasympathetic ganglia, sympathetic postganglionic fibers.
- **Muscular layer:** double smooth muscle layer deep to submucosa — internal circular, external longitudinal. Mechanically processes/propels materials; movements coordinated by **myenteric neural plexus** (between the two muscular layers) — parasympathetic ganglia + sympathetic postganglionic fibers. Parasympathetic stimulation increases muscle tone/contractions; sympathetic decreases tone, promotes relaxation. Localized thickenings of circular layer form **sphincters** (restrict movement, ensure one-way passage).
- **Serosa:** serous membrane covering muscular layer within the peritoneal cavity. No serosa around pharynx, esophagus, rectum — instead an **adventitia** (dense collagen fiber network) firmly attaches these to adjacent structures.

Muscular Layers and Movement of Digestive Materials

- Smooth muscle cells: 5–10 µm diameter, 30–200 µm length; surrounded by connective tissue (no tendons/aponeuroses, unlike skeletal muscle); contractile proteins not organized into sarcomeres → nonstriated, but contractions as strong as skeletal/cardiac muscle.
- **Plasticity** = ability to tolerate extreme stretching and adapt to new length while retaining contraction ability — important for organs with large volume changes (e.g., stomach).
- Digestive smooth muscle cells are involuntary; many lack autonomic motor innervation. **Pacemaker cells** in muscularis mucosae/muscular layer spontaneously depolarize, triggering contraction; gap junctions electrically connect adjacent muscle cells so stimulation of one cell (or local chemical/hormonal/O₂-CO₂/stretch/irritation stimuli) can trigger a coordinated regional response.
- **Peristalsis:** contractions moving a **bolus** (food mass) along the tract — circular muscles contract behind the bolus, longitudinal muscles contract next (shortening the segment), circular muscle wave forces material forward. (Analogy: squeezing a toothpaste tube.)
- **Segmentation:** churning/fragmenting movements in small intestine (and some large intestine) that mix contents with secretions but don't produce net directional movement.
- Peristalsis/segmentation triggered by hormones, chemicals, physical stimulation; afferent/efferent fibers of glossopharyngeal, vagus, or pelvic nerves initiate peristaltic waves.
- **Short reflexes:** sensory receptors trigger local peristalsis limited to a few cm; afferent fibers synapse within myenteric plexus, produce localized **myenteric reflexes** not involving CNS. **Enteric nervous system** = neural network coordinating these reflexes — has roughly as many neurons/neurotransmitters as the spinal cord.
- **Long reflexes:** sensory info reaches CNS, involves interneurons/motor neurons, provides higher-level control of large-scale digestive/glandular activity; may involve glossopharyngeal, vagus, pelvic nerve motor fibers.

The Peritoneum

- Serous membrane, two parts: **visceral peritoneum (serosa)** — continuous with **parietal peritoneum** lining body wall inner surfaces.
- Organ-peritoneum relationships:
 - **Intraperitoneal organs:** covered on all sides by visceral peritoneum, lie within peritoneal cavity (e.g., stomach, liver, ileum).
 - **Retroperitoneal organs:** covered by parietal peritoneum on anterior surface only, lie outside peritoneal cavity; typically don't develop from embryonic gut (e.g., kidneys, ureters, abdominal aorta).

- **Secondarily retroperitoneal organs:** form as intraperitoneal organs but become retroperitoneal during development as visceral peritoneum fuses with opposing parietal peritoneum (e.g., pancreas, distal 2/3 of duodenum, ascending/descending colon, rectum).
- Peritoneum continually produces watery lubricating fluid; ~7 L secreted/reabsorbed daily, but peritoneal cavity volume at any time is very small. Abnormal fluid accumulation (liver disease, heart failure, electrolyte imbalance) → dangerous blood volume reduction and visceral distortion.
- Clinical: **Ascites** = buildup of serous fluid in peritoneal cavity (normal 100-400 mL exceeded), causing abdominal swelling, discomfort, heartburn, indigestion, shortness of breath, lower back pain. Most common cause = advanced liver disease or hepatitis B/C infection; also kidney failure, heart failure, advanced cancer. Treatment: sodium restriction, diuretics.

Mesenteries

- **Mesenteries** = fused double sheets of peritoneal membrane connecting parietal to visceral peritoneum; areolar connective tissue between the two surfaces provides passage for blood vessels, nerves, lymphatics; stabilize organ position, prevent entanglement during movement/position changes.
- Development: dorsal and ventral mesenteries suspend digestive tract/accessory organs. Ventral mesentery disappears except in 2 locations: **lesser omentum** (stomach-liver, ventral surface of stomach) and **falciform ligament** (liver-anterior abdominal wall/diaphragm).
- **Greater omentum:** dorsal mesentery of stomach enlarges into a pouch extending inferiorly between body wall and small intestine anterior surface; contains thick adipose tissue layer — functions: (1) energy reserve, (2) insulation minimizing heat loss across anterior abdominal wall; also contains numerous lymph nodes protecting against foreign proteins/toxins/pathogens.
- **Mesentery proper:** thick mesentery suspending all but the first 25 cm of small intestine — provides stability while permitting independent movement.
- **Mesocolon:** mesentery attached to large intestine. **Transverse mesocolon** suspends transverse colon; **sigmoid mesocolon** suspends sigmoid colon. During development, dorsal mesentery of ascending colon, descending colon, and rectum fuses to posterior body wall, becoming secondarily retroperitoneal (visceral peritoneum covers only anterior/lateral surfaces).

25.2 The Oral Cavity

- Functions: (1) sensory analysis of food before swallowing, (2) mechanical digestion (teeth, tongue, palate), (3) lubrication (mixing with mucus/saliva), (4) limited chemical digestion of carbohydrates (salivary amylase) and lipids (lingual lipase).
- **Oral mucosa:** nonkeratinized stratified squamous epithelium, protects against abrasion. **Buccal fat pads** and **buccinator** support cheek mucosa (lateral oral cavity walls); anteriorly continuous with lips. **Oral vestibule** = space between cheeks/lips and teeth. **Gingivae** (gums) = mucosal ridges surrounding tooth bases on alveolar processes.
- **Tongue:** forms oral cavity floor; **mylohyoid** (inferior to tongue) provides additional floor support. Functions: (1) mechanical digestion (compression, abrasion, distortion), (2) manipulation for chewing/swallowing, (3) sensory analysis (touch, temperature, taste), (4) secretion of mucin and lingual lipase. Divided into anterior **body** and posterior **root**; dorsum has **lingual papillae** (friction; taste buds along edges); V-shaped **circumvallate papillae** mark body-root boundary. Inferior epithelium thinner; **frenulum of the tongue** (midline mucosal fold) connects tongue body to oral floor mucosa — overly restrictive frenulum = **ankyloglossia** (tongue-tie), correctable surgically. Muscles: **extrinsic** (hyoglossus, styloglossus, genioglossus, palatoglossus — gross movements) and **intrinsic** (shape changes, precise movements e.g. speech); both controlled by **hypoglossal nerve (XII)**.
- **Hard palate:** separates oral/nasal cavity; formed by palatine processes of maxilla + horizontal plates of palatine bones.
- **Soft palate:** posterior to hard palate; separates oral cavity from nasopharynx, closes nasopharynx during swallowing. **Uvula** dangles from posterior soft palate margin, prevents premature food entry into pharynx. **Palatal arches** (2 pairs): **palatoglossal arches** (soft palate to tongue base) and **palatopharyngeal arches** (soft

palate to pharynx side). **Palatine tonsils** lie between these arches. **Fauces** = space between oral cavity and pharynx (bounded by soft palate, tongue base).

- **Salivary glands** (3 pairs):

1. **Parotid glands** — largest, ~20 g each; between zygomatic arch and sternocleidomastoid anteriorly, extends from mastoid process across masseter posteriorly; drains via **parotid duct** into oral vestibule at 2nd upper molar. Serous-cell-dominated (enzyme-rich saliva).

2. **Sublingual glands** — floor of mouth mucosa; numerous ducts open along the tongue frenulum. Mucous-cell-dominated (mucin-rich).

3. **Submandibular glands** — floor of mouth along mandible medial surface, inferior to mylohyoid line; ducts open beside the frenulum, posterior to teeth. Both serous and mucous cells (enzymes + mucin).

- Saliva composition: ~70% from submandibular, ~25% from parotid, ~5% from sublingual glands. ~1.0–1.5 L/day; 99.4% water, remainder ions, buffers, metabolites, enzymes. **Mucin** glycoproteins give lubricating action.
- Regulation: ANS controls secretion (each gland has parasympathetic + sympathetic innervation). Parasympathetic stimulation (via trigeminal sensory input or taste buds via CN VII, IX, X) speeds secretion, producing large volumes; sympathetic role unclear, may provoke small amounts of thick saliva.
- Clinical: **Mumps** — mumps virus infection, most common ages 5-9; humans are sole reservoir, spread via respiratory droplets/saliva; targets salivary glands (usually parotid, swollen/painful); post-adolescent males may get testicular infection (sterility risk) or pancreatic infection (temporary/permanent diabetes); severe cases can affect CNS; first exposure confers permanent immunity; vaccine has nearly eliminated US disease, still present in developing countries.
- **Teeth**: movements of tongue pass food across teeth for **mastication** (chewing) — breaks down connective tissue/plant fiber, saturates food with saliva/enzymes.
 - Tooth structure: **dentine (dentin)** = mineralized matrix similar to bone but without living cells (cells in central **pulp cavity** extend processes into dentine); pulp cavity is spongy, highly vascular; **root canal** = narrow tunnel at tooth root/base for vessels/nerves; **apical foramen** at root tip = entry point for dental artery/vein/nerve.
 - Root sits in **tooth socket (alveolus)**; **periodontal ligament** (collagen fibers, dentine-to-bone) creates a **gomphosis** joint. **Cement** covers root dentine (bone-like, less erosion-resistant than dentine), protects/anchors periodontal ligament.
 - **Neck** = root-crown boundary; **crown** = visible portion above gingiva. **Gingival sulcus** = shallow groove around tooth neck, prevents bacterial access to lamina propria/cement; breakdown → bacterial infection = **gingivitis**.
 - **Enamel** covers crown dentine, forms **occlusal surface** (biting surface with **cusps**); densely packed calcium phosphate crystals = hardest biologically manufactured substance; adequate calcium/phosphate/vitamin D essential in childhood for complete, decay-resistant enamel.
 - 4 tooth types: **Incisors** (blade-shaped, front of mouth, clip/cut, single root); **Canines/cuspids** (conical, sharp ridge/point, tear/slash, single root); **Premolars/bicuspids** (flattened crown, 2 rounded cusps, crush/mash/grind, 1-2 roots); **Molars** (large flattened crown, 4-5 rounded cusps, crush/grind; upper jaw typically 3 roots, lower jaw usually 2 roots).
 - **Deciduous teeth** (first set): 20 total (per side of upper/lower jaw: 2 incisors, 1 canine, 2 deciduous molars). Replaced by **permanent teeth** of larger adult jaws as periodontal ligaments/roots of deciduous teeth erode and teeth fall out/are pushed aside. Adult premolars replace deciduous molars; adult molars extend rows as jaw enlarges; 3 additional molars appear per side with age, bringing permanent count to 32. **Third molars (wisdom teeth)** may not erupt before age 21; may fail to erupt from inappropriate position or insufficient space.
 - Mastication: muscles of mastication close/rock jaw; buccal, labial, lingual muscles assist positioning; food forced across occlusal surfaces until compacted into a swallowable bolus by the tongue.

25.3 The Pharynx

- Common passageway for food, liquids, air; divisions (from Ch 24): nasopharynx, oropharynx, laryngopharynx. Lamina propria has dense elastic fiber layer bound to underlying skeletal muscle.
- Pharyngeal muscles (from Ch 10) involved in swallowing: superior/middle/inferior pharyngeal constrictors (push bolus toward esophagus); palatopharyngeus and stylopharyngeus (elevate larynx); palatal muscles (raise soft palate/pharyngeal wall).

- **Deglutition (swallowing)** — 3 phases:

1. **Buccal phase** — voluntary; bolus compressed against hard palate; tongue retracts, forcing bolus into pharynx, helping elevate soft palate, isolating nasopharynx.

2. **Pharyngeal phase** — involuntary; begins when bolus contacts palatal arches/posterior pharyngeal wall; palatopharyngeus/stylopharyngeus elevate larynx; epiglottis folds, directing bolus past closed glottis; pharyngeal constrictors propel bolus into esophagus in <1 second; respiratory centers inhibited, breathing ceases during this phase.

3. **Esophageal phase** — upper esophageal sphincter opens; peristaltic waves push bolus through esophagus; lower esophageal sphincter opens as bolus approaches, bolus enters stomach.

25.4 The Esophagus

- Hollow muscular tube transporting food/liquid to stomach; posterior to trachea, slightly left of midline; passes along dorsal mediastinal wall; enters peritoneal cavity through the **esophageal hiatus** (diaphragm opening); empties into stomach. Begins at cricoid cartilage level (anterior to vertebra C6), ends anterior to vertebra T7.
- Blood supply: esophageal arteries (from thyrocervical trunk and external carotid arteries — neck), bronchial arteries and esophageal arteries (mediastinum), inferior phrenic artery, left gastric artery (abdomen). Venous drainage: esophageal, inferior thyroid, azygos, gastric veins. Innervation: vagus and sympathetic trunks via the esophageal plexus.
- Terms "upper/lower esophageal sphincter" (lower = "cardiac sphincter") used functionally though not histologically distinct sphincter structures.
- Histology: mucosa (nonkeratinized stratified squamous, abrasion-resistant), submucosa, muscular layer comparable to general plan, but distinctive features:
 - Mucosa/submucosa form large longitudinal folds allowing expansion for a large bolus; muscle tone keeps esophagus closed except during swallowing.
 - Muscularis mucosae very thin/absent near pharynx, thickens to 200-400 µm near stomach junction; composed of a single longitudinal smooth muscle layer (unlike rest of digestive tract).
 - **Submucosal esophageal glands** — one of only 2 digestive tract regions with submucosal glands (other = duodenum); simple branched tubular glands producing mucous secretion (lubrication, protection).
 - Muscular layer: superior 1/3 mostly skeletal muscle + some smooth muscle; middle 1/3 = even mix skeletal/smooth; inferior 1/3 = only smooth muscle. Visceral reflexes (involuntary) control both muscle types here.
 - **Adventitia** anchors esophagus to posterior body wall; esophagus is retroperitoneal between diaphragm and stomach; peritoneum covers anterior and left lateral surfaces only.
- Clinical: **Esophagitis** = inflammation of esophageal stratified squamous epithelium; most common cause = lower esophageal sphincter failure allowing acid reflux ("heartburn"); can lead to asthma if acid is inhaled; chronic recurrent reflux esophagitis = **gastroesophageal reflux disease (GERD)**. Worsened by obesity, pregnancy, age, citrus, tomato products, strong seasonings. Treatment: healthy weight, quit smoking, dietary changes, smaller meals, staying upright 2+ hrs post-meal. **Hiatal hernia** = stomach/other abdominal organs pushing upward through weakened esophageal hiatus; common in elderly, not always symptomatic, can contribute to GERD.

25.5 The Stomach

- Functions: (1) temporary food storage, (2) mechanical digestion, (3) chemical digestion (acids/enzymes) → produces **chyme** (viscous, strongly acidic, soupy mixture).
 - Intraperitoneal, J-shaped; smooth rounded anterior/posterior surfaces; extends between vertebrae T7–L3.
 - 4 regions:
1. **Cardia** — superior/medial, surrounds esophagus junction; **cardiac orifice** = esophageal lumen opening.
 2. **Fundus** — projects superior to esophagus-stomach junction, contacts diaphragm's inferior/posterior surface.
 3. **Body** — between fundus and J-curve; largest region; mixing tank for food and gastric secretions.
 4. **Pyloric part** — between body and duodenum; subdivided into **pyloric antrum** (connects to body), **pyloric canal** (empties to duodenum), **pylorus** (muscular tissue around pyloric orifice/outlet). **Pyloric sphincter** = thickened circular muscle regulating chyme release into duodenum.
- **Lesser curvature** = short medial surface; **greater curvature** = long lateral surface. **Omenta** = stomach mesenteries connecting it to adjacent organs (lesser omentum: hepatogastric ligament connects liver to lesser curvature, hepatoduodenal ligament connects liver to duodenum; greater omentum: apron-like pouch from greater curvature).
 - **Gastric folds (gastric rugae)** — prominent when stomach empty, flatten as stomach fills/distends.
 - Blood supply — 3 branches of celiac trunk: **left gastric artery** (lesser curvature, cardia); **splenic artery** (fundus directly, greater curvature via left gastro-epiploic artery); **common hepatic artery** → right gastric, right gastro-epiploic, gastroduodenal arteries (greater/lesser curvature at pylorus). Venous drainage: gastric and gastro-epiploic veins → hepatic portal vein.
 - Musculature: 3 smooth muscle layers (vs. usual 2) — outer longitudinal, middle circular, inner oblique — extra layer strengthens wall, aids mixing/churning for chyme formation.
 - Histology: simple columnar epithelium lines all regions; surface = **mucous surface cells** (protective mucus layer). **Gastric pits** = shallow depressions with **regenerative stem cells** at base (continually divide to replace shed superficial cells — additional defense).
 - Each gastric pit connects to several **gastric glands** (simple branched tubular glands) with 4 secretory cell types (parietal + chief cells secrete ~1500 mL gastric juice/day):
 - **Mucous neck cells** — found in all stomach regions; columnar; secrete water-soluble lubricating mucus.
 - **Parietal cells** — large, round/pyramid-shaped; numerous in proximal gastric gland portions. Secrete **intrinsic factor** (glycoprotein aiding vitamin B12 absorption in intestine) and **hydrochloric acid (HCl)** — lowers gastric juice pH, kills microorganisms, breaks down plant cell walls/meat connective tissue, denatures proteins.
 - **Enteroendocrine cells / G cells** — numerous in fundus (scattered among parietal/chief cells), occasional in cardia/pyloric part. **G cells** (in pyloric part gastric pits) secrete **gastrin** — released when food enters stomach, stimulates parietal/chief cell secretory activity, promotes gastric wall smooth muscle mixing/churning.
 - **Chief cells** — found only in fundus, most abundant at gastric gland base; columnar; secrete **pepsinogen** (inactive proenzyme) — gastric acid converts it to **pepsin** (active proteolytic enzyme, works best at pH 1.5–2.0). Newborn stomachs (not adults) also produce **rennin** (coagulates milk proteins) and **gastric lipase** (digests milk fats).
 - Stomach lining does NOT directly absorb nutrients, but indirectly absorbs some salts, water, lipid-soluble substances (e.g., alcohol, aspirin, NSAIDs enter lamina propria after damaging surface epithelium, then enter blood via lamina propria capillaries).
 - Regulation of gastric activity: direct control via CNS (vagus nerve = parasympathetic; celiac plexus branches = sympathetic) — sight/thought of food triggers parasympathetic motor output to parietal/chief/mucous cells; arrival of food stimulates stretch receptors and mucosal chemoreceptors → reflexive muscular contractions + gastrin release from G cells; parietal and chief cells respond to gastrin by accelerating secretion (parietal cells especially sensitive — acid production rises more dramatically than enzyme secretion). Sympathetic activation

inhibits gastric activity. Small intestine hormones **secretin** and **cholecystokinin (CCK)** inhibit gastric secretion while stimulating pancreatic/liver secretion (complementary secondary effect).

- Clinical: **Gastritis** = stomach lining inflammation/irritation. **Acute gastritis** — sudden onset, often viral ("stomach flu") or food poisoning; fever, cramping, nausea, vomiting, diarrhea within 2-4 hrs of eating contaminated food. **Chronic gastritis** — often caused by **Helicobacter pylori (H. pylori)** bacterial infection; untreated can lead to peptic ulcer. **Peptic ulcers** = alimentary mucosa sores from acidic gastric secretion exposure; found in stomach (gastric ulcers) or duodenum (duodenal ulcers); causes: excess alcohol, smoking/chewing tobacco, aspirin/NSAIDs, caffeine, severe illness, radiation. Diagnosed/treated via endoscopy; H. pylori treated with antibiotics; acid-neutralizing/blocking medications help; avoid alcohol, tobacco, aspirin, NSAIDs. Untreated ulcer can erode through epithelium exposing lamina propria (pain, bleeding) or perforate the stomach (contents leak into peritoneum = surgical emergency).
- Clinical note: **Endoscope** = flexible fiberoptic tube with light/camera for direct digestive tract visualization; **esophagogastroduodenoscopy (EGD)** = upper GI endoscopy via mouth; **colonoscopy** = endoscope via anus to examine colon.

25.6 The Small Intestine

- 90% of nutrient absorption occurs here (most remainder in proximal large intestine). Length averages 6 m (20 ft; range 5.0–8.3 m/5-25 ft); diameter ~4 cm (1.6 in.) at stomach junction, ~2.5 cm (1 in.) at large intestine junction. Found in all abdominal regions except left hypochondriac and epigastric.
- Held by mesenteries attached to dorsal body wall; specialized anatomy maximizes surface area: **circular folds (plicae circulares)** — permanent (unlike gastric folds), ~800 folds (~2/cm) along duodenum, jejunum, proximal ileum half; **intestinal villi** on mucosa; **microvilli** on each epithelial cell's apical surface. Combined, these increase surface area >600-fold, to ~2 million cm².
- 3 subdivisions:
 - **Duodenum**: shortest/widest, ~25 cm (10 in.); connects to pylorus via pyloric sphincter; curves in a C around the pancreas; proximal 2.5 cm (1 in.) intraperitoneal, rest secondarily retroperitoneal, located between vertebrae L1-L4. "Mixing bowl" receiving chyme from stomach. No supporting mesentery (proximal portion intraperitoneal/movable, rest fixed retroperitoneal).
 - **Jejunum**: begins at **duodenojejunal flexure** (sharp bend, small intestine re-enters peritoneal cavity, becomes intraperitoneal); ~2.5 m (8 ft) long; bulk of chemical digestion and nutrient absorption occurs here; thicker wall, larger diameter than ileum; found in umbilical region; vasculature from mesentery is more straightforward/fewer branches than ileum's.
 - **Ileum**: longest segment, ~3.5 m (12 ft); intraperitoneal; ends at **ileocecal valve** (sphincter controlling flow into cecum, protrudes into cecum); tends to lie in hypogastric region; mesentery thicker, more fat than jejunum's.
 - Jejunum/ileum supported by fan-shaped **mesentery proper**; blood vessels/lymphatics/nerves pass through it — branches of superior mesenteric artery/vein; vagus nerve = parasympathetic innervation, superior mesenteric ganglion = sympathetic innervation.
- Histology: **intestinal villi** (fingerlike mucosal projections) covered by simple columnar epithelium; apical cell surfaces have **microvilli**. **Goblet cells** between columnar cells secrete mucin. **Intestinal glands (crypts)** at villi bases extend into lamina propria; stem cells near gland base continually divide, producing new epithelial cells that migrate to villus tip within days, renewing epithelium and adding intracellular enzymes to chyme. **Paneth cells** at intestinal gland bases — innate immunity role, release defensins and lysozyme, killing some bacteria while allowing others to establish normal intestinal flora. Intestinal glands also contain enteroendocrine cells producing intestinal hormones.
- **Lamina propria** (villus core): loose areolar connective tissue with lymphatic cells, occasional lymphoid nodules, extensive capillary network absorbing/carrying nutrients to hepatic portal circulation. Each villus has a central **lacteal** (lymphatic vessel) transporting materials too large for blood capillaries — e.g., absorbed fatty acids assembled into **chylomicrons** (protein-lipid packages), which reach venous circulation via the thoracic duct.
- Regional specializations:

- **Duodenum:** submucosa has **duodenal submucosal glands (Brunner's glands)** — produce large mucus quantities when chyme arrives, protecting epithelium from chyme acidity, and secrete bicarbonate ions raising chyme pH (from 1-2 to 7-8 roughly halfway along duodenum). Most abundant proximally, decrease toward jejunum. **Duodenal ampulla** = muscular chamber where bile duct (liver/gallbladder) and pancreatic duct join within duodenal wall; opens into duodenum at **duodenal papilla**; **hepatopancreatic sphincter** encircles the bile duct/pancreatic duct/duodenal ampulla lumen.
- **Jejunum/ileum:** circular folds/villi prominent in proximal jejunum half (most nutrient absorption here); diminish toward ileum end (parallels reduced absorptive activity). Distal ileum lacks circular folds; lamina propria has 20-30 masses of **aggregated lymphoid nodules (Peyer's patches)** — most abundant in terminal ileum near large intestine entrance; protect small intestine from large-intestine bacteria.
- Regulation: weak peristaltic contractions move materials along as absorption occurs; controlled by neural reflexes (submucosal, myenteric plexuses); parasympathetic stimulation increases reflex sensitivity, accelerates peristalsis/segmentation (mixing within a few cm of stimulus). Food entering stomach triggers coordinated intestinal movements pushing chyme toward large intestine; ileocecal valve opens simultaneously. Hormonal/CNS controls regulate secretory output; secretions = **intestinal juice**; triggered by local short reflexes + parasympathetic (vagal) stimulation; sympathetic stimulation inhibits secretion. Duodenal enteroendocrine cells produce **secretin** and **cholecystokinin (CCK)**, coordinating secretory activities of stomach, duodenum, liver, pancreas.

25.7 The Large Intestine

- Horseshoe-shaped; begins at ileum junction, ends at anus; inferior to stomach/liver, forms near-complete frame around small intestine. Also called "large bowel"; ~1.5 m (5 ft) long, 7.5 cm (3 in.) wide. 3 parts: **cecum**, **colon**, **rectum**.
- Functions: (1) absorb water/electrolytes, compact contents into feces, (2) absorb bacteria-produced vitamins, (3) store feces before defecation.
- Blood supply: superior and inferior mesenteric arteries; venous drainage: superior and inferior mesenteric veins.
- **Cecum and appendix:** material from ileum enters expanded pouch = **cecum** (ileum attaches at medial surface, opens via ileocecal valve); collects/stores material, begins compaction. **Appendix (vermiform appendix)** = slender hollow structure (~9 cm/3.6 in., variable) attached to cecum's posteromedial surface; **meso-appendix** connects it to ileum/cecum; primarily a lymphatic system organ; inflammation = **appendicitis**. Both cecum and appendix are intraperitoneal.
- **Colon:** wall forms pouches = **haustra** (permit distension/elongation; creases extend into mucosal lining as internal folds). **Teniae coli** = 3 separate longitudinal smooth muscle bands along outer colon surface (= outer longitudinal muscular layer equivalent elsewhere); muscle tone in teniae coli creates the haustra. **Omental appendices (fatty appendices of the colon)** = teardrop-shaped fat sacs in serosa.
- 4 colon regions:
 - **Ascending colon:** begins at cecum's superior border, ascends along right lateral/posterior abdominal wall to liver's inferior surface; turns left at **right colic (hepatic) flexure**; secondarily retroperitoneal (visceral peritoneum covers lateral/anterior surfaces only).
 - **Transverse colon:** curves anteriorly from right colic flexure, crosses abdomen right to left; initial segment intraperitoneal (supported by transverse mesocolon, separated from anterior wall by greater omentum layers); becomes secondarily retroperitoneal as it passes inferior to stomach's greater curvature. **Gastrocolic ligament** attaches it to stomach's greater curvature. Turns right near spleen at **left colic (splenic) flexure**, becomes descending colon.
 - **Descending colon:** secondarily retroperitoneal; proceeds inferiorly along left abdomen side; curves at iliac fossa to become sigmoid colon.
 - **Sigmoid colon:** S-shaped, ~15 cm (6 in.) long; posterior to urinary bladder, suspended by sigmoid mesocolon; empties into rectum.
- **Rectum:** forms last 15 cm (6 in.) of digestive tract; expandable, temporary feces storage; fecal movement into rectum triggers urge to defecate.

- **Anal canal** = last rectum portion; **anal columns** = small longitudinal folds; distal column margins joined by transverse folds marking boundary between rectal columnar epithelium and stratified squamous epithelium (like oral cavity). **Anus** = anal canal exit; epidermis becomes keratinized, identical to skin surface.
- **Internal anal sphincter** = circular muscle layer thickening, involuntary (not voluntary control). **External anal sphincter** = skeletal muscle ring around distal anal canal, voluntary control. Lamina propria/submucosa of anal canal has a venous network — elevated venous pressure (straining during defecation, pregnancy) can distend these veins, producing **hemorrhoids**.
- Histology vs. small intestine: (1) wall much thinner despite ~3x larger diameter; (2) lacks villi; (3) more abundant goblet cells; (4) deeper intestinal glands with more goblet cells (mucus lubricates compacted waste); (5) large lymphoid nodules scattered in lamina propria, extending into submucosa; (6) longitudinal muscle layer reduced to the 3 teniae coli bands (though mixing/propulsive contractions resemble small intestine).
- Regulation: movement cecum → transverse colon is slow (water absorption converts material to sludgy paste); peristaltic waves move material along colon length; **haustral churning** = segmentation-like movements mixing adjacent haustra contents. Stomach/duodenum distension triggers **mass movements** (powerful peristaltic contractions via intestinal neural plexuses) moving material from transverse colon through the rest of the large intestine, forcing feces into rectum. Rectal distension stimulates stretch receptors, initiating the **defecation reflex** — relaxes internal anal sphincter, feces move into anal canal; voluntary relaxation of external anal sphincter permits defecation.

25.8 Accessory Digestive Organs

The Liver

- Functions: (1) regulates metabolism, (2) regulates blood composition (removes old/damaged RBCs, synthesizes plasma proteins), (3) produces/secreted bile.
- Largest visceral organ; firm, reddish-brown; ~1.5 kg (3.3 lb); mostly right hypochondriac/epigastric regions, may extend into left hypochondriac/umbilical regions.
- **Metabolic regulation**: regulates circulating carbohydrate, lipid, amino acid levels. Blood from all absorptive digestive surfaces (stomach, small/large intestine) enters via hepatic portal system before reaching systemic circulation — hepatocytes extract nutrients/toxins first. Hepatocytes monitor/adjust metabolite levels (store excess, mobilize reserves or synthesize to correct deficiencies); remove/inactivate/store/excrete toxins and metabolic wastes; absorb/store fat-soluble vitamins A, D, K, E.
- **Hematological regulation**: largest blood reservoir in the body, receives ~25% of cardiac output; phagocytic cells remove old/damaged RBCs, cellular debris, pathogens as blood passes through; liver cells synthesize/secrete plasma proteins (osmotic concentration, nutrient transport, clotting/complement systems).
- **Bile production**: synthesized by liver cells, stored in gallbladder, excreted into duodenal lumen. Bile = mostly water + ions + **bilirubin** (hemoglobin-derived pigment) + **bile salts** (lipids). Water/ions dilute/buffer chyme acids entering small intestine; bile salts enable enzymes to break lipids into absorbable fatty acids.
- Liver estimated to have >200 functions (partial list in text): synthesize somatomedins; synthesize/secrete bile; store glycogen/lipid reserves; maintain normal blood glucose/amino acid/fatty acid concentrations; synthesize/convert nutrients (transaminate amino acids, convert carbs to lipids); synthesize/release cholesterol bound to transport proteins; inactivate toxins; store iron reserves; store fat-soluble vitamins; synthesize plasma proteins; synthesize clotting factors; synthesize inactive hormone angiotensinogen; phagocytose damaged RBCs (via stellate macrophages); store blood (major venous reserve contributor); absorb/break down circulating hormones (including insulin, epinephrine) and immunoglobulins; absorb/inactivate lipid-soluble drugs.
- Liver cells have limited regeneration ability after injury, but full function recovery requires vascular pattern regeneration too.
- Anatomy: intraperitoneal; fibrous capsule deep to visceral peritoneum. **Falciform ligament** (ventral mesentery) marks division between left/right lobes on anterior surface. **Round ligament (ligamentum teres)** = thickening of falciform ligament's inferior margin, marks degenerated fetal umbilical vein path. **Coronary ligament** suspends liver from diaphragm's inferior surface.
- Classical 4-lobe description: **left lobe**, **right lobe**, small **caudate lobe** (marked off by IVC impression, dividing right lobe from it), **quadrate lobe** (inferior to caudate lobe, between left lobe and gallbladder). Newer

terminology subdivides lobes into segments based on hepatic artery/portal vein/hepatic duct subdivisions (more useful for surgery).

- Surfaces: **diaphragmatic surfaces** (superior, anterior, posterior — relate to diaphragm; anterior surface follows body wall curve; posterior surface has stomach/small intestine/right kidney/large intestine impressions); **visceral surface** (inferior).
- **Blood supply**: 2 vessels — **hepatic artery proper** (~1/3 of hepatic blood flow, oxygen-rich) and **hepatic portal vein** (~2/3, nutrient-rich from intestines). Blood returns via **hepatic veins** → inferior vena cava.
- Histology: liver divided by connective tissue into ~100,000 **liver lobules** (basic functional units); hexagonal in cross section. **Hepatocytes** form irregular plates (spoke-like); up to age 7, plates 2 cells thick; after 7, single-cell thick. Apical/basal hepatocyte surfaces covered with short microvilli. Blood from hepatic artery proper + hepatic portal system drains into fenestrated **hepatic sinusoids** surrounding hepatocyte plates. Sinusoidal lining includes **stellate macrophages (Kupffer cells)** — phagocytic, part of monocyte-macrophage system; engulf pathogens, cell debris, damaged blood cells; also store iron, some lipids, heavy metals (tin, mercury) absorbed by digestive tract.
- Blood enters sinusoids from small portal vein/hepatic artery proper branches; each hexagonal lobule has 6 **portal triads** (one per corner) — each contains: (1) interlobular vein, (2) interlobular artery, (3) interlobular bile duct. Blood flows through sinusoids (hepatocytes absorb solutes, secrete plasma proteins) → **central vein** of the lobule → central veins merge into hepatic veins → inferior vena cava.
- **Bile secretion/transport**: hepatocytes form bile, secreted into **bile canaliculi** (narrow channels between adjacent liver cells) → extend outward from central vein → connect to **bile ductules** → carry bile to **interlobular bile duct** in nearest portal triad → right/left hepatic ducts collect bile from all lobe bile ducts → unite to form **common hepatic duct** leaving the liver → bile flows either into the bile duct emptying at the duodenal ampulla, or into the **cystic duct** leading to the gallbladder.

The Gallbladder

- Hollow, pear-shaped muscular sac; stores/concentrates bile before release into small intestine. Located in a fossa on the posterior surface of the liver's right lobe; intraperitoneal.
- 3 regions: **fundus**, **body**, **neck**. **Cystic duct** exits gallbladder, unites with common hepatic duct to form the **bile duct**.
- **Sphincter of ampulla** surrounds the bile duct/duodenal ampulla lumen at the duodenum; opens into duodenum at the **duodenal papilla**. Sphincter contraction seals the passage, preventing bile entry into small intestine.
- Functions: storing and modifying bile. When sphincter of ampulla closed, bile flows into cystic duct → gallbladder for storage. Filled gallbladder holds 40–70 mL bile; composition changes during storage — water absorbed, bile salts/other components become concentrated.
- **Bile ejection**: stimulated by **cholecystokinin (CCK)**, released into blood at the duodenum when lipid/partially-digested-protein-rich chyme arrives; CCK relaxes sphincter of ampulla and contracts the gallbladder.
- Histology: wall = mucosa, lamina propria (areolar connective tissue), muscular layer (2 interlacing smooth muscle layers — inner mostly longitudinal, outer mostly circular), serosa; NO muscularis mucosae or submucosa.

The Pancreas

- Posterior to stomach, extends laterally from duodenum toward spleen; elongated, pinkish-gray; ~15 cm (6 in.) long, ~80 g (3 oz). 3 subdivisions: **head**, **body**, **tail** (short, rounded). Secondarily retroperitoneal, firmly bound to posterior abdominal wall. Lumpy/nodular surface texture; thin transparent connective tissue capsule.
- Mixed gland — both exocrine (digestive enzymes/buffers) and endocrine (Ch 19 covered endocrine islets) functions.
- **Pancreatic duct** delivers exocrine secretions to the duodenal ampulla; sometimes an **accessory pancreatic duct (duct of Santorini)** branches off before the main duct leaves the pancreas — when present, usually empties at a separate duodenal papilla outside the duodenal ampulla.
- Blood supply: splenic, superior mesenteric, common hepatic arteries (via pancreatic arteries and superior/inferior pancreaticoduodenal arteries); drained by splenic vein and branches.

- Histology: connective tissue septa divide pancreas into lobules (blood vessels/duct tributaries run within septa); compound tubulo-acinar gland. **Pancreatic acini** (simple cuboidal epithelium) = exocrine secretory units, beginning of the duct system; drain into progressively larger ducts → main pancreatic duct (runs pancreas length).
- **Pancreatic acini** produce **pancreatic juice** = water, ions, digestive enzymes, released into duodenum to break down materials into absorbable molecules. Pancreatic ducts also secrete buffers (primarily sodium bicarbonate) — neutralize chyme acid, stabilize intestinal pH.
- **Pancreatic islets** scattered among acini — ~1% of pancreatic cell population, responsible for endocrine function (covered in Ch 19: insulin, glucagon, etc.).
- **Pancreatic enzymes** by target: **lipases** (digest lipids); **carbohydrases** (e.g., pancreatic amylase — digest sugars/starches); **nucleases** (break down nucleic acids); **proteolytic enzymes** — includes **proteinases** (break apart large protein complexes) and **peptidases** (break small peptide chains into individual amino acids).
- Regulation: duodenal hormones trigger pancreatic juice secretion. **Secretin** (released when acid chyme arrives) triggers watery, buffer-rich pancreatic juice production. **Cholecystokinin (CCK)** stimulates pancreatic enzyme production/secretion.

25.9 Aging and the Digestive System

- Digestion/absorption essentially normal in elderly, but changes parallel other systems' age-related changes:
 - Cumulative damage becomes apparent (e.g., tooth loss from decay/gingivitis); toxins (alcohol, heavy metals, chemicals) absorbed by digestive tract are processed/stored by the liver — chronic exposure can cause cirrhosis or other liver disease.
 - Epithelial stem cell division rate declines → less efficient tissue repair → digestive epithelium more susceptible to abrasion/acid/enzyme damage; peptic ulcers more likely; mouth/esophagus/anus stratified epithelium thins, becomes more fragile.
 - Smooth muscle tone decreases, motility decreases, peristaltic contractions weaken → slower chyme movement, promotes constipation; sagging haustra walls in colon → **diverticulitis** symptoms; straining to eliminate compacted feces stresses vessel walls → hemorrhoids; weakened cardiac sphincter → esophageal reflux/frequent heartburn.
 - Cancer rates increase, especially in organs with stem-cell-maintained epithelium — colon and stomach cancer rates rise in elderly; oral/pharyngeal cancers particularly common in elderly smokers.
 - Changes in other systems affect digestion directly/indirectly — e.g., reduced bone mass/calcium → tooth socket erosion, tooth loss; declining olfactory/gustatory sensitivity → dietary changes affecting the whole body.

Clinical Correlations Summary (Ch 25)

- **Ascites**: peritoneal fluid buildup (liver disease, hepatitis, kidney/heart failure, cancer).
- **Mumps**: viral salivary gland infection, can cause sterility (testes) or diabetes (pancreas) in post-adolescent males.
- **Gingivitis**: gingival sulcus bacterial infection.
- **Esophagitis/GERD/hiatal hernia**: acid reflux-related conditions.
- **Gastritis (acute/chronic)** and **peptic ulcer disease**: H. pylori-related, NSAID/alcohol/tobacco-related.
- **Appendicitis**: appendix inflammation.
- **Hemorrhoids**: distended anal canal veins from straining/pregnancy.
- **Diverticulitis**: sagging colon haustra walls with age.
- Clinical Case (Tayvian): **Clostridium difficile (C. diff) colitis** — antibiotics kill normal beneficial colonic microbiome bacteria, allowing C. diff overgrowth; C. diff produces toxins causing colitis (colon inflammation), watery/bloody diarrhea, cramping, fever, weight loss. Treatment: **fecal microbiota transplant** — donor feces (containing healthy bacteria) infused via NG tube through stomach into duodenum, recolonizing the large intestine, crowding out C. diff, restoring normal digestion/vitamin production. Bristol stool chart referenced (types 1-7, hard lumps to liquid, correlating with constipation to inflammation).

- Related clinical terms: **Crohn disease** (incurable chronic inflammatory bowel disease, any GI part mouth-to-anus, strictures/fistulas/fissures common); **esophageal varices** (swollen fragile esophageal veins from portal hypertension); **gastrectomy** (surgical stomach removal, usually for advanced stomach cancer); **periodontal disease** (tooth loosening from periodontal ligament erosion by bacterial acids); **irritable bowel syndrome (IBS)** (common large intestine disorder — cramping, abdominal pain, bloating, gas, diarrhea, constipation); **pancreatic cancer** (asymptomatic early, late detection, ~4% survival rate); **polyps** (small usually-benign stalked mucosal growths); **pyloric stenosis** (uncommon — lower stomach muscle enlarges, blocks food entry to small intestine).

Urinary (Ch 26)

Introduction — Urinary System Overview

- Digestive, cardiovascular, respiratory, urinary systems together prevent internal "pollution." Urinary system removes most metabolic wastes generated by body cells (digestive and integumentary systems assist minimally).
- Additional urinary system functions:
 - Regulates blood plasma concentrations of sodium, potassium, chloride, calcium, other ions (via excretion control).
 - Regulates blood volume/pressure by (a) adjusting water lost in urine, (b) releasing erythropoietin, (c) releasing renin.
 - Works with respiratory system to regulate blood pH.
 - Conserves valuable nutrients (prevents their excretion).
 - Eliminates metabolic wastes — especially urea, uric acid, toxic substances.
 - Synthesizes calcitriol (vitamin D hormone derivative, stimulates intestinal calcium absorption).
 - Helps liver detoxify poisons.
 - During starvation, deaminates amino acids for use by other tissues.
- Components: kidneys (produce urine), ureters, urinary bladder (temporary storage), urethra. **Micturition (urination)** = muscular bladder contraction forcing urine through urethra out of body.

26.1 The Kidneys

- Location: lateral to vertebral column, between last thoracic and 3rd lumbar vertebrae; right kidney sits slightly lower than left due to liver. Anterior right kidney surface covered by liver, right colic (hepatic) flexure, duodenum; anterior left kidney surface covered by spleen, stomach, pancreas, jejunum, left colic (splenic) flexure. Adrenal (suprarenal) gland sits atop each kidney. Kidneys, adrenal glands, ureters are all retroperitoneal.
- Position maintained by (1) overlying peritoneum, (2) adjacent visceral organ contact, (3) supporting connective tissue — 3 concentric protective/stabilizing layers:
 1. **Fibrous capsule** — covers outer kidney surface; collagen fibers maintain shape, provide physical protection.
 2. **Perinephric (perirenal) fat capsule** — adipose tissue layer surrounding fibrous capsule.
 3. **Renal fascia** — collagen fibers extend from fibrous capsule through perinephric fat to this dense outer connective tissue layer; anchors kidney to surrounding structures/deep fascia of posterior body wall muscles. **Pararenal fat body** = additional adipose layer separating posterior/lateral renal fascia from body wall. Anteriorly, renal fascia attaches to peritoneum and contralateral anterior renal fascia.
 - This arrangement cushions kidneys against daily jolts; stretched suspensory fibers or reduced fat padding increases traumatic injury vulnerability.
 - Kidney shape/size: kidney-bean shaped, brownish-red; ~10 cm (4 in.) long, 5.5 cm (2.2 in.) wide, 3 cm (1.2 in.) thick; ~150 g (5.30 oz) average. **Hilum** = medial surface indentation; fibrous capsule inner layer folds inward

here, lining the **renal sinus** (internal cavity). Renal blood vessels, lymphatics, nerves, ureter pass through hilum, branch within renal sinus; capsule's thick outer layer extends across hilum, stabilizing these structures.

- Internal structure (superficial to deep): **renal cortex** (granular, reddish-brown, deep to capsule) → **renal medulla** (deep to cortex) → **renal sinus**.
 - Medulla contains 6–18 triangular **renal pyramids**; base faces cortex, tip (**renal papilla**) projects into renal sinus; fine grooves converge at papilla.
 - **Renal columns** = cortical tissue bands extending into medulla, separating adjacent pyramids.
 - **Kidney lobe** = one renal pyramid + overlying cortex area + adjacent renal column tissue.
 - Urine produced in kidney lobes; ducts within each renal papilla empty into a **minor calyx**; 4–5 minor calyces merge into a **major calyx**; major calyces combine into the funnel-shaped **renal pelvis**, connected to the ureter.
- **Nephrons**: microscopic tubular structures where urine production begins. Two types: **cortical nephrons** (in renal cortex) and **juxtamedullary nephrons** (closer to renal medulla). ~1.25 million nephrons/kidney, combined length ~145 km (90 miles).

Blood Supply to the Kidneys

- 20–25% of total cardiac output (~1200 mL/min) flows through the kidneys. Each kidney receives blood via a **renal artery** (branches from abdominal aorta's lateral surface near the superior mesenteric artery) → enters renal sinus → branches into **segmental arteries** → **interlobar arteries** (radiate through renal columns between pyramids, into medulla) → **arcuate arteries** → **cortical radiate (interlobular) arteries** (supply kidney lobe portions) → numerous **afferent arterioles** (supply individual nephrons).
- Venous return mirrors arterial path: nephrons → venules/small veins → **cortical radiate (interlobular) veins** → **arcuate veins** → **interlobar veins** → merge to form the **renal vein** (no segmental veins exist).

Innervation of the Kidneys

- Urine production regulated via **autoregulation** (local blood flow regulation) — reflexive arteriole diameter changes altering blood flow/filtration rates.
- **Renal nerves** innervate kidneys/ureters; both ANS branches present, but mostly sympathetic postganglionic fibers from celiac and inferior mesenteric ganglia. Renal nerve enters at hilum, follows renal artery branches to nephrons. Sympathetic innervation: (1) adjusts urine formation rate by changing nephron blood flow, (2) influences urine composition by stimulating renin release.

Histology of the Kidney — The Nephron and Collecting System

- **Nephron** = basic structural/functional kidney unit; consists of **renal corpuscle** + **renal tubule**.
- **Renal corpuscle**: spherical structure = **glomerular capsule** (cup-shaped chamber) + **glomerulus** (capillary network). Renal tubule begins here, empties into the collecting system (tubes carrying tubular fluid away).
- Blood arrives at glomerulus via afferent arteriole, produces **filtrate**. Filtrate flows through 3 renal tubule subdivisions: (1) **proximal convoluted tubule (PCT)**, (2) **nephron loop (loop of Henle)**, (3) **distal convoluted tubule (DCT)**.
- Each nephron → **connecting tubule** → **collecting duct** (leaves cortex, descends into medulla) → **papillary duct** → drains into renal pelvis.
- **Cortical nephrons**: ~85% of all nephrons; almost entirely within superficial cortex; relatively short nephron loop; efferent arteriole → **peritubular capillaries** (surround entire renal tubule) → drain to small venules → interlobular veins. Perform most reabsorptive/secretory functions (being more numerous).
- **Juxtamedullary nephrons**: remaining ~15%; located closer to medulla, longer nephron loops extending deep into renal pyramids; create conditions necessary for producing concentrated urine.
- Urine at renal pelvis differs greatly from initial filtrate — filtration alone (passive, size-based) can't distinguish wastes from useful small molecules (water, ions, glucose, fatty acids, amino acids); distal nephron segments therefore: reabsorb useful metabolic substrates, reabsorb >80% of filtrate water, secrete additional wastes missed by filtration.

The Renal Corpuscle

- Diameter ~150-250 μm ; includes glomerulus capillaries + glomerular capsule (Bowman's capsule). Glomerulus projects into capsule like heart into pericardial cavity.
- Capsule outer wall = simple squamous epithelium = **capsular outer (parietal) layer**; ends at **visceral layer** covering glomerular capillaries, composed of **podocytes** (large cells with complex processes wrapping glomerular capillaries). **Capsular space** separates parietal/visceral epithelia.
- Parietal/visceral epithelia connect at the **vascular pole** — glomerular capillaries connect to afferent (blood in) and efferent (blood out, smaller diameter) arterioles here.
- Filtration: blood pressure forces fluid/solutes from glomerular capillaries into capsular space; filtrate $\approx$ plasma minus proteins. 5 filtration barriers:

1. **Endothelial surface layer** — thick carbohydrate-rich glycocalyx on capillary luminal surface, limits large plasma protein filtration.

2. **Capillary endothelium** — fenestrated capillaries, pores 60-100 nm; block blood cells but allow solute diffusion.

3. **Basement membrane** — several times thicker than typical capillary basement membrane; blocks larger plasma proteins, allows small plasma proteins/amino acids/glucose/ions; unique among cardiovascular basement membranes (specific detail truncated in extraction but implies specialized composition).

4. **Glomerular epithelium** — podocyte secondary processes ("feet") wrap glomerular capillary basement membrane, separated by narrow **filtration slits**; filtrate at this point = water + dissolved ions/small molecules, few/no plasma proteins.

5. **Subpodocyte space** — ~60% of glomerulus filtration space; lies between podocyte secondary processes and podocyte cell bodies; assists filtration slits.

- **Mesangial cells**: located between endothelial cells of adjacent glomerular capillaries; provide physical capillary support, regulate glomerular blood flow/filtration by engulfing organic materials and regulating capillary diameters.

- Filtrate also contains glucose, free fatty acids, amino acids, vitamins — reabsorbed by the proximal convoluted tubule.

Proximal Convoluted Tubule (PCT)

- First renal tubule part; begins at tubular pole (opposite vascular pole) of renal corpuscle. Simple cuboidal epithelium with apical microvilli (increase reabsorptive surface area).
- Primary function = reabsorption: actively reabsorbs organic nutrients, ions, plasma proteins (if any); reabsorbs ~60% of sodium, chloride ions, water; also actively reabsorbs potassium, calcium, magnesium, bicarbonate, phosphate, sulfate ions. Osmotic forces pull water across the PCT wall into peritubular fluid as solutes are reabsorbed.

The Nephron Loop

- Begins where proximal straight tubule enters the medulla; has **descending limb** and **ascending limb** (fluid flows toward cortex in ascending limb). Both limbs lined by simple squamous epithelium in the deeper medulla; contain thin and thick segments (referring to epithelial height, not lumen diameter).
- Ascending limb has active transport mechanisms pumping Na^+/Cl^- out of tubular fluid $\rightarrow$ medullary interstitial fluid gains high solute concentration (near loop base, deepest medulla: interstitial solute concentration = 4x plasma, i.e., 1200 mOsm vs 300 mOsm).
- Descending thin and ascending thick limbs: freely permeable to water, impermeable to ions/other solutes; high surrounding osmotic concentration $\rightarrow$ osmotic water flow out of the nephron. **Vasa recta** (slender capillary group) absorb this water, return it to general circulation.
- Net effect: nephron loop reabsorbs additional 25% of tubular fluid water, and even higher % of sodium/chloride. PCT + nephron loop together reclaim: all organic nutrients, 85% of water, >90% of sodium/chloride. Remaining water, ions, all organic wastes continue to the distal convoluted tubule.

Distal Convoluted Tubule (DCT)

- Ascends out of medulla into cortex; passes between afferent/efferent arterioles at vascular pole. Differs from PCT: (1) smaller diameter, (2) epithelial cells lack microvilli, (3) cell boundaries easily visible. Reflects function: PCT = reabsorption, DCT = secretion.

- DCT functions: (1) active secretion of ions, acids, other materials, (2) reabsorption of sodium and calcium ions from tubular fluid, (3) reabsorption of water (concentrates tubular fluid). Sodium transport activity controlled by circulating **aldosterone** (adrenal cortex).

The Juxtaglomerular Complex

- Regulates blood pressure and filtrate formation. Renin and erythropoietin (secreted here) elevate blood volume, hemoglobin levels, blood pressure, restore normal filtrate production. 3 specialized cell types:
1. **Macula densa** — thought to monitor Na⁺ concentration in tubular fluid, regulates glomerular filtration rate and renin release from juxtaglomerular cells.
 2. **Juxtaglomerular cells** — modified smooth muscle cells in afferent arteriole wall, secrete renin.
 3. **Extraglomerular mesangial cells** — located in triangular space between afferent/efferent arterioles, provide feedback control between macula densa and juxtaglomerular cells.

The Collecting System

- DCT (last nephron segment) opens into collecting system = connecting tubules, collecting ducts, papillary ducts. Every nephron connects to a connecting tubule; several connecting tubules → one collecting duct; several collecting ducts converge → larger papillary duct → empties into a minor calyx.
- Epithelium: begins simple cuboidal (connecting tubules), changes to columnar (collecting/papillary ducts).
- Transports tubular fluid from nephron to renal pelvis, makes final osmotic concentration/volume adjustments. **ADH (antidiuretic hormone)** controls collecting system permeability — significant because collecting ducts pass through the medulla where the nephron loop created high solute concentrations. Low permeability → dilute urine (most fluid flows to renal pelvis). High permeability → osmotic water flow out of duct into medulla → small volume, highly concentrated urine. Higher circulating ADH = more water reabsorbed = more concentrated urine.

26.2 Structures for Urine Transport, Storage, and Elimination

- Filtrate modification/urine production ends when fluid enters the minor calyx. Remaining structures (ureters, bladder, urethra) transport/store/eliminate urine.
- **Transitional epithelium** lines minor/major calyces, renal pelvis, ureters, urinary bladder, proximal urethra — tolerates distension/relaxation cycles without damage.

The Ureters

- Paired retroperitoneal muscular tubes, kidneys → urinary bladder; ~30 cm (12 in.) long each; begin as funnel-shaped renal pelvis continuation; pass inferiorly/medially over psoas major. Anatomical locations differ by sex (reproductive organ variation).
- Penetrate posterior bladder wall at oblique angle; **ureteral orifices** are slit-like (not rounded) — prevents urine backflow toward kidneys when bladder contracts.
- Histology: 3 layers — (1) inner mucosa (transitional epithelium), (2) middle muscular layer (inner longitudinal + outer circular smooth muscle), (3) outer adventitia (continuous with kidney's fibrous capsule and abdominal wall parietal peritoneum). Peristaltic contractions (~every 30 sec) triggered by ureteral wall stretch receptor stimulation; travel renal pelvis → bladder, "milking" urine along.
- Clinical: kidney/dialysis facts — kidneys filter all body blood every 30 minutes, removing wastes/excess fluid, balancing electrolytes/minerals/pH; also produce renin (BP control), erythropoietin (RBC production), calcitriol (blood calcium control). One healthy kidney usually sufficient for healthy life. **End-stage kidney disease** = 90% of kidney function lost. ~500,000 Americans on dialysis; ~200,000 have functioning transplants; >100,000 waiting for transplant, <20,000 transplants performed/year.
 - **Acute renal failure** — sudden kidney function loss, often from multiple trauma, MI, or reduced renal blood supply; often reversible, may need temporary dialysis during recovery.
 - **Chronic renal failure** — gradual loss, may be asymptomatic until significantly impaired; leading causes: diabetes, hypertension; also genetic diseases, infections, kidney stones, autoimmune disease, certain drugs (legal/illegal, including some antibiotics). First-line management: strict fluid/protein/mineral (sodium, potassium, phosphorus) intake limits.

- **Dialysis:** performs some healthy-kidney functions; required once 90% of kidney function lost; removes wastes/salt/extra water, balances electrolytes/minerals/pH — but CANNOT produce renin, erythropoietin, or calcitriol. Performed at dialysis center ~3x/week, few hours each; requires venous access (surgical forearm shunt). **Peritoneal dialysis:** uses peritoneal cavity as dialysis membrane via permanent intraperitoneal catheter; **continuous ambulatory peritoneal dialysis** = manual, 4-5x/day (~2 quarts dialysate infused, drained after 4-5 hrs); **automated peritoneal dialysis** = machine cycles dialysate overnight during sleep.
- **Kidney transplant** = best renal failure treatment; from living or deceased (within 48 hrs of death) donor; recipient's nonfunctioning kidney(s) may be removed if infected; transplanted kidney/ureter placed extraperitoneally in pelvic cavity (iliac fossa), ureter connected to recipient's bladder; 2-year survival rate >90% (much better than dialysis-only survival).

The Urinary Bladder

- Hollow muscular organ, temporary urine storage. In males, base lies between rectum and pubic symphysis; in females, base sits inferior to uterus, anterior to vagina. Full bladder holds ~1 L urine.
- Peritoneum covers superior surface when empty; as bladder fills, displaces parietal peritoneum from anterior abdominal wall, becomes intraperitoneal. Stabilizing peritoneal folds: **median umbilical ligament** (anterior/superior bladder border to umbilicus) and **lateral umbilical ligaments** (bladder sides to umbilicus) — remnants of the two fetal umbilical arteries. Connective tissue bands anchor posterior/inferior/anterior bladder surfaces to pelvic bones.
- Mucosa has folds (**rugae**) that disappear as bladder stretches/fills.
- **Trigone** = smooth, thick mucosa area functioning as a funnel channeling urine into the urethra during bladder contraction. Urethral entrance = most inferior bladder point; surrounding area = **neck of the urinary bladder**, containing the **internal urethral sphincter** (smooth muscle, involuntary control via pelvic nerves).
- Histology: mucosa (transitional epithelium with rugae) → submucosa (connective tissue) → muscular layer (3 layers: inner longitudinal, middle circular, outer longitudinal smooth muscle — together form the **detrusor**) → adventitia (outermost). Detrusor contraction compresses bladder, expels urine into urethra.
- Clinical: **Transitional cell carcinoma** = most common bladder cancer type, begins in innermost transitional epithelium layer; smoking = most important risk factor (toxins filtered by lungs/kidneys can still concentrate in stored urine); hallmark sign = **hematuria** (bloody urine); diagnosed/treated via **cystoscopy** (lighted endoscope through urethra into bladder).

The Urethra

- Extends from bladder neck to exterior. Differs by sex in length/function.
- **Female urethra:** short, 3-5 cm (1-2 in.), bladder to vestibule; **external urethral orifice** near anterior vaginal wall.
- **Male urethra:** 18-20 cm (7-8 in.), bladder neck to penis tip; 3 subdivisions: (1) **prostatic urethra** (through prostate center), (2) **membranous urethra (intermediate part)** (short segment through pelvic cavity's muscular floor), (3) **spongy (penile) urethra** (membranous urethra to external urethral orifice at penis tip).
- Both sexes: **external urethral sphincter** = skeletal muscle band acting as a valve; branches of hypogastric plexus control both external/internal sphincters, but only external sphincter has voluntary control (via pudendal nerve's perineal branch). Sphincter has resting tone, must be voluntarily relaxed for urination; autonomic innervation of external sphincter becomes important only when voluntary control lacking (infants, spinal cord injury adults).
- Histology: female urethral lining = transitional epithelium near bladder neck, stratified squamous for the rest; lamina propria has extensive venous network, surrounded by concentric smooth muscle layers.
- Male urethral histology varies along length: transitional (near bladder) → pseudostratified/stratified columnar → stratified squamous (near external orifice). Thick, elastic lamina propria; longitudinal mucosal creases; mucus-secreting cells in epithelium; males have epithelial mucous glands forming tubules into lamina propria. Connective tissue anchors urethra to surrounding structures.

Urinary Reflexes: Urine Storage and Urine Voiding

- Urine reaches bladder via ureteral peristalsis. Urge to urinate typically appears around 200 mL bladder volume. Urination (micturition) involves interplay between spinal reflexes and higher brain centers (conscious control).
- **Urine storage reflex:** low-frequency afferent impulses from bladder stretch receptors → (1) increase sympathetic activity (inhibits detrusor contraction, stimulates internal urethral sphincter contraction), (2) stimulate external urethral sphincter contraction. **Pontine storage center** also stimulates somatic motor neurons to the external urethral sphincter, causing contraction. Together these promote continence (urine storage).
- **Urine voiding reflex:** high-frequency afferent impulses from stretch receptors stimulate interneurons relaying to the **pontine micturition center** → initiates sacral spinal reflexes: (1) increased parasympathetic activity (detrusor contracts, internal urethral sphincter relaxes), (2) decreased sympathetic activity (internal urethral sphincter relaxes), (3) decreased somatic motor activity (external urethral sphincter relaxes). Result: voiding (urination).
- At end of typical urination, <10 mL urine remains in bladder.
- Clinical: **Urinary tract infections (UTIs)** — bacteria/fungi/virus colonization, most often **Escherichia coli**; women more susceptible (short urethra, proximity to anus); sexual intercourse can push bacteria toward bladder. Symptoms: **dysuria** (painful urination), fever, foul/strong-smelling urine, cloudy/bloody urine, frequent urination, persistent urge; in elderly, confusion is often the first symptom. Diagnosed via urine bacteria/blood cell presence. Inflamed urethral wall reduces water/sodium reabsorption → more frequent urination, increased fluid needs. Inflammation of urethra = **urethritis**; of bladder lining = **cystitis**. Treated with antibiotics; mainstay therapy = dilute urine (drink fluids), keep bladder empty (urinate often). Untreated: bacteria may migrate up ureters to renal pelvis → **pyelitis** (renal pelvis inflammation); if bacteria invade cortex/medulla too → **pyelonephritis** (high fever, intense affected-side pain, vomiting, diarrhea, blood cells/pus in urine).

26.3 Aging and the Urinary System

- Aging generally increases kidney problems. 4 typical age-related changes:
1. **Decline in functional nephrons** — between ages 25-85, total nephron number drops 30-40%.
 2. **Reduced glomerular filtration** — from decreased glomeruli count, cumulative filtration apparatus damage in remaining glomeruli, reduced renal blood flow.
 3. **Reduced ADH sensitivity** — distal nephron/collecting system become less responsive to ADH with age; less water/sodium reabsorption → more frequent urination, increased daily fluid requirements.
 4. **Problems with urinary reflexes** — 3 factors: (a) external sphincter loses muscle tone, less effective voluntary urine retention → incontinence (often slow leakage); (b) urination control loss from stroke, Alzheimer's disease, other CNS problems affecting cerebral cortex/hypothalamus; (c) in males, urinary retention from enlarged prostate compressing urethra.

Clinical Correlations Summary (Ch 26)

- **Acute/chronic renal failure, dialysis** (hemodialysis and peritoneal), **kidney transplant**: described above.
- **Transitional cell carcinoma** of bladder: smoking is the biggest risk factor; hematuria is the hallmark sign.
- **UTI / urethritis / cystitis / pyelitis / pyelonephritis**: ascending infection spectrum.
- Clinical Case (Jack): **Kidney stone (renal calculus)** — small, hard, sharp-edged mineral deposit (usually calcium/phosphate) that normally stays dissolved in urine; formed when concentrated urine (from dehydration/fluid loss via sweating, inadequate water intake) causes minerals to crystallize in the renal pelvis. As the stone moves into the ureter, it obstructs urine flow, stretches the collecting system behind it, causing flank/lumbar pain (**renal colic**); abraded ureteral epithelium bleeds, causing **hematuria**; peristaltic ureteral contractions trying to push the stone cause additional pain; pain radiates along the stone's path (crossing psoas major, down to testicle/groin as stone nears the bladder, then to the urethral tip as it passes through). Treatment/prevention: increased fluid intake ("drink, drink, drink") to keep urine dilute and prevent further stone formation. Non-invasive treatment option: **shock-wave lithotripsy** (pulverizes kidney stones using high-pressure shock waves through a water-filled tub).
- Related clinical terms: **azotemia** (excessive urea/nitrogen compounds in blood); **bacteriuria** (bacteria in urine); **cystocele** (bladder bulging into vagina from weakened supportive tissue); **enuresis** (involuntary urination,

especially in sleeping children); **hydroureter** (ureter dilation from urine flow obstruction); **nephroptosis** ("floating kidney" — kidney displaced downward from normal position); **nephrotic syndrome** (kidney disorder causing excessive urine protein excretion); **nephrotoxin** (kidney-specific harmful toxin); **polycystic kidney disease** (inherited kidney tubule development/structure abnormality); **shock-wave lithotripsy** (noninvasive kidney stone pulverization); **urologist** (physician specializing in urinary system function/disorders).

Reproductive (Ch 27)

27.1 Organization of the Reproductive System

- Reproductive system produces, stores, nourishes, transports functional gametes; combination of sperm (father) + oocyte (mother) at **fertilization/conception** produces a **zygote** (single cell); ~9 months of growth/division produces an infant. Reproductive system also produces sex hormones affecting all other systems.
- 4 components: **Gonads** (produce gametes and hormones); **reproductive tract** (ducts that receive/store/transport gametes); **accessory glands/organs** (secrete fluids into reproductive/other ducts); **external genitalia** (perineal structures).
- Male gonads = **testes** — secrete androgens (mainly testosterone), produce ~half a billion sperm/day. Mature sperm mix with accessory gland secretions to form **semen**; expelled during **ejaculation**.
- Female gonads = **ovaries** — typically produce one immature oocyte/month; oocyte travels within **uterine tubes** opening into the muscular **uterus**; **vagina** connects uterus to exterior. Male ejaculation introduces semen into vagina during intercourse; sperm ascend the tract, may encounter oocyte for fertilization.

27.2 Anatomy of the Male Reproductive System

- Duct pathway: testes → **epididymis** → **ductus deferens (vas deferens)** → **ejaculatory duct** → **urethra** → exit body. Accessory organs: **seminal glands, prostate, bulbo-urethral glands** — secrete into ejaculatory ducts/urethra. External genitalia: **scrotum** (encloses testes), **penis** (erectile organ, urethra passes through distal portion).

The Testes

- ~5 cm (2 in.) long, 3 cm (1.2 in.) wide, 2.5 cm (1 in.) thick; 10-15 g (0.35-0.53 oz) each; hang within scrotum (skin pouch inferior to perineum, anterior to anus).
- **Descent of the testes**: form in-body-cavity adjacent to kidneys during development; **gubernaculum testis** (connective tissue/muscle cord) extends from inferior testis to a small peritoneal pocket's posterior wall — doesn't elongate as fetus grows, holding testes in relative position while body grows around them. 7th developmental month: rapid growth + hormone-stimulated gubernaculum muscle contraction → testes move through inguinal canal into scrotum outside the body = **descent of the testes**; small peritoneal cavity pockets develop as testes move.
- As each testis moves, ductus deferens + testicular vessels/nerves/lymphatics accompany it, forming the **spermatic cord** body.
- **Spermatic cords**: paired, extend abdominopelvic cavity to testes; begin at **deep inguinal ring**, through **inguinal canal**, exit at **superficial inguinal ring**, descend to scrotum. Contains: ductus deferens, **deferential artery**, **testicular artery**, **pampiniform plexus** (testicular vein), **ilio-inguinal** and **genitofemoral nerves** (lumbar plexus).
- **Inguinal canals**: link scrotal chambers to peritoneal cavity; usually close, but spermatic cord presence creates permanent abdominal wall weak points → **inguinal hernias** relatively common in males; rare in females (inguinal canals present but hernias rare).
- **Scrotum**: internally divided into 2 chambers by the **raphe** (surface ridge); each testis in a separate scrotal cavity. **Tunica vaginalis** (serous membrane) lines scrotal cavity, reduces friction. Layers deep to dermis: **dartos muscle** (smooth muscle — resting tone elevates testes, wrinkles scrotal surface) and **cremaster muscle** (skeletal muscle, deep to dartos — contraction via **cremasteric reflex** tenses scrotum, pulls testes toward body).

during sexual arousal/cold exposure). Sperm development requires temperature $\sim 1.1^{\circ}\text{C}$ (2°F) below core body temperature; cremaster adjusts testis position to maintain this.

- Scrotal innervation: hypogastric plexus, ilio-inguinal, genitofemoral, pudendal nerve branches. Blood supply: internal pudendal arteries (from internal iliac), external pudendal arteries (from femoral), cremasteric branch of inferior epigastric arteries (from external iliac).
- **Tunica albuginea**: dense fibrous capsule surrounding testis (covered by tunica vaginalis); collagen fibers continuous with epididymis; extend into testis interior forming **septa testis**, converging at the **mediastinum of the testis** (supports vessels/lymphatics/sperm-collecting ducts).
- Histology: septa testis partition testis into **lobules**; ~ 800 slender coiled **seminiferous tubules** distributed among lobules (each $\sim 80\text{ cm}/31\text{ in.}$ long; total $\sim$ half mile per testis) — site of sperm production. Each U-shaped tubule connects to a single **straight tubule** entering the mediastinum $\rightarrow$ interconnect forming the **rete testis** (network of passageways) $\rightarrow$ 15-20 **efferent ductules** connect rete testis to epididymis.
- Between tubules: loose connective tissue, blood vessels, **interstitial endocrine cells (Leydig cells)** — produce androgens (testosterone = most important). Testosterone functions: stimulate spermatogenesis; promote sperm physical/functional maturation; maintain male reproductive tract accessory organs; cause secondary sexual characteristic development (e.g., facial hair); stimulate body growth/metabolism; influence brain development/sexual behavior/drive.
- **Spermatogenesis**: begins at outermost seminiferous tubule cell layer. **Spermatogonia** (stem cells, form during embryonic development, dormant until puberty) continuously divide from puberty onward — one daughter cell remains an undifferentiated stem cell, other becomes a **primary spermatocyte** entering **meiosis**.
- Meiosis produces haploid gametes (23 individual chromosomes) vs. mitosis's diploid daughter cells (23 pairs). Primary spermatocyte $\rightarrow$ meiosis I $\rightarrow$ pair of **secondary spermatocytes** $\rightarrow$ meiosis II $\rightarrow$ pair of **spermatids** each = 4 spermatids per primary spermatocyte.
- Spermatogenesis directly stimulated by testosterone, indirectly by FSH; interstitial endocrine cells produce testosterone in response to LH.
- **Spermiogenesis**: spermatids differentiate into physically mature sperm while embedded in **nurse cells (Sertoli cells)** cytoplasm (attached to basal lamina, extend toward lumen between spermatocytes undergoing meiosis). Sperm detach from nurse cell, released into tubule lumen — marks end of spermiogenesis. Whole spermatogenesis process (start to finish) takes ~ 9 weeks.
- **Nurse (Sertoli) cell functions (5)**:

1. **Maintain blood-testis barrier** — comparable to blood-brain barrier; tight junctions between nurse cells isolate tubule lumen from general circulation; tubule lumen fluid (high androgens, estrogens, potassium, amino acids) differs greatly from interstitial fluid; barrier prevents immune system from attacking sperm-specific antigens (not found on somatic cells) on developing sperm.

2. **Support mitosis/meiosis** — FSH + testosterone stimulate nurse cells, which promote spermatogonial division and spermatocyte meiotic division.

3. **Support spermiogenesis** — surround/enfold spermatids, providing nutrients/chemical stimuli.

4. **Secrete inhibin** — depresses pituitary FSH and hypothalamic GnRH production; negative feedback control of spermatogenesis.

5. **Secrete androgen-binding protein (ABP)** — binds androgens in tubule fluid, elevates local androgen concentration, stimulates spermiogenesis (FSH stimulates ABP production).

- **Sperm anatomy** — 3 regions:

1. **Head** — densely packed chromosomes; tip has the **acrosome** (membrane-bound vesicle with fertilization enzymes).

2. **Middle piece** — short neck attaches head to middle piece, contains both centrioles of original spermatid; distal centriole connects to microtubules continuous through middle piece/tail; mitochondria (spiral arrangement around microtubules) provide ATP for tail movement.

3. **Tail** — only flagellum in the human body; whip-like, corkscrew motion (unlike predictable ciliary beating).

- Mature sperm lack ER, Golgi apparatus, lysosomes, peroxisomes, inclusions, and most other organelles; no glycogen/energy reserves — must absorb nutrients from surrounding fluid.

The Male Reproductive Tract

- **Epididymis:** start of reproductive tract, along testis posterior border; firm, palpable through scrotum; ~7 m (23 ft) long but tightly coiled. 3 regions: **head** (receives sperm via efferent ducts), **body** (distal to last efferent duct, along testis posterior margin), **tail** (near testis inferior border, reverses direction, epithelium changes — main sperm storage site). Pseudostratified columnar epithelium with **stereocilia** (increase surface area for absorption/secretion) in head/body; stereocilia disappear in tail (epithelium becomes similar to ductus deferens).
- Epididymis functions: (1) monitors/adjusts seminiferous tubule fluid composition; (2) recycles damaged sperm/cellular debris (products released into interstitial fluid for blood vessel pickup); (3) stores sperm, facilitates functional maturation — takes ~2 weeks to transit and mature. Sperm leaving epididymis are mature but immobile; **capacitation** (2 steps) makes them motile/fertilization-capable: (a) become motile when mixed with seminal gland secretions, (b) become fertilization-capable upon exposure to female reproductive tract conditions (alters sperm plasma membrane). An epididymal secretion prevents premature capacitation while sperm remain in the male tract. Rhythmic stereocilia movement + epididymal smooth muscle peristalsis push sperm toward ductus deferens.
- Clinical: **Testicular cancer** — disease of young/middle-aged men, almost always develops in rapidly-dividing spermatogonia stem cells; often first symptom = painless lump; best defense = regular testicular self-examination; often curable if caught early; treatment = **orchiectomy** (surgical testis removal) + chemotherapy.
- **Ductus deferens (vas deferens):** 40-45 cm (16-18 in.) long; begins at epididymal tail, ascends through inguinal canal as part of spermatic cord. Within abdomen: passes posteriorly/inferiorly along bladder's lateral surface, medial/anterior to ureters, descends along bladder's posterior surface toward prostate; enlarges near prostate forming the **ampulla**. Wall has thick smooth muscle layer; pseudostratified columnar epithelium with stereocilia; peristaltic contractions propel sperm/fluid; also stores sperm for months (low metabolic rate, temporary inactivity). Each ampulla joins the seminal gland's excretory duct → forms the **ejaculatory duct** (short, ~2 cm/<1 in., penetrates prostate's muscular wall, empties into prostatic urethra).
- **The Urethra:** extends bladder to penis tip, 15-20 cm (6-8 in.); 3 regions: **prostatic**, **membranous (intermediate part)**, **spongy**. Serves both urinary and reproductive systems in males.

The Accessory Glands

- Functions: (1) activate sperm, (2) provide motility nutrients, (3) produce buffers counteracting urethral/vaginal acidity.
- **Seminal glands (seminal vesicles):** paired, embedded in connective tissue between bladder posterior wall and rectum anterior wall; each ~15 cm (6 in.) tubular gland with short side branches, coiled/folded into compact ~5×2.5 cm mass. Pseudostratified columnar epithelium, extremely active secretory; contribute ~60% of semen volume; secretion has same osmotic concentration as plasma but different composition — contains prostaglandins, clotting proteins, high fructose (sperm ATP production). Discharged into ductus deferens during ejaculation via sympathetically-controlled peristaltic contractions (ductus deferens, seminal glands, prostate); mixing with seminal gland secretions activates sperm flagellar beating (mobility).
- **Prostate:** small, muscular, rounded, ~4 cm (1.6 in.) diameter; encircles prostatic urethra; contains 30-50 compound tubulo-alveolar glands (simple columnar epithelium, varying to squamous/cuboidal/pseudostratified columnar), wrapped in thick smooth muscle. Produces **prostatic fluid** — weakly acidic, ~20-30% of semen volume; rich in enzymes nourishing sperm/preventing coagulation; contains **seminalplasmin** (antibiotic, may prevent male UTIs). Peristaltic wall contractions eject secretions into prostatic urethra.
- **Bulbo-urethral glands (Cowper's glands):** paired, at penis base, ~10 mm diameter; ducts parallel spongy urethra 3-4 cm before emptying into urethral lumen; simple cuboidal-to-columnar epithelium; compound tubulo-alveolar mucous glands secreting thick, sticky, alkaline mucus — neutralizes residual urinary acids, lubricates penis tip.

Semen

- Typical ejaculation = 2-5 mL, containing:

1. **Sperm** — normal count 20-100 million/mL.

2. **Seminal fluid** — glandular secretion mixture: seminal glands (60%), prostate (30%), nurse cells/epididymis (5%), bulbo-urethral glands (5%).

3. **Enzymes** — (a) protease (dissolves vaginal mucus), (b) seminalplasmin (antibiotic, kills bacteria incl. E. coli), (c) prostatic enzyme converting fibrinogen to fibrin (coagulates semen within minutes post-ejaculation), (d) fibrinolysin (liquefies clotted semen after 15-30 minutes).

The Penis

- Tubular organ for intercourse and urine conduction. 3 main structures: **root** (attaches to ischial rami, within urogenital triangle inferior to pubic symphysis), **body/shaft** (tubular, movable, contains erectile tissue), **glans penis** (expanded distal end around external urethral orifice).
- Skin: hairless, more pigmented than elsewhere; dermis has smooth muscle layer; underlying loose connective tissue allows movement; subcutaneous layer has superficial vessels/lymphatics, few fat cells.
- **Foreskin (prepuce)**: skin fold surrounding penis tip, attaches at glans neck; **preputial glands** in glans/foreskin inner surface secrete **smegma** (waxy material, good bacterial nutrient source → mild inflammation/infection common if not washed); **circumcision** = surgical foreskin removal.
- Erectile tissue: 3 parallel cylindrical columns — maze of vascular channels separated by elastic connective tissue/smooth muscle partitions. Resting state: constricted arterial branches, tense muscular partitions restrict blood flow. Erection: penis stiffens, elevates.
- **Corpora cavernosa** (2, posterior surface): encircled by dense collagenous sheath, separated by thin septum; diverge at base forming the **crus of the penis** (bound to ischial ramus/pubis by ligaments); extend to penis neck; each surrounds a central **deep artery of the penis**.
- **Corpus spongiosum**: surrounds spongy urethra; extends superficial fascia to penis tip, expanding into the **glans penis**; sheath has more elastic fibers than corpora cavernosa; contains a pair of small arteries. **Bulb of the penis** = thickened proximal end of corpus spongiosum.
- Semen release = 2-step process: **Emission** — sympathetic nervous system coordinates peristaltic contractions sweeping ductus deferens → seminal glands → prostate → bulbo-urethral glands, mixing semen components. **Ejaculation** — powerful rhythmic contractions of **ischiocavernosus** (inserts along penis sides, stiffens organ) and **bulbospongiosus** (wraps penis base, pushes semen toward external urethral orifice) muscles; controlled by reflexes involving inferior lumbar/superior sacral spinal cord segments.

27.3 Anatomy of the Female Reproductive System

- Oocytes form in ovaries → travel along **uterine tubes** (fertilization site) → reach **uterus** → uterus opens to **vagina** → external opening surrounded by external genitalia. Accessory glands secrete into reproductive tract (as in males).
- **Broad ligament**: large mesentery enclosing ovaries, uterine tubes, uterus. Uterine tubes along broad ligament's superior border, open into pelvic cavity lateral to ovaries. **Mesosalpinx** = free edge of broad ligament attached to each uterine tube. **Mesovarium** = thickened broad ligament fold supporting/stabilizing each ovary's position.
- Broad ligament attaches to pelvic cavity sides/floor, continuous with parietal peritoneum; subdivides pelvic cavity: **recto-uterine pouch** (between posterior uterine wall and anterior colon) and **vesico-uterine pouch** (between anterior uterine wall and posterior bladder wall).
- Additional stabilizing ligaments within broad ligament sheet: limit side-to-side movement/rotation (broad ligament) vs. superior-inferior movement (others, detailed below).

The Ovaries

- Flattened oval, ~5 cm long × 2.5 cm wide × 8 mm thick (2×1×0.33 in.); 6-8 g each (~0.25 oz). Produce oocytes, secrete hormones.
- No peritoneal covering; stabilized by mesovarium, **ovarian ligament** (extends from lateral uterine wall near uterine tube attachment to ovary's medial surface), and **suspensory ligament** (extends from ovary's lateral surface past uterine tube's open end to pelvic wall; contains ovarian artery and vein, connecting to ovary at the **ovarian hilum**).

- Pink/yellowish, nodular; surface covered by single cuboidal epithelium layer = **germinal epithelium**, overlying dense connective tissue = **tunica albuginea**. Interior (stroma): superficial **cortex** (gamete production) and deeper **medulla**.

Ovarian Cycle and Oogenesis

- 3 interrelated processes: **oogenesis** (oocyte production), **ovarian cycle** (monthly oocyte maturation events), **uterine cycle** (uterine preparation for implantation).
- Oogenesis begins before birth, accelerates at puberty, ends at menopause. Fetal development: **oogonia** (female reproductive stem cells) → diploid **primary oocytes (2n)**. **Ovarian follicles** in ovarian cortex = sites of oocyte growth/oogenesis.
- Puberty: rising FSH triggers start of ovarian cycle; monthly FSH stimulates some primordial follicles to develop further.
- Follicle development stages:

1. **Primordial ovarian follicles**: primary oocytes in outer ovarian cortex near tunica albuginea, in clusters called **egg nests**; single squamous follicle cell layer surrounds each primary oocyte.

2. **Primary ovarian follicles**: activated primordial follicles develop; follicle cells become cuboidal, divide repeatedly forming several layers = **granulosa cells**; oocyte secretes proteins forming the **zona pellucida** (extracellular coat); granulosa cell microvilli + oocyte microvilli penetrate zona pellucida, providing nutrients. Adjacent stromal cells form a **thecal endocrine cell** layer around the follicle; thecal + granulosa cells together produce **estrogens** (**estradiol** = most important, dominant pre-ovulation hormone). Estrogen functions: (a) stimulate bone/muscle growth, (b) maintain female secondary sex characteristics, (c) affect CNS activity (sex behaviors/drives), (d) maintain reproductive gland/organ function, (e) initiate uterine lining repair/growth.

3. **Secondary ovarian follicles**: only a few primary follicles mature further (apparently controlled by an oocyte-produced growth factor); follicle wall thickens; primary oocyte continues slow growth.

4. **Tertiary (vesicular) ovarian follicle**: over 2-3 months, follicle wall grows, deeper follicular cells secrete follicular fluid (accumulates in pockets between inner/outer cellular layers); follicle doubles in size, creates prominent ovarian surface bulge; oocyte + follicular cell covering project into the expanded central chamber = **antrum** (surrounded by granulosa cell mass); granulosa cells around the secondary oocyte form the **corona radiata** (protective layer). Primary oocyte (suspended in prophase of meiosis I) completes that division — unlike spermatogenesis, cytoplasm divides unevenly: produces one **secondary oocyte** + one small nonfunctional **polar body**. Secondary oocyte proceeds to metaphase of meiosis II, but this stage isn't completed unless fertilization occurs (if it does, meiosis II completes, producing an ovum + second polar body). Net result: oogenesis produces a single functional ovum (most cytoplasm) + polar bodies (extra chromosome containers).

5. **Ovulation**: tertiary follicle releases the secondary oocyte; distended follicular wall ruptures, releasing contents (including secondary oocyte) into pelvic cavity; sticky follicular fluid keeps corona radiata attached near the ruptured follicle wall; oocyte moved into uterine tube via fimbriae contact or ciliary fluid currents. Usually only 1 oocyte released per cycle, though multiple follicles may have developed partway — undeveloped ones undergo **follicular atresia**.

6. **Corpus luteum formation**: empty tertiary follicle collapses, ruptured vessels bleed; LH stimulates remaining granulosa cells to invade/proliferate, creating the **corpus luteum**.

7. **Corpus albicans formation**: corpus luteum degenerates ~12 days post-ovulation (unless fertilization occurs); progesterone/estrogen decline; fibroblasts invade, forming a pale scar tissue knot = **corpus albicans**; marks end of the ovarian cycle. New cycle begins with new primordial follicle activation.

- Declining progesterone/estrogen at cycle's end stimulates hypothalamic **GnRH** production → rise in pituitary FSH/LH → new follicle development period. These ovarian hormonal changes also drive the uterine cycle.

The Uterine Tubes

- ~13 cm (5 in.) long hollow muscular tubes, 5 regions: (1) **fimbriae** (fingerlike projections draping over ovary surface), (2) **infundibulum** (ovary-closest end, expanded funnel, ciliated lining beating toward tube's middle), (3) **ampulla** (middle region, fertilization typically occurs here), (4) **isthmus** (short region connecting ampulla to uterine wall), (5) **uterine part** (isthmus continuation, opens into uterine cavity).

- Histology: ciliated + nonciliated simple columnar epithelium; concentric smooth muscle layers. Transport = ciliary movement + peristaltic contractions; a few hours before ovulation, sympathetic/parasympathetic nerves from hypogastric plexus "turn on" ciliary beating.
- Oocyte transit time: 3-4 days infundibulum → uterine cavity; for fertilization, oocyte must encounter sperm within first 12-24 hours of transit (typically in the ampulla). Tube provides nutritive environment (lipids, glycogen) for sperm and developing pre-embryo. Unfertilized oocytes degenerate in tube's terminal portions or in the uterus.

The Uterus

- Empty uterus: small, pear-shaped, ~7.5 cm (3 in.) long, 5 cm (2 in.) max diameter, 30-40 g (1-1.4 oz). Normal position: bends anteriorly near base = **anteflexion** (body lies across bladder's posterior/superior surfaces). Backward bend toward sacrum = **retroflexion** (occurs in ~20% of women, no clinical significance).
- Divisions: **body** (largest region), **fundus** (rounded portion superior to uterine tube attachment), **isthmus of uterus** (constriction where body ends), **cervix** (inferior portion, isthmus to vagina) — projects ~1.25 cm (0.5 in.) into vagina; distal cervix forms curving surface around the **external os** (external opening) leading to the **cervical canal** → opens at the **internal os** into the uterine cavity.
- Cervical mucus (viscous, fills canal/covers external os) prevents vaginal bacteria passage; becomes watery near ovulation (facilitates sperm entry) — infertility treatment may include mucus-thinning drugs.
- Blood supply: uterine and ovarian artery/vein branches; numerous lymphatics. Innervation: hypogastric plexus (sympathetic) + sacral S3-S4 (parasympathetic); sensory afferents enter spinal cord via T11-T12 dorsal roots.
- **Suspensory ligaments** (3 pairs, beyond broad ligament): **uterosacral ligaments** (lateral uterus to anterior sacrum — prevent inferior/anterior uterine movement), **round ligaments** (lateral uterus below uterine tube bases, through inguinal canal to external genitalia connective tissue — restrict posterior movement), **cardinal (transverse cervical) ligaments** (uterus/vagina base to lateral pelvic walls — prevent inferior movement). Pelvic floor skeletal muscle/fascia provide additional support.
- **Uterine wall**: outer muscular **myometrium**, inner glandular **endometrium**; outer fundus/anterior-posterior body surfaces covered by serous **perimetrium** (continuous with peritoneum, incomplete layer).
 - Uterine glands open onto endometrial surface, extend deep into lamina propria nearly to myometrium; estrogen-driven changes through the uterine cycle.
 - **Myometrium** = thickest wall layer; 3 poorly-defined smooth muscle layers (middle = thickest, circular/spiral arrangement; inner/outer = longitudinal); provides force to expel fetus during birth.
 - Blood supply within myometrium: uterine artery branches → **arcuate arteries** (encircle endometrium) → **radial arteries** → **straight arteries** (basal layer) and **spiral arteries** (functional layer).
 - Endometrium subdivides into **functional layer** (closest to uterine cavity, most glands, cyclical changes) and **basal layer** (attaches to myometrium, contains gland terminal branches, structurally constant).
- **Uterine (menstrual) cycle**: averages 28 days (range 21-35); 3 phases:

1. **Menstrual phase**: functional layer degeneration; nourishing arteries constrict, weakening/rupturing, blood pours into functional layer connective tissue; degenerating tissue/blood shed through external os/vagina = **menstruation (menses)**; lasts 1-7 days; ~35-50 mL blood lost. Occurs as progesterone/estrogen levels decline at prior cycle's end.

2. **Proliferative phase**: basal layer circulation stays constant (survives menstruation); rising estrogen stimulates gland epithelial cell multiplication/spread, restoring endometrial epithelium and functional layer, occurring alongside primary/secondary follicle enlargement in the ovary — estrogens from developing follicles stimulate/sustain this phase. By ovulation, functional layer several mm thick, highly vascularized, glycogen-rich mucus (essential for fertilized ovum's earliest survival).

3. **Secretory phase**: begins at ovulation, persists while corpus luteum intact; endometrial glands enlarge, accelerate secretion; endometrial arteries elongate/spiral through functional layer tissue; combined progesterone + estrogen (from corpus luteum) stimulation. Secretory activity peaks ~12 days post-ovulation; activity declines as corpus luteum stops producing hormones; cycle ends with menstruation onset (functional layer disintegration). Secretory phase usually lasts 14 days — ovulation date can be back-calculated as 14 days before first day of next menstruation.

- **Menarche** = first uterine cycle (age 11-12 typically). **Menopause** = last uterine cycle (age 45-55). Cycles continue between these except for illness/stress/starvation/pregnancy interruptions; irregular cycles typical for first 2 years after menarche and last 2 years before menopause.
- Clinical: **Cervical cancer** — one of few cancers known to result from viral infection (certain **HPV** types, sexually transmitted); HPV vaccination could theoretically eliminate many cases; early cervical cancer usually asymptomatic, detectable via pelvic exam + **Pap smear** screening (named after Dr. George Papanicolaou); precancerous lesions/early cancers have good prognosis. **Uterine fibroids** — most common benign uterine tumors, non-cancerous, form in myometrium; can interfere with fertility, cause heavy/painful periods or intermenstrual bleeding. **Endometrial cancer** — most common uterine cancer type, arises within endometrium, most cases ages 50-70; estrogen replacement therapy without progesterone increases risk; postmenopausal bleeding = most common sign (always investigate); Pap smear may miss it; good prognosis if caught before spread beyond uterus, treated via hysterectomy ± radiation.

The Vagina

- Average length 7.5-9 cm (3-3.5 in.), highly distensible (variable actual dimensions). Cervix projects into proximal vagina; **vaginal fornix** = shallow recess around cervical protrusion. Lies parallel to/in close posterior contact with rectum. Urethra travels along superior vaginal wall anteriorly (bladder to vestibule).
- Blood supply: vaginal branches of internal iliac arteries (sometimes from uterine arteries); venous drainage via vaginal vein. Innervation: hypogastric plexus, sacral nerves S2-S4, pudendal nerve branches.
- 3 major functions: (1) passageway for menstrual fluid elimination, (2) receives penis during intercourse, holds sperm before uterine entry, (3) forms lower birth canal portion during childbirth.
- Walls moistened by cervical gland secretions + water movement across permeable epithelium. Muscular layer + outer adventitia have extensive vessel/lymphatic networks. **Hymen** = elastic epithelial fold separating vagina/vestibule, may partially/completely block vaginal entrance. **Bulbospongiosus muscles** flank the vaginal orifice, contraction constricts entrance; cover the **bulb of vestibule** (erectile tissue mass each side of vaginal entrance) — embryologically homologous to male corpus spongiosum (homologous = similar structure/origin, can have different functions).
- Vagina has normal resident bacteria population supported by cervical mucus nutrients; bacterial metabolic activity creates acidic environment restricting pathogen growth (also inhibits sperm motility — seminal fluid buffers counteract this for successful fertilization).
- Histology: mucosal stratified squamous epithelium, forms **rugae** when relaxed; thick elastic lamina propria (small vessels, nerves, lymph nodes); inner circular + outer longitudinal smooth muscle layers. Near uterus: serosal covering continuous with pelvic peritoneum; rest of vagina: highly elastic adventitia blending with surrounding connective tissue.

The External Genitalia

- **Vulva (puddendum)** = region enclosing female external genitalia. Vagina opens into the **vestibule** (central space bounded by **labia minora** — smooth hairless skin folds). Urethra opens into vestibule anterior to vaginal entrance; **urethral glands** discharge near external urethral orifice. **Clitoris** projects into vestibule anterior to urethral orifice — contains erectile tissue homologous to male corpora cavernosa; engorges during arousal; small erectile **glans** atop it; labia minora extensions encircle clitoral body forming its **prepuce**.
- **Lesser vestibular glands** (variable number) moisten the exposed vestibule surface. **Greater vestibular glands** (pair) discharge secretions near posterolateral vaginal entrance margins during arousal — resemble male bulbo-urethral glands.
- Vulva anterior limit = **mons pubis** (adipose tissue bulge anterior to pubic symphysis); lateral limits = **labia majora** (also adipose-tissue-containing, homologous to male scrotum) — encircle/partially conceal labia minora and vestibular structures; outer surface has coarse hair (like mons pubis), inner surface relatively hairless; sebaceous glands + scattered apocrine sweat glands moisten/lubricate inner surface.

The Breasts

- Milk production (**lactation**) occurs in mammary glands (specialized female reproductive accessory organs) within the breasts, nourishing the newborn.

- Breasts lie in subcutaneous tissue of the pectoral fat pad, either side of chest. **Nipple** = small conical projection where mammary gland ducts open onto body surface. **Areola** = reddish-brown skin region around nipple; large sebaceous glands give it a granular texture.
- Glandular tissue (apocrine) = several separate **lobes**, each with multiple secretory **lobules**; lobule ducts converge into a single **lactiferous duct** per lobe; near nipple, duct expands into a **lactiferous sinus** (15-20 sinuses typically open onto each nipple's surface). Dense connective tissue surrounds duct system, forms inter-lobe/lobule partitions = **suspensory ligaments of the breast** (originate in overlying skin dermis). Loose connective tissue separates mammary complex from underlying pectoralis muscles. Blood supply: internal thoracic artery branches.
- Resting (non-pregnant) mammary gland dominated by duct system, not active glandular cells; breast size reflects adipose tissue amount, not glandular tissue; secretory apparatus doesn't develop until pregnancy.
- Clinical: **Breast cancer** — 1 in 8 American women develop it in her lifetime; only ~10% due to inherited gene mutations, but BRCA1/BRCA2 mutation carriers have significantly elevated lifetime risk; can (rarely) affect men too. Cause unknown; healthy weight, physical activity, limited alcohol help reduce risk; regular self-exam + mammography screening important (annual mammograms recommended starting mid-40s). Signs: breast lump/thickening feeling different from surrounding tissue, bloody nipple discharge, size/shape/appearance change, skin changes (dimpling, redness, flaking, orange-peel pitting), newly inverted nipple. Most begin in lactiferous ducts (**ductal carcinoma**); some in lobules (**lobular carcinoma**). Staged by tumor size, local lymph node spread evidence, distant metastasis evidence (spreads via lymphatics — draining lymph nodes sampled for tumor cells); graded by growth/spread speed; tested for hormone receptors (estrogen/progesterone receptor-positive tumors may respond to hormone therapy); gene pattern research ongoing to guide treatment/prognosis.
- **Mammary gland development during pregnancy**: requires anterior pituitary hormones (**prolactin/PRL**, **growth hormone/GH**) + placental **human placental lactogen (HPL)**. Ducts become mitotically active, gland cells appear; by end of 6th pregnancy month, glands fully developed, producing secretions stored in duct system. Milk released via infant suckling stimulation → posterior pituitary releases **oxytocin** → contracts smooth muscle in lactiferous duct/sinus walls → milk ejection.

27.4 Aging and the Reproductive System

- Female reproductive system: most striking age-related changes occur at **menopause**; male changes are gradual, over a longer period.
- **Menopause**: ovulation/menstruation cessation, typically age 45-55; preceding years show gradually irregular cycles; premature menopause = before age 40 (underlying cause: shortage of primordial follicles).
- Hormonal changes at menopause: rapid sustained rise in GnRH, FSH, LH production; declining circulating estrogen/progesterone. Estrogen decline effects: (1) reduces uterus size, (2) reduces breast size, (3) thins urethral/vaginal walls, (4) weakens connective tissue supporting ovaries/uterus/vagina. Reduced estrogen also linked to osteoporosis (brittle bones) and cardiovascular/neural effects including hot flashes, anxiety, depression.
- **Male climacteric**: gradual changes over ages 50-60; declining circulating testosterone with increasing FSH/LH. Sperm production continues (men can father children into their 80s), but gradual reduction in sexual activity occurs, possibly linked to declining testosterone. Testosterone replacement therapy sometimes suggested to enhance libido in older men (and women).

Clinical Correlations Summary (Ch 27)

- **Ovarian cancer**: uncommon but deadly; usually develops in germinal epithelium; no screening test exists, no early symptoms, often undetected until spread within peritoneum; advanced disease hard to treat (surgery + chemo). Signs of advanced disease: lower abdominal pain/heaviness, abnormal vaginal bleeding, abnormal periods, weight gain/loss, gas/nausea/appetite loss. Risk reduced by early full-term pregnancy (and with each subsequent pregnancy), oral contraceptives, tubal ligation, hysterectomy (without ovary removal). Risk increases with age, family history, BRCA1/BRCA2 mutations (BRCA1 = tumor suppressor gene; normal-gene lifetime ovarian cancer risk <2%).
- **Testicular torsion** (Clinical Case — Liam): testis twists on its spermatic cord, cutting off blood supply (testicular artery inflow, pampiniform plexus outflow) and twisting ilio-inguinal/genitofemoral nerves; often

from an inherited trait allowing free testicular rotation in the scrotum; can occur during sleep, after injury, or vigorous activity. Signs: tender, elevated (higher-riding) testis, swelling, epididymis displaced anteromedially (near scrotal raphe instead of normal posterior position), absent cremasteric reflex on affected side (stroking inner thigh doesn't retract the testis). Diagnosed via bedside ultrasonography confirming absent blood flow; emergency surgical detorsion required to save the testis.

- **Cervical cancer, uterine fibroids, endometrial cancer:** described above.
- **Breast cancer:** described above.
- Related clinical terms: **amenorrhea** (menstruation not starting by 16, or stopping ≥ 6 months during reproductive years); **cervical dysplasia** (abnormal cervical epithelial cell growth, may progress to cancer); **endometriosis** (endometrial tissue growth outside the uterus, causes intermittent pain especially if within uterine tubes/peritoneal cavity); **genital herpes** (STD, herpes virus, painful genital blisters); **gonorrhea** (STD, *Neisseria gonorrhoeae*, "the clap"); **gynecology** (medicine branch for female reproductive function/disease); **gynecomastia** (male breast tissue development, e.g. from adrenal/chemotherapy/marijuana causes); **hydrocele** (serous fluid accumulation in tunica vaginalis or along spermatic cord); **menorrhagia** (extremely heavy menstrual bleeding); **oophorectomy** (surgical ovary removal); **oophoritis** (ovarian inflammation); **orchitis** (testicular inflammation); **polycystic ovary syndrome (PCOS)** (most common female endocrinopathy, most common cause of anovulatory infertility); **premature ejaculation** (~30% of men regularly affected); **premenstrual dysphoric disorder (PMDD)** (physical/emotional symptoms 5-11 days before menstruation, >150 associated signs/symptoms); **premenstrual syndrome (PMS)** (physical/mood disturbances after ovulation, ending with menstrual flow onset); **pseudohermaphrodites** (external genitalia not matching gonads); **salpingitis** (uterine tube inflammation); **uterine prolapse** (weakened pelvic floor muscles/ligaments allow uterus to descend into vaginal canal); **vasectomy** (surgical removal of a ductus deferens segment, sterilization); **vulvovaginal candidiasis** (yeast infection, usually *Candida albicans*).

Embryology and Development (Ch 28)

28.1 An Overview of Development

- **Development** = gradual modification of anatomical structures from conception to maturity. **Differentiation** = formation of specialized cell types during development, via selective changes in genetic activity.
- Development is a continuum, divided into 2 periods:
 - **Prenatal development** — begins at fertilization/conception (sperm + oocyte fuse), continues to delivery. **Embryology** = study of developmental events during prenatal development.
 - **Postnatal development** — begins at birth, continues to maturity (full development/completed growth).
- Prenatal development subdivisions:
 - **Pre-embryonic development** — first 2 weeks after fertilization; produces a pre-embryo that implants in the uterine lining.
 - **Embryonic development** — 3rd through 8th weeks.
 - **Fetal development** — begins at start of 9th developmental week, continues until birth.

28.2 Fertilization

- Typically occurs near the junction between the ampulla and isthmus of the uterine tube, usually within a day of ovulation. Oocyte travels only a few cm during this ~24-hour window while sperm travel from vagina to ampulla.
- Sperm arriving in the vagina are already motile but must undergo **capacitation** within the female reproductive tract before fertilizing an oocyte. Sperm transit time (vagina to fertilization site): 30 minutes to 2 hours; aided by uterine muscle contractions and ciliary currents in the uterine tubes.
- Of ~200 million sperm in a typical ejaculate, only ~10,000 enter the uterine tube, and fewer than 100 reach the isthmus. Sperm count <20 million/mL = functional sterility (too few survive to fertilize). Although only one sperm fertilizes the oocyte, dozens are required for successful fertilization (to break down the corona radiata).

- Fertilization is complicated by 3 factors: (1) oocyte surrounded by the **corona radiata** (follicle cell layer), (2) oocyte's metabolic operations discontinued, (3) oocyte is in metaphase of the second meiotic division when it leaves the follicle.
- Corona radiata protects the oocyte through the ruptured follicular wall and infundibulum. Sperm must penetrate the corona radiata via the **acrosome reaction** — releases acrosomal enzymes including **hyaluronidase**, which breaks down bonds between follicle cells; dozens of sperm must release hyaluronidase before an intact sperm can reach the oocyte.
- The penetrating sperm (with intact acrosome) binds to sperm receptors in the **zona pellucida**, triggering acrosome rupture. Sperm contacts oocyte surface, cell membranes fuse, sperm enters the **ooplasm** (oocyte cytoplasm) — triggers **oocyte activation**: metabolic rate rises suddenly, plasma membrane changes prevent additional sperm fertilization (blocks **polyspermy** — if it occurs, normal development cannot proceed).
- Oocyte completes meiosis after activation; nuclear material reorganizes as the **female pronucleus**. Sperm nucleus swells, becomes the **male pronucleus**, migrates to cell center; pronuclei fuse = **amphimixis**. Resulting cell = **zygote** with normal diploid 46 chromosomes = "moment of conception."
- Fertilization = fusion of 2 haploid gametes (23 chromosomes each) → zygote (46 chromosomes, diploid). Sperm delivers paternal chromosomes only; oocyte provides all nourishment/genetic programming to support embryonic development for ~1 week post-conception — hence oocyte diameter is more than twice the entire sperm length.
- First cleavage yields 2 daughter cells; cleavage divisions subdivide the zygote's cytoplasm.

28.3 Prenatal Development

- Gestation period = 3 trimesters, ~3 months each.
 - **First trimester**: pre-embryonic, embryonic, early fetal development; beginnings of all major organ systems appear.
 - **Second trimester**: fetal organ/organ system development; body proportions change; by end, fetus looks distinctively human.
 - **Third trimester**: rapid fetal growth; most major organ systems become fully functional early in this trimester; infant born 1-2 months premature has reasonable survival chance.

The First Trimester — 4 general processes:

1. **Cleavage**: cell division sequence beginning immediately after fertilization, ending at first zygote-uterine wall contact; zygote → pre-embryo → **blastocyst** (multicellular complex).
2. **Implantation**: begins when blastocyst attaches to uterine endometrium, continues as it burrows into the uterine wall.
3. **Placentation**: blood vessel formation around blastocyst edges — first step in placenta formation (links maternal/embryonic systems, provides respiratory/nutritional support).
4. **Embryogenesis**: formation of a viable embryo, establishing foundations for all major organ systems.
 - By end of first trimester (12th week): fetus ~75 mm (3 in.) long, ~14 g (0.5 oz). Only ~40% of conceptions survive the first trimester due to process complexity — pregnant women advised to avoid drugs/alcohol/disruptive stress during this period.
 - **Cleavage/blastocyst formation**: series of cell divisions subdividing zygote cytoplasm, producing increasingly smaller daughter cells (**blastomeres**). First cleavage completes ~30 hours post-fertilization; subsequent divisions every 10-12 hours (less predictable as blastomere number rises). After 3 days: solid ball of cells = **morula**. After 5 days: hollow ball = **blastocyst**, with inner cavity = **blastocoele**. Outer layer = **trophoblast** (provides food to developing embryo; only pre-embryo cells contacting uterine wall). Second cell group = **inner cell mass** (clustered at one end, exposed to blastocoele on one side, insulated by trophoblast on the other) — stem cells producing all body cell types.
 - **Implantation**: zygote reaches uterus ~4 days post-fertilization as a morula; blastocyst forms over next 2-3 days, absorbing nutrients from glycogen-rich uterine cavity fluid (endometrial gland secretions). Blastocyst contacts endometrium, implants (usually fundus or body of uterus).

- Trophoblast cells at contact point divide rapidly (multi-layer thick); near endometrial wall, plasma membranes between trophoblast cells disappear, creating a multinucleated cytoplasm layer = **syncytiotrophoblast**.
- Syncytiotrophoblast erodes a path through uterine epithelium by secreting hyaluronidase (dissolves proteoglycans between epithelial cells — same mechanism as sperm penetration of corona radiata). Gap initially forms, then maternal epithelial cells repair the surface; further blastocyst development occurs entirely within the endometrium's functional layer.
- Syncytiotrophoblast enlarges/spreads into endometrium, disrupting/digesting uterine glands; released nutrients absorbed by syncytiotrophoblast, diffuse across underlying **cytotrophoblast** to the inner cell mass (energy for early embryo formation). Trophoblast cytoplasmic extensions grow around endometrial capillaries, destroying capillary walls; maternal blood flows slowly through trophoblastic channels = **lacunae**. Fingerlike **primary villi** (syncytiotrophoblast extension + cytotrophoblast core) extend into surrounding endometrium; trophoblast begins breaking down larger endometrial veins/arteries, increasing lacunar blood flow.
- **Formation of the blastodisc:** inner cell mass separates from trophoblast, creating the fluid-filled **amniotic cavity**. Inner cell mass organizes into a 2-layer-thick oval sheet = **blastodisc: epiblast** (faces amniotic cavity) and **hypoblast** (exposed to blastocoele contents).
- **Gastrulation and germ layer formation:** third layer forms via cell migration (**gastrulation**) — epiblast cells migrate toward the blastodisc center, toward the **primitive streak**, then move between epiblast and hypoblast, creating 3 distinct **germ layers: ectoderm** (remains in contact with amniotic cavity), **endoderm** (from hypoblast), **mesoderm** (new middle layer). Gastrulation produces the 2-layer **embryonic disc**, forming the embryo's body; all other blastocyst cells become extra-embryonic membranes.
- **Extra-embryonic membranes (4):** (1) **yolk sac** (endoderm + mesoderm), (2) **amnion** (ectoderm + mesoderm), (3) **allantois** (endoderm + mesoderm), (4) **chorion** (mesoderm + trophoblast). Support development by maintaining stable environment, providing O₂/nutrient access via maternal blood; leave few traces in adult systems.
 - **Yolk sac:** first to develop; migrating hypoblast cells spread around blastocoele's outer edge forming a complete pouch below the blastodisc (visible 10 days post-fertilization); mesodermal cells migrate around it during gastrulation, completing formation; blood vessels appear within mesoderm — yolk sac becomes an important early blood cell formation site.
 - **Amnion:** ectodermal layer expands, spreads over amniotic cavity's inner surface; mesodermal cells follow, forming a second outer layer; combination = amnion. Expands as embryo/fetus enlarges; amniotic cavity contains **amniotic fluid** (surrounds/cushions developing embryo/fetus).
 - **Allantois:** begins as an endoderm outpocketing near the yolk sac base (week 3); free endodermal tip grows toward the blastocyst wall, surrounded by mesodermal cell mass; base later gives rise to the urinary bladder.
 - **Chorion:** mesoderm associated with the allantois spreads around the trophoblast's inside, forming a mesodermal layer beneath the trophoblast; combination of mesoderm + trophoblast = chorion. As embryo/trophoblastic complex enlarge, simple diffusion can no longer meet developing embryo's demands; blood vessels developing within chorionic mesoderm link embryo with trophoblast — circulation through chorionic vessels begins early in week 3, when the heart starts beating.
- **Placentation:** chorionic blood vessel appearance = first functional placenta formation step. By week 3, mesoderm extends along each trophoblastic villus core, forming **chorionic villi** in contact with maternal tissue; villi enlarge/branch, forming an intricate endometrial network; maternal vessels continue eroding, maternal blood flows through syncytiotrophoblast-lined lacunae; diffusion occurs between maternal blood (lacunae) and fetal blood (chorionic villi vessels).
 - Entire blastocyst initially surrounded by chorionic villi; chorion enlarges like a balloon within the endometrium. **Capsular decidua (decidua capsularis)** = thin endometrium portion covering the embryo, separating it from the uterine cavity — no longer exchanges nutrients, chorionic villi disappear here. **Basal decidua (decidua basalis)** = disc-shaped region in deepest endometrium where placental function concentrates. **Parietal decidua (decidua parietalis)** = rest of endometrium, no chorion contact.

- Germ layer contributions table (partial): **Ectodermal**: integumentary system (epidermis, hair follicles/hairs, nails, glands), pharyngeal cartilage skeletal derivatives (portion of sphenoid, auditory ossicles, styloid processes, hyoid horns/superior rim — via neural crest), all nervous tissue (brain, spinal cord), pituitary gland and adrenal medullae, nasal passageway mucous epithelium, oral cavity/anus mucous epithelium, salivary glands. **Mesodermal**: dermis, all skeletal system except some pharyngeal derivatives, all muscular system, adrenal cortex and endocrine tissues of heart/kidneys/gonads, all cardiovascular system (incl. bone marrow), all lymphatic system, kidneys (nephrons, initial collecting system), gonads and adjacent duct portions, body cavity linings and connective tissue supporting all organ systems. **Endodermal**: thymus, thyroid, pancreas (endocrine), respiratory epithelium (except nasal passageways) and mucous glands, digestive mucous epithelium (except mouth/anus), exocrine glands (except salivary), liver, pancreas, urinary bladder and distal duct system, distal reproductive duct portions and gamete-producing stem cells.
- **Embryogenesis**: shortly after gastrulation begins, embryo body separates from the rest of the embryonic disc — folding/differential growth produces a **head fold** (bulge into amniotic cavity) and later a **tail fold** (weeks 3-4); embryo becomes physically/developmentally separate from the blastodisc and extra-embryonic membranes; anterior/posterior surfaces and left/right sides become visible. First trimester = critical period establishing organ formation basis = **organogenesis**.
 - Connection between embryo and chorion = **body stalk** (contains distal allantois portions, placental blood vessels). Narrow connection between embryo endoderm and yolk sac = **yolk stalk**.
 - Clinical: **Congenital malformations (birth defects)** = structural/functional anomalies present at birth, occurring during intrauterine life. Common examples: congenital heart disease, Down syndrome, cleft lip/palate, spina bifida, clubfoot deformity, phenylketonuria (PKU — enzyme deficiency processing phenylalanine). 94% of severe congenital malformations occur in low/middle-income countries worldwide. Up to half of all conceptions produce zygotes not surviving cleavage; ≥10% of fertilizations produce chromosomally abnormal zygotes (most don't complete development). Of implanted embryos, ~20% fail to complete 5 months of development (average survival ~8 weeks) = **spontaneous abortion (miscarriage)**. <1% of newborns show chromosomal abnormalities from spontaneous mutations. Of 2.75 million annual worldwide neonatal deaths, 10% are due to congenital malformations. **Consanguinity** (biological parent blood relation) increases rare genetic condition prevalence, nearly doubles neonatal death risk. Maternal infections (rubella, syphilis, Zika) can cause malformations; maternal nutrition (folate/iodine deficiency, obesity, diabetes) linked to some defects — folate deficiency increases neural tube defect (spina bifida, anencephaly) risk. Environmental factors (alcohol, tobacco, certain medications, street drugs, radiation) are risk factors. Advanced maternal/paternal age increases chromosomal condition risk (e.g., Down syndrome/trisomy 21 — extra chromosome 21). Screening: preconception screening, amniocentesis (amniotic fluid testing), chorionic villus sampling (1st/2nd trimester), neonatal screening (blood disorders, metabolism/hormone defects, congenital deafness, heart defects). Public health approach needs: surveillance, causal research, prevention research/programs.
 - Clinical: **Teratogen** = any agent causing embryo/fetal abnormality during prenatal development; fetus most vulnerable during first trimester. Types: physical agents (high fever), maternal health factors (diabetes, obesity, hypertension), environmental chemicals (herbicides, industrial solvents), infectious agents (rubella, Zika). Most frequent preventable cause of abnormal fetal development in the US = maternal tobacco/alcohol/drug use. Smoking nearly doubles low-birth-weight risk (poor prenatal growth and/or premature delivery). **Fetal alcohol syndrome (FAS)** = most common known non-genetic cause of growth deficiency, functional deficits, intellectual impairment; alcohol crosses placenta, broken down more slowly in fetus than adult (prolonged high fetal alcohol levels); effects: small heads, characteristic facial abnormalities, short stature, low weight, poor motor coordination, speech/language delays, attention/memory deficits, vision/hearing problems, heart/kidney/bone abnormalities — irreversible, severity varies by child. Maternal methamphetamine/heroin/codeine/oxycodone/methadone use → fetal addiction → **neonatal abstinence syndrome** after birth (hyperreflexia, increased muscle tone, irritability, poor feeding, vomiting, diarrhea, sleep difficulties, seizures; lifelong CNS/learning/behavioral effects).

Placental Circulation and Hormones

- Blood flows fetus → placenta via paired **umbilical arteries**, returns via single **umbilical vein**. Active/passive exchange between fetal/maternal bloodstreams occurs at chorionic villi.
- Placenta synthesizes hormones: **Human chorionic gonadotropin (HCG)** production begins days after implantation — stimulates corpus luteum to continue progesterone production through early pregnancy (basis of

pregnancy tests). During 2nd/3rd trimesters, placenta also releases progesterone, estrogens, **human placental lactogen (HPL)**, and **relaxin** — synthesized/released into maternal circulation by the trophoblast.

- Mother must support both herself and fetus — breathe, eat, excrete "for two"; demands increase significantly in later trimesters.
- Uterine growth: 7.5 cm (3 in.) to 30 cm (12 in.) long by end of gestation, containing ~5 L fluid; uterus + contents weigh ~10 kg (22 lb); expansion occurs via enlargement/elongation of existing smooth muscle fibers (not new fiber formation). Full-term pregnancy pushes many abdominal organs out of normal position.

Second and Third Trimesters

- By end of first trimester, foundations of all major organ systems have formed; second trimester completes their functional development. By end of 2nd trimester, fetus weighs ~0.64 kg (1.4 lb); fetus (encircled by amnion) grows faster than surrounding placenta; amnion's mesodermal outer covering fuses with chorion's inner lining.
- Third trimester: all fetal organ systems become functional; growth rate slows but this trimester has greatest weight gain — fetus gains ~2.6 kg (5.7 lb) in the last 3 months, reaching full-term weight ~3.2 kg (7.05 lb).
- Developmental milestones table highlights (by gestational month, "(b)" = begins, "(c)" = completes): Month 1 (5mm, 0.02g) — somite formation, neural tube formation, eye/ear formation begin, heartbeat begins, trachea/lung formation begins, allantois complete. Month 2 (1.6cm, 1.0g) — nail beds/hair follicles/sweat glands form, axial/appendicular musculature forms, CNS/PNS organization, taste buds/olfactory epithelium form, thymus/thyroid/lymph node formation begins, major blood vessels form, bronchial subdivisions/salivary glands form, kidney formation (adult form) begins, mammary gland formation begins. Month 3 (5.4cm, 14g) — epidermal layers appear, ossification centers spread, basic spinal cord/brain structure complete, thymus/thyroid complete, tonsils/blood formation in marrow begins, gallbladder/pancreas complete, definitive gonads/ducts/genitalia form. Month 4 (11.6cm, 100g) — hair/sebaceous glands form, facial/palatal organization completes, fetus starts moving, rapid cerebrum expansion, basic eye/ear structure complete, lymphocyte migration to lymphoid organs, embryonic kidney degeneration. Month 5 (16.4cm, 300g) — keratin/nail production, spinal cord myelination begins, tonsils complete, nostrils open, intestinal subdivisions complete. Month 6 (30cm, 600g) — perineal muscles complete, CNS tract formation begins, adrenal glands complete, spleen/liver/bone marrow complete, alveolar formation begins, epithelial organization/testicular descent begins. Month 7 (37.6cm, 1.005kg) — keratinization/nail complete, eyelids open/retina complete, intestinal plicae complete. Month 8 (42.4cm, 1.702kg) — epiphyseal cartilage formation complete, taste receptors complete, complete pulmonary branching/function, complete nephron formation, testicular descent complete at/near birth.

28.4 Labor and Delivery

- **Parturition (childbirth)** = forcible expulsion of the fetus from the uterus through cervix and vagina, via strong rhythmic uterine contractions = **labor**. Triggered by increased oxytocin levels + increased uterine oxytocin sensitivity, stimulating myometrial contractions. True labor contractions begin near the uterine fundus, sweep toward the cervix, strong and regularly-timed (vs. occasional false labor spasms).
- **Stages of Labor (3):**

1. **Dilation stage:** begins with true labor onset; cervix dilates, fetus moves down cervical canal; highly variable length, typically ≥8 hours; contractions initially up to 30 sec, every 10-30 min; amnion usually ruptures late in process ("water breaking").

2. **Expulsion stage:** begins as cervix (pushed open by approaching fetus) completes dilation to ~10 cm (4 in.); continues until fetus completely emerges from vagina; usually <2 hours. Arrival of newborn = **delivery/birth**. If vaginal canal too small (perineal tearing risk), an **episiotomy** (perineal musculature incision) may be performed, later repaired with sutures (simpler than repairing a jagged tear). Relative fetal skull/maternal pelvic outlet size affects delivery ease; slow progress/complications may require **cesarean section (C-section)** (abdominal wall + uterine incision to remove infant).

3. **Placental stage:** muscle tension builds in the partially empty uterine walls (shrinking uterus), tearing endometrium-placenta connections; placenta (**afterbirth**) ejected usually within an hour of delivery; ~500-600 mL blood loss accompanies this (easily tolerated due to pregnancy-increased maternal blood volume).

- **Premature labor:** true labor begins before fetus completes normal development. Newborn survival chances directly relate to birth weight. Most fetuses born at 25-27 weeks (birth weight <600g/21.1 oz) die despite

intensive care; survivors have high developmental abnormality risk. **Premature delivery** generally = birth at 28-36 weeks (birth weight >1 kg/2.2 lb) — good survival/normal development chances with care.

- Clinical: **Common complications of parturition** — **fetal distress** (inadequate fetal oxygen, manifests as decreased fetal heart rate; may require emergency C-section or forceps/vacuum-assisted delivery). **Breech position (malpresentation)** — baby positioned buttocks/feet first; risk of getting "stuck" (head is largest part) — indication for C-section. **Placenta previa** — placenta covers the cervix; natural labor risks early placental separation and dangerous blood loss for mother/baby; C-section often scheduled before natural labor. **Meconium aspiration** — fetus inhales amniotic fluid contaminated with meconium (first feces) in utero, causing breathing difficulty; risk factors: post-date "aging" placenta, decreased fetal oxygen, maternal diabetes/hypertension, long/difficult labor. **Cephalopelvic disproportion (CPD)** — fetal head too large for maternal pelvis, preventing vaginal delivery; most common C-section indication.
- Clinical: **Multiple births** — increasing due to fertility drugs, assisted reproductive technology, rising maternal age at birth (older women more likely to release multiple oocytes per cycle). US twin birth rate: 33.7 per 1,000 deliveries (per CDC). **Dizygotic ("fraternal") twins** — 2 separate oocytes ovulated/fertilized; due to meiotic chromosome shuffling, odds against 2 zygotes from the same parents having identical genes exceed 1 in 8.4 million. **Monozygotic ("identical") twins** — result from blastomere separation early in cleavage, or inner cell mass splitting before gastrulation; genetically identical (same gamete pair origin). Triplets/quadruplets/larger multiples: multiple ovulations, blastomere splitting, or combination. Natural multiple birth rate pattern: twins 1/89 births, triplets 1/7,921 births, quadruplets 1/704,969 births. Fertility drugs stimulating abnormally high follicle maturation increase multiple birth incidence.

28.5 The Neonatal Period

- **Neonatal period** = birth to 1 month after. Newborn (**neonate**) has few mature-adult anatomical/functional/physiological characteristics; developmental processes continue post-delivery.
- Before delivery, gas/nutrient/waste/hormone/immunoglobulin transfer occurred across the placental interface; at birth, the newborn must become relatively self-sufficient. Transition summary:
 1. At birth, lungs are collapsed and fluid-filled; filling with air requires a massive, powerful first inhalation.
 2. Lung expansion alters cardiovascular blood pressure/flow patterns: **ductus arteriosus** closes (isolating pulmonary/systemic trunks), **foramen ovale** closes (separating atria) — completes pulmonary/systemic circuit separation.
 3. Normal neonatal vitals considerably higher than adult: heart rate 120-140 bpm, respiratory rate 30 breaths/min.
 4. Digestive system relatively inactive before birth, though it accumulates **meconium** (bile secretions, mucus, epithelial cells mixture) — excreted in the first few days of life; newborn begins nursing during this period.
 5. As arterial blood wastes build up, kidneys begin excreting them; glomerular filtration is normal, but neonatal kidneys cannot significantly concentrate urine — high urinary water losses mean proportionally much greater neonatal fluid requirements than adults.
 6. In the first days post-delivery, neonate has little body temperature control ability; as the infant grows and its subcutaneous adipose "blanket" thickens, metabolic rate rises; daily/hourly body temperature fluctuations continue throughout childhood.

28.6 Embryology of Organ Systems (Embryology Summaries)

- Embryonic primordia and their derivatives maintain definite anatomical relationships that persist into the newborn and adult.

Integumentary System

- Epidermal columns grow into the dermis (months 3-4), forming hair follicles, sebaceous glands, sweat glands. Melanocytes migrate into the epidermis, giving it an adult-like appearance. Embryonic connective tissue differentiates into the dermis (fibroblasts form collagen networks) and the subcutaneous layer. Nails begin as epidermal thickenings near finger/toe tips (month 4); by birth, nail production is restricted to the nail root. Hair follicles develop around a papilla (connective tissue mass, month 5); at birth, a hair projects from the follicle, sebaceous secretions lubricate the shaft; sweat gland ducts function at birth. Mammary glands develop similarly

to sweat glands with broader branching; at birth, mammary glands haven't completed development — in females, further duct/gland elaboration occurs at puberty, functional maturity only late in pregnancy.

Skull (Cranium)

- Pharyngeal arches (5 rounded cartilage ridges either side of pharynx) form the face and neck. Cartilages around the developing brain/nose/eyes/ears fuse (weeks 5-9) into the cartilaginous **chondrocranium**. Mandibular arch's dorsal portion fuses with the chondrocranium; fused cartilages don't ossify — osteoblasts cover them in dermal bone, fusing with nasal cavity entrance bone to form the maxillae; ossification centers in the mouth roof form the palatine processes, fusing with maxillae. Second (hyoid) pharyngeal arch forms near temporal bones — fusion of its superior tips with temporals forms the styloid processes; ventral hyoid arch portion ossifies as the hyoid bone; third arch fuses with hyoid; fourth/sixth arches form laryngeal cartilages. By 12 weeks, ossification well underway in cranium/face; cranial roof remains incomplete at birth (frontal/parietal bones via intramembranous ossification). Clinical: **cleft lip** (skin fusion failure, ~1/1000 births) and **cleft palate** (split into orbit/palate, half as common as cleft lip) — both surgically correctable.

Vertebral Column and Ribs (Skeletal — axial)

- Notochord lies anterior/deep to the spinal cord. Week 4: mesoderm forms **somites** on either side of notochord/spinal cord; medial somite mesenchyme (**sclerotome**) produces the vertebral column, contributes to cranial floor. Sclerotomal cells migrate around the notochord, differentiate into chondroblasts, forming cartilaginous blocks (future vertebral centra) around the notochord. Vertebral centra cartilages grow around the spinal cord, forming a complete vertebral model; articulations develop between adjacent cervical/thoracic/lumbar cartilage blocks; sacral/coccygeal cartilages fuse. Notochord is eventually eliminated by vertebral centra expansion, except between vertebrae, where it becomes the **nucleus pulposus** of intervertebral discs; surrounding mesenchyme forms the **anulus fibrosus**. Ribs form at every vertebra but remain small/fuse with growing vertebrae in cervical/lumbar/sacral/coccygeal regions; thoracic vertebra ribs enlarge, following body wall curvature, fusing with sternal cartilage at the ventral midline. At birth: vertebrae/ribs ossified but cartilage remains (e.g., anterior rib portions = **costal cartilages**); neural arch bases enlarge until ages 3-6; vertebral processes/bodies grow until ages 18-25.

Appendicular Skeleton

- Week 4: paired mesodermal ridges appear along the embryo's abdomen (throat to anus); mesoderm accumulates at ridge ends forming 2 pairs of **limb buds**. By week 5, pectoral limb buds have a cartilaginous core with scapular cartilages developing in trunk mesenchyme. Pelvic girdle/limb formation parallels the pectoral pattern (knee faces posteriorly, elbow faces anteriorly due to differing limb bud rotation). By week 8, cartilaginous models of major skeletal components are well-formed; endochondral ossification begins ~week 10; hip bone ossification begins at 3 separate centers. Joints form where cartilages contact — joint cavity surfaces remain cartilaginous while the rest of the bone ossifies.

Muscular System

- Mesoderm forms skeletal muscle associated with pharyngeal arches (near the head) and limb buds. Somite medial portion = **myotome** (forms skeletal muscles). Lateral plate mesoderm's peripheral portion forms; a cavity (**coelom**) appears within it, dividing it into inner visceral and outer parietal layers. Myotomal muscles organize around the developing vertebral column into **epaxial muscles** (above spinal axis — remain segmented, e.g., intervertebral muscles, erector spinae) and **hypaxial muscles** (below spinal axis — form oblique/transverse/rectus muscle groups). Pharyngeal arch mesoderm: 2nd (hyoid) arch → muscles of facial expression; 3rd/4th/6th arches → pharyngeal/intrinsic laryngeal muscles; mandibular arch mesoderm → muscles of mastication; also gives rise to trapezius/sternocleidomastoid. Limb bud myoblasts (from nearby myotomal segments) migrate in — dorsal surface migration creates limb extensors, ventral migration creates flexors.

Nervous System

- Week 2: ectoderm near the midline thickens, forming the elevated **neural plate** (largest near the future head). A crease (**neural groove**) develops along the plate's axis; edges (**neural folds**) move together, first contacting near the plate's midpoint (~end of week 3). Folds fuse into a cylindrical **neural tube**, losing connection with superficial ectoderm — this process = **neurulation**, completed in <1 week. **Neural crest** cells (at fold tips) don't form the neural tube; they migrate to various locations, forming much of the PNS (dorsal root ganglia,

sensory neurons, Schwann cells, satellite cells, autonomic ganglia, also contribute to teeth, laryngeal cartilages, melanocytes, skull, eye connective tissue/intrinsic muscles, adrenal medullae).

- Neural tube thickens via repeated mitosis (mid-week 5): 3 layers form — **ependymal layer** (lines neural cavity, continues mitosis), **mantle layer** (surrounding, produces gray matter — first-arriving cells become neurons, last-arriving become astrocytes/oligodendrocytes), **marginal layer** (superficial, contains axons — becomes white matter as axon tracts crowd together).
- Spinal cord: dorsal root ganglia form from neural crest cells by week 7; ventral roots form from motor neuron axons; dorsal (sensory) + ventral (motor) roots combine into spinal nerves distal to each dorsal root ganglion — stereotyped branching pattern accounts for dermatome distribution. Neural cavity becomes laterally compressed; **dorsolateral plate** (sensory relay) and **ventrolateral plate** (motor) enlarge rapidly; roof plate/floor plate remain thin.
- Brain: cephalic neural tube portion enlarges even before neural tube formation completes, forming 3 primary regions: **prosencephalon** (forebrain), **mesencephalon** (midbrain), **rhombencephalon** (hindbrain) — by week 4. Brain vs. spinal cord development differences: (1) early breakdown of mantle/marginal organization, (2) neural cortex appearance, (3) differential regional growth, (4) characteristic bends/folds (flexures), (5) loss of segmental organization. Rhombencephalon subdivides into **metencephalon** and **myelencephalon**. Prosencephalon forms the **telencephalon** and **diencephalon**. Mesencephalon develops the **aqueduct of the midbrain**. By week 11, expanding cerebral hemispheres overgrow the diencephalon; metencephalon cortical formation/expansion produces the **cerebellum** (overlying pons nuclei/tracts). Migrating neuroblasts form the cerebral cortex; underlying gray matter masses form the basal nuclei.
- Clinical: **Spina bifida** — developing vertebral laminae fail to unite from abnormal neural tube formation; neural arch incomplete, meninges bulge under back skin; severity ranges from unnoticed to extensive vertebral column involvement. **Neural tube defect (NTD)** — developmental error in spinal cord formation (broad plate instead of hollow tube), often associated with spina bifida; affects ~1/1000 individuals; prenatal testing detects with 80-85% success.

Special Sense Organs

- All develop from epithelium-nervous system interaction. **Eye**: optic vesicles (paired bulges in prosencephalon lateral walls) become indented, forming **optic cups** connected to diencephalon via optic stalks; overlying epidermis forms a **lens placode** → thickens → **lens vesicle** → becomes the lens; mesoderm around this complex forms the choroid and scleral coats; anterior/posterior chambers develop as mesodermal cavities.
- **Olfactory**: paired thickened areas (**nasal placodes**) form in front of the prosencephalon (week 5); later enfolded/protected by developing facial structures.
- **Taste**: gustatory receptors = least specialized special sense organ; taste buds develop as nerve fibers grow into the developing mouth/pharynx; nerve endings contacting epithelial cells cause differentiation into gustatory cells; cut sensory nerves cause taste bud degeneration; relocated nerves stimulate new taste bud development at the new site.
- **Equilibrium/hearing**: paired **otic placodes** appear either side of the rhombencephalon (late week 3) → **otic vesicle** → forms the **membranous labyrinth**; will later be surrounded by ossifying **bony labyrinth**. Developing ossicles and sensory terminals grow toward developing hair cells.

Endocrine System

- All secretory glands (exocrine/endocrine) derive from epithelia: (1) covering the embryo exterior, (2) lining the digestive tract, (3) lining the neural cavity.
- **Pharyngeal pouches** (5, extending laterally toward pharyngeal clefts) play a key endocrine developmental role. 3rd/4th pouch dorsal masses form the **parathyroid glands**; ventral masses fuse to create the **thymus**. 5th pouch wall cells incorporate into the thyroid gland, differentiating into **C thyrocytes** — completed by end of month 3.
- **Thyroid gland**: begins as a ventral midline pharyngeal pocket; branches, thickens, loses surface connection; shifts caudally to a position near the thyroid cartilage as the embryo enlarges, incorporating C thyrocytes from the 5th pouch en route.
- **Pituitary gland** (compound origin): ectodermal pocket forms in the pharynx's dorsal midline, loses pharyngeal connection, becomes a hollow cell ball inferior to the diencephalon floor (posterior to optic chiasm) — this

becomes the **anterior lobe (adenohypophysis)**. **Posterior lobe (neurohypophysis)** begins as a hypothalamic floor depression, grows toward the developing anterior lobe.

- **Adrenal glands** (compound origin): neural crest cells migrate from the CNS shortly after neural tube formation, aggregating on either side of the neural cavity to become the **adrenal medulla**; overlying epithelial cells divide, surrounding the neural crest cells to form the thick **adrenal cortex**.
- **Heart interatrial septa/foramen ovale**: two overlapping interatrial septa develop (week 5); gap between them (**foramen ovale**) permits right-to-left atrial blood flow; a one-way valve flap prevents left-to-right backflow; diverts blood from the pulmonary circuit until birth. At birth, the foramen ovale closes, separating pulmonary/systemic circuits; the remaining depression = **fossa ovalis**.

Cardiovascular System (Heart Development)

- Week 2: heart = pair of thin-walled muscular tubes beneath the pharynx floor. By week 3, heart is pumping/circulating blood; cardiac tubes fuse into a single central chamber; two large veins deliver blood, a single large artery (**truncus arteriosus**) carries blood to general circulation. Heart elongates, curves into an S-shape (increasingly pronounced) as embryo grows; atrial/ventricular regions differ in thickness by week 4.
- Pericardial cavity forms as parietal/visceral layers split (week 3).

Major Vessel Development

- Three vessel complexes tracked: **aortic arches**, **venae cavae**, **hepatic portal/umbilical vessels**.
- **Aortic arches**: each carries arterial blood through a pharyngeal arch, numbered 1-6; common carotid arteries fuse to create the dorsal aortae distributing blood throughout the body; some arches disintegrate as development proceeds; **ductus arteriosus** provides a short circuit between pulmonary/systemic circuits (most right-atrial blood bypasses the lungs via the ductus arteriosus or foramen ovale before birth). Left half of arch 4 becomes the definitive aortic arch (carries blood from the left ventricle).
- **Venae cavae**: body wall/limb/head venous drainage centers on paired anterior cardinal veins; interconnections form, fusion/disintegration produces more direct larger-diameter right atrial connections, ultimately producing the superior and inferior venae cavae.
- **Umbilical/hepatic portal vessels**: paired umbilical veins (week 4) return placental blood to liver capillary networks; veins draining the digestive tract fuse, forming the hepatic portal vein. By week 12, the right umbilical vein disintegrates — placental blood travels along a single (left) umbilical vein; **ductus venosus** allows some venous blood to bypass the liver, reaching the inferior vena cava.
- At birth: chest expansion/first breath drops pleural cavity pressure, pulmonary vessels dilate, pulmonary blood flow increases; right atrial pressure falls, higher left atrial pressure closes the foramen ovale valve; smooth muscle contracts the ductus arteriosus, which converts to the fibrous **ligamentum arteriosum**.

Lymphatic System

- Development closely tied to blood vessel formation. Paired **jugular lymph sacs** form near the neck midline; by week 7, lymphatic vessels extend. A large **median lymph sac** marks the future **cisterna chyli** location. As limb buds enlarge, lymphatic vessels grow into the area alongside developing arteries/veins. **Lymph nodes** gradually appear along lymphatic vessels by end of month 6. At birth, the **thymus** is relatively large, filling much of the anterior mediastinum.

Respiratory System

- A shallow **pulmonary groove** appears in the pharynx's midventral floor (~3.5 weeks), near the last pharyngeal arch level, gradually deepening. By week 4, the groove becomes a closed pocket extending caudally, anterior to the esophagus — becomes the **trachea**; its tip branches, forming paired **lung buds**, which branch repeatedly forming **bronchioles**. Over the following 3 months (and continuing variably after birth), each bronchiole gives rise to several hundred **alveoli**.
- Lung buds elongate into the mediastinum, projecting into the ventral body cavity posterior to the developing heart. Week 6: thin **pleuropericardial membrane** forms between heart and developing lungs. By week 8, the pericardial sac is complete, isolating the pericardial cavity from the rest of the ventral body cavity. By week 9, the diaphragm attaches to the pericardial sac and mediastinum tissues, separating the abdominopelvic cavity from the pleural cavities.

Digestive System

- Week 3: endodermal cells migrate around the blastocyst's inside, completing the **yolk sac** pouch. As the embryo forms on the embryonic shield, two endoderm pockets form: **foregut** and **hindgut**, connected to the yolk sac via the **yolk stalk**. The digestive tube is suspended in the ventral body cavity by dorsal and ventral mesenteries; the ventral mesentery disintegrates except where major vessels/organs grow into it (path of umbilical arteries, and where the umbilical vein/liver develop) — becomes the **greater omentum** and **lesser omentum**.
- Stomach and liver form as pockets budding from the foregut into the ventral mesentery; pancreas grows from adjacent digestive epithelium into the dorsal/ventral mesenteries respectively.
- Hindgut extends into the tail, forming the **cloaca**; a tubular allantois extension projects into the body stalk; fusion of yolk stalk and body stalk creates the **umbilical stalk (umbilical cord)**.
- A partition grows across the cloaca (~week 10), dividing it into a posterior **rectum** and anterior **urogenital sinus** (retains allantois connection).
- Intestines elongate, push into the umbilical stalk (temporary physiological herniation), continue elongating/coiling outside the embryo body, then by week 10 begin moving back into the visceral cavity while continuing to grow longer.

Urinary System

- Kidneys develop in 3 successive stages along the **urogenital ridge** (thickened mesoderm ridge): **pronephros** (~7 tubule pairs in the nephrotome — small, nonfunctional, disintegrate almost immediately; only lasting contribution = paired **pronephric ducts** growing caudally to connect with the cloaca), **mesonephros** (~4 weeks — ~70 tubules develop per side, grow toward and fuse with the pronephric duct, now called the **mesonephric duct**; does not persist — degenerates as its last segments are still forming), **metanephros** (becomes the functional adult kidney).
- A **ureteric bud (metanephric diverticulum)** forms in each mesonephric duct wall, elongates/branches within the adjacent metanephros; metanephros tubules connect to the ureteric bud's terminal branches, forming large nephrons with enormous glomeruli.
- Ureteric bud branches within the metanephros form the calyces and collecting system; nephrons form within metanephros mesoderm, tapping into collecting tubules. Kidneys begin producing filtrate by month 3 — the filtrate contains no waste products (waste excreted by the placenta via maternal kidneys); sterile filtrate mixes with amniotic fluid, is swallowed by the fetus, reabsorbed across the digestive tract lining.
- Most fetal metabolic wastes cross the placenta into maternal circulation; small kidney-produced urine amounts accumulate in the cloaca/allantois (endoderm-lined sac extending into the umbilical stalk).
- Near end of month 2, cloaca subdivides into dorsal rectum and ventral urogenital sinus; proximal allantois portions persist as the urinary bladder; the connection between bladder and body surface forms the urethra.

Reproductive System

- Week 3: **primordial germ cells** migrate from the yolk sac wall (near the allantois) to the dorsal abdominal wall, entering the **genital ridges** paralleling the mesonephros. Each ridge has thick epithelium continuous with **primary sex cords** extending into the ridge's medulla. Anterior to each mesonephric duct, the **paramesonephric (Müllerian) duct** forms (no kidney connection), extending along the genital ridge toward the cloaca. At this "sexually indifferent" stage, male and female embryos cannot be distinguished.
- Both sexes initially have mesonephric AND paramesonephric ducts. Unless exposed to androgens, the embryo (regardless of genetic sex) develops as female. In normal male embryos, genital ridge medulla cells begin producing testosterone after week 6, triggering duct system/external genitalia changes.
- **External genitalia** (indifferent stage, ~4 weeks): mesenchymal swellings = **cloacal folds** around the cloacal membrane; **genital tubercle** forms (→ penis glans in males, clitoris in females). ~2 weeks later, cloaca subdivides — cloacal membrane separates into posterior **anal membrane** (bounded by anal folds) and anterior **urogenital membrane** (bounded by urethral folds); a **genital swelling** forms lateral to each urethral fold.
- **Male reproductive development**: primary sex cords proliferate, germ cells migrate into the sex cords → **testis cords** → form the seminiferous tubules. By month 4, testis cords connect to mesonephric tubule remnants via the **rete testis**; paramesonephric duct degenerates. Mesonephric tubules/duct persist, becoming (after testicular descent into scrotum) the definitive male duct system (epididymis, ductus deferens, etc.) and prostate. Genital tubercle enlarges; urethral fold tips move together forming the spongy urethra (10 weeks); genital swellings form paired **scrotal swellings**. In the newborn male, the urethral fold fusion line is visible on the penis.

- **Female reproductive development:** primary sex cords degenerate; primordial germ cells migrate into the genital ridge's outer region (cortex) → ovary. Mesonephric tubules/duct degenerate; paramesonephric (Müllerian) duct develops a broad peritoneal cavity opening, fuses in the midline forming a common chamber (uterus), separate from the urogenital sinus. Urethral folds do NOT fuse — develop into the labia minora; genital swellings form the labia majora; genital tubercle develops into the clitoris; urethra opens immediately posterior to the clitoris; hymen = urogenital membrane elaboration.

Clinical Correlations Summary (Ch 28)

- **Congenital malformations, teratogens, fetal alcohol syndrome, neonatal abstinence syndrome:** described above (28.3).
- **Spina bifida, neural tube defects, cleft lip/palate:** described above (28.6, nervous system/cranium).
- **Complications of parturition:** fetal distress, breech position, placenta previa, meconium aspiration, cephalopelvic disproportion — described above (28.4).
- **Multiple births** (dizygotic vs monozygotic twins): described above (28.4).
- Clinical Case (Mayu and Ruchira): "least-alike twins" — conceived via **in vitro fertilization (IVF)** with hormonally-stimulated maternal oocytes harvested via laparoscopic surgery, fertilized in the lab; two blastocysts transferred to the uterus on post-fertilization day 6 (coinciding with when a naturally fertilized blastocyst would complete its uterine tube-to-uterus journey); both implanted, developed, and were born successfully. Despite different appearance, the twins were **dizygotic (fraternal)** — a lab technician error caused one oocyte to be fertilized with sperm from a different (previously-tested) man due to a reused pipette, so the twins had different biological fathers (superfecundation-type scenario) — different racial/ethnic sperm donors led to visibly different-appearing twins despite being carried and born together.
- Related clinical terms: **eclampsia** (convulsions in a pregnant woman with high blood pressure, may progress to coma, threatens mother/baby); **infertility** (inability to achieve pregnancy after 1 year of appropriately-timed intercourse); **in vitro fertilization** (fertilization outside the body, generally in a Petri dish); **gamete intrafallopian transfer (GIFT)** (assisted reproduction — eggs/sperm mixed and placed directly into the uterine tube rather than fertilized in the lab); **oligodactyly** (fewer than 5 digits on a limb at birth); **placenta abruptio** (placental separation from the uterine implantation site before delivery); **placenta previa** (placenta abnormally covers the cervix, totally or partially — most common pregnancy complication, affecting ~5% of pregnancies); **polydactyly** (more than 5 fingers/toes on a limb); **therapeutic cloning** (patient skin cell nucleus inserted into an enucleated fertilized egg, divides to form a blastocyst for genetically-matched stem cell/tissue extraction).